

THE MACHINE IN THE GHOST

RATIONALITY: FROM AI TO ZOMBIES

in six volumes

- I Map and Territory
- II How to Actually Change Your Mind
- III The Machine in the Ghost
- IV Mere Reality
- V Mere Goodness
- VI Becoming Stronger

ELIEZER YUDKOWSKY

Rationality

From AI to Zombies

BOOK III

The Machine in the Ghost

PERSONAL PRINT EDITION

Rationality: From AI to Zombies, Book III: *The Machine in the Ghost*

Written by Eliezer Yudkowsky.

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Minds: An Introduction

by Rob Bensinger

You're a mind, and that puts you in a pretty strange predicament.

Very few things get to be minds. You're that odd bit of stuff in the universe that can form predictions and make plans, weigh and revise beliefs, suffer, dream, notice ladybugs, or feel a sudden craving for mango. You can even form, *inside your mind*, a picture of your whole mind. You can reason about your own reasoning process, and work to bring its operations more in line with your goals.

You're a mind, implemented on a human brain. And it turns out that a human brain, for all its marvelous flexibility, is a lawful thing, a thing of pattern and routine. Your mind can follow a routine for a lifetime, without ever once noticing that it is doing so. And these routines can have great consequences.

When a mental pattern serves you well, we call that "rationality."

You exist as you are, hard-wired to exhibit certain species of rationality and certain species of irrationality, because of your ancestry. You, and all life on Earth, are descended from ancient self-replicating molecules. This replication process was initially clumsy and haphazard, and soon yielded replicable *differences* between the replicators. "Evolution" is our name for the change in these differences over time.

Since some of these reproducible differences impact reproducibility—a phenomenon called "selection"—evolution has resulted in organisms suited to reproduction in environments like the ones their ancestors had. Everything about you is built on the echoes of your ancestors' struggles and victories.

And so here you are: a mind, carved from weaker minds, seeking to understand your own inner workings, that they can be improved upon—improved upon relative to *your* goals, and not those of your designer, evolution. What useful policies and



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insights can we take away from knowing that this is our basic situation?

Ghosts and Machines

Our brains, in their small-scale structure and dynamics, look like many other mechanical systems. Yet we rarely think of our minds in the same terms we think of objects in our environments or organs in our bodies. Our basic mental categories—belief, decision, word, idea, feeling, and so on—bear little resemblance to our physical categories.

Past philosophers have taken this observation and run with it, arguing that minds and brains are fundamentally distinct and separate phenomena. This is the view the philosopher Gilbert Ryle called “the dogma of the Ghost in the Machine.”¹ But modern scientists and philosophers who have rejected dualism haven’t necessarily replaced it with a better predictive model of how the mind works. *Practically* speaking, our purposes and desires still function like free-floating ghosts, like a magisterium cut off from the rest of our scientific knowledge. We can talk about “rationality” and “bias” and “how to change our minds,” but if those ideas are still imprecise and unconstrained by any overarching theory, our scientific-sounding language won’t protect us from making the same kinds of mistakes as those whose theoretical posits include spirits and essences.

Interestingly, the mystery and mystification surrounding minds doesn’t just obscure our view of *humans*. It also accrues to systems that seem mind-like or purposeful in evolutionary biology and artificial intelligence (AI). Perhaps, if we cannot readily glean what we are from looking at ourselves, we can learn more by using obviously *inhuman* processes as a mirror.

There are many ghosts to learn from here—ghosts past, and present, and yet to come. And these illusions are real cognitive events, real phenomena that we can study and explain. If there *appears* to be a ghost in the machine, that appearance is itself the hidden work of a machine.

1. Gilbert Ryle, *The Concept of Mind* (University of Chicago Press, 1949).

The first sequence of *The Machine in the Ghost*, “The Simple Math of Evolution,” aims to communicate the dissonance and divergence between our hereditary history, our present-day biology, and our ultimate aspirations. This will require digging deeper than is common in introductions to evolution for non-biologists, which often restrict their attention to surface-level features of natural selection.

The third sequence, “A Human’s Guide to Words,” discusses the basic relationship between cognition and concept formation. This is followed by a longer essay introducing Bayesian inference.

Bridging the gap between these topics, “Fragile Purposes” abstracts from human cognition and evolution to the idea of minds and goal-directed systems at their most general. These essays serve the secondary purpose of explaining the author’s general approach to philosophy and the science of rationality, which is strongly informed by his work in AI.

Rebuilding Intelligence

Yudkowsky is a decision theorist and mathematician who works on foundational issues in Artificial General Intelligence (AGI), the theoretical study of domain-general problem-solving systems. Yudkowsky’s work in AI has been a major driving force behind his exploration of the psychology of human rationality, as he noted in his very first blog post on *Overcoming Bias*, The Martial Art of Rationality:

Such understanding as I have of rationality, I acquired in the course of wrestling with the challenge of Artificial General Intelligence (an endeavor which, to actually succeed, would require sufficient mastery of rationality to build a complete working rationalist out of toothpicks and rubber bands). In most ways the AI problem is enormously more demanding than the personal art of rationality, but in some ways it is actually easier. In the martial art of mind, we need to acquire the real-

time procedural skill of pulling the right levers at the right time on a large, pre-existing thinking machine whose innards are not end-user-modifiable. Some of the machinery is optimized for evolutionary selection pressures that run directly counter to our declared goals in using it. Deliberately we decide that we want to seek only the truth; but our brains have hardwired support for rationalizing falsehoods. [...]

Trying to synthesize a personal art of rationality, using the science of rationality, may prove awkward: One imagines trying to invent a martial art using an abstract theory of physics, game theory, and human anatomy. But humans are not reflectively blind; we do have a native instinct for introspection. The inner eye is not sightless; but it sees blurrily, with systematic distortions. We need, then, to apply the science to our intuitions, to use the abstract knowledge to correct our mental movements and augment our metacognitive skills. We are not writing a computer program to make a string puppet execute martial arts forms; it is our own mental limbs that we must move. Therefore we must connect theory to practice. We must come to see what the science means, for ourselves, for our daily inner life.

From Yudkowsky's perspective, I gather, talking about human rationality without saying anything interesting about AI is about as difficult as talking about AI without saying anything interesting about rationality.

In the long run, Yudkowsky predicts that AI will come to surpass humans in an "intelligence explosion," a scenario in which self-modifying AI improves its own ability to productively redesign itself, kicking off a rapid succession of further self-improvements. The term "technological singularity" is sometimes used in place of "intelligence explosion;" until January 2013, MIRI was named "the Singularity Institute for Artificial

Intelligence” and hosted an annual Singularity Summit. Since then, Yudkowsky has come to favor I.J. Good’s older term, “intelligence explosion,” to help distinguish his views from other futurist predictions, such as Ray Kurzweil’s exponential technological progress thesis.²

Technologies like smarter-than-human AI seem likely to result in large societal upheavals, for the better or for the worse. Yudkowsky coined the term “Friendly AI theory” to refer to research into techniques for aligning an AGI’s preferences with the preferences of humans. At this point, very little is known about when generally intelligent software might be invented, or what safety approaches would work well in such cases. Present-day autonomous AI can already be quite challenging to verify and validate with much confidence, and many current techniques are not likely to generalize to more intelligent and adaptive systems. “Friendly AI” is therefore closer to a menagerie of basic mathematical and philosophical questions than to a well-specified set of programming objectives.

As of 2015, Yudkowsky’s views on the future of AI continue to be debated by technology forecasters and AI researchers in industry and academia, who have yet to converge on a consensus position. Nick Bostrom’s book *Superintelligence* provides a big-picture summary of the many moral and strategic questions raised by smarter-than-human AI.³

For a general introduction to the field of AI, the most widely used textbook is Russell and Norvig’s *Artificial Intelligence: A Modern Approach*.⁴ In a chapter discussing the moral and philosophical questions raised by AI, Russell and Norvig note the technical difficulty of specifying good behavior in strongly adaptive AI:

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2. Irving John Good, “Speculations Concerning the First Ultraintelligent Machine,” in *Advances in Computers*, ed. Franz L. Alt and Morris Rubinoff, vol. 6 (New York: Academic Press, 1965), 31–88, doi:10.1016/S0065-2458(08)60418-0.
 3. Nick Bostrom, *Superintelligence: Paths, Dangers, Strategies* (Oxford University Press, 2014).
 4. Stuart J. Russell and Peter Norvig, *Artificial Intelligence: A Modern Approach*, 3rd ed. (Upper Saddle River, NJ: Prentice-Hall, 2010).

[Yudkowsky] asserts that friendliness (a desire not to harm humans) should be designed in from the start, but that the designers should recognize both that their own designs may be flawed, and that the robot will learn and evolve over time. Thus the challenge is one of mechanism design—to define a mechanism for evolving AI systems under a system of checks and balances, and to give the systems utility functions that will remain friendly in the face of such changes. We can’t just give a program a static utility function, because circumstances, and our desired responses to circumstances, change over time.

Disturbed by the possibility that future progress in AI, nanotechnology, biotechnology, and other fields could endanger human civilization, Bostrom and Čirković compiled the first academic anthology on the topic, *Global Catastrophic Risks*.⁵ The most extreme of these are the *existential risks*, risks that could result in the permanent stagnation or extinction of humanity.⁶

People (experts included) tend to be *extraordinarily bad* at forecasting major future events (new technologies included). Part of Yudkowsky’s goal in discussing rationality is to figure out which biases are interfering with our ability to predict and prepare for big upheavals well in advance. Yudkowsky’s contributions to the *Global Catastrophic Risks* volume, “Cognitive biases potentially affecting judgement of global risks”⁷ and “Artificial intelligence as a positive and negative factor in global

5. Bostrom and Čirković, *Global Catastrophic Risks* (Oxford University Press, 2008).

6. An example of a possible existential risk is the “grey goo” scenario, in which molecular robots designed to efficiently self-replicate do their job too well, rapidly outcompeting living organisms as they consume the Earth’s available matter.

7. Available at sharov.me/r/3-1



risk,”⁸ tie together his research in cognitive science and AI. Yudkowsky and Bostrom summarize near-term concerns along with long-term ones in a chapter of the *Cambridge Handbook of Artificial Intelligence*, “The ethics of artificial intelligence.”⁹

Though this is a book about *human* rationality, the topic of AI has relevance as a source of simple illustrations of aspects of human cognition. Long-term technology forecasting is also one of the more important applications of Bayesian rationality, which can model correct reasoning even in domains where the data is scarce or equivocal.

Knowing the design can tell you much about the designer; and knowing the designer can tell you much about the design.

We’ll begin, then, by inquiring into what our own designer can teach us about ourselves.

8. Available at sharov.me/r/3-2



9. Nick Bostrom and Eliezer Yudkowsky, “The Ethics of Artificial Intelligence,” in *The Cambridge Handbook of Artificial Intelligence*, ed. Keith Frankish and William Ramsey (New York: Cambridge University Press, 2014).

Interlude: The Power of Intelligence



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In our skulls we carry around three pounds of slimy, wet, grayish tissue, corrugated like crumpled toilet paper.

You wouldn't think, to look at the unappetizing lump, that it was some of the most powerful stuff in the known universe. If you'd never seen an anatomy textbook, and you saw a brain lying in the street, you'd say "Yuck!" and try not to get any of it on your shoes. Aristotle thought the brain was an organ that cooled the blood. It doesn't *look* dangerous.

Five million years ago, the ancestors of lions ruled the day, the ancestors of wolves roamed the night. The ruling predators were armed with teeth and claws—sharp, hard cutting edges, backed up by powerful muscles. Their prey, in self-defense, evolved armored shells, sharp horns, toxic venoms, camouflage. The war had gone on through hundreds of eons and countless arms races. Many a loser had been removed from the game, but there was no sign of a winner. Where one species had shells, another species would evolve to crack them; where one species became poisonous, another would evolve to tolerate the poison. Each species had its private niche—for who could live in the seas and the skies and the land at once? There was no ultimate weapon and no ultimate defense and no reason to believe any such thing was possible.

Then came the Day of the Squishy Things.

They had no armor. They had no claws. They had no venoms.

If you saw a movie of a nuclear explosion going off, and you were told an Earthly life form had done it, you would never in your wildest dreams imagine that the Squishy Things could be responsible. After all, Squishy Things aren't radioactive.

In the beginning, the Squishy Things had no fighter jets, no machine guns, no rifles, no swords. No bronze, no iron. No hammers, no anvils, no tongs, no smithies, no mines. All the Squishy Things had were squishy fingers—too weak to break a tree, let alone a mountain. Clearly not dangerous. To cut stone you would need steel, and the Squishy Things couldn't excrete steel. In the environment there were no steel blades

for Squishy fingers to pick up. Their bodies could not generate temperatures anywhere near hot enough to melt metal. The whole scenario was obviously absurd.

And as for the Squishy Things manipulating DNA—that would have been beyond ridiculous. Squishy fingers are not that small. There is no access to DNA from the Squishy level; it would be like trying to pick up a hydrogen atom. Oh, *technically* it's all one universe, *technically* the Squishy Things and DNA are part of the same world, the same unified laws of physics, the same great web of causality. But let's be realistic: you can't get there from here.

Even if Squishy Things could *someday* evolve to do any of those feats, it would take thousands of millennia. We have watched the ebb and flow of Life through the eons, and let us tell you, a year is not even a single clock tick of evolutionary time. Oh, sure, *technically* a year is six hundred trillion trillion trillion Planck intervals. But nothing ever happens in less than six hundred million trillion trillion trillion Planck intervals, so it's a moot point. The Squishy Things, as they run across the savanna now, will not fly across continents for at least another ten million years; *no one* could have that much sex.

Now explain to me again why an Artificial Intelligence can't do anything interesting over the Internet unless a human programmer builds it a robot body.

I have observed that someone's flinch-reaction to "intelligence"—the thought that crosses their mind in the first half-second after they hear the word "intelligence"—often determines their flinch-reaction to the notion of an intelligence explosion. Often they look up the keyword "intelligence" and retrieve the concept *booksmarts*—a mental image of the Grand Master chess player who can't get a date, or a college professor who can't survive outside academia.

"It takes more than intelligence to succeed professionally," people say, as if charisma resided in the kidneys, rather than the brain. "Intelligence is no match for a gun," they say, as if guns had grown on trees. "Where will an Artificial Intelligence get money?" they ask, as if the first *Homo sapiens* had found

dollar bills fluttering down from the sky, and used them at convenience stores already in the forest. The human species was not born into a market economy. Bees won't sell you honey if you offer them an electronic funds transfer. The human species *imagined* money into existence, and it exists—for *us*, not mice or wasps—because we go on believing in it.

I keep trying to explain to people that the archetype of intelligence is not Dustin Hoffman in *Rain Man*. It is a human being, period. It is squishy things that explode in a vacuum, leaving footprints on their moon. Within that gray wet lump is the power to search paths through the great web of causality, and find a road to the seemingly impossible—the power sometimes called creativity.

People—venture capitalists in particular—sometimes ask how, if the Machine Intelligence Research Institute successfully builds a true AI, the results will be *commercialized*. This is what we call a framing problem.

Or maybe it's something deeper than a simple clash of assumptions. With a bit of creative thinking, people can imagine how they would go about travelling to the Moon, or curing smallpox, or manufacturing computers. To imagine a trick that could accomplish *all these things at once* seems downright impossible—even though such a power resides only a few centimeters behind their own eyes. The gray wet thing still seems mysterious to the gray wet thing.

And so, because people can't quite see how it would all work, the power of intelligence seems less real; harder to imagine than a tower of fire sending a ship to Mars. The prospect of visiting Mars captures the imagination. But if one should promise a Mars visit, and also a grand unified theory of physics, and a proof of the Riemann Hypothesis, and a cure for obesity, and a cure for cancer, and a cure for aging, and a cure for stupidity—well, it just sounds wrong, that's all.

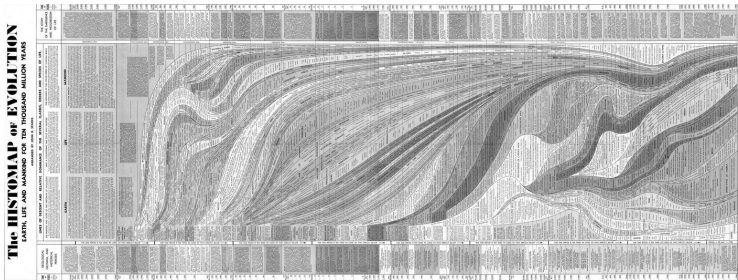
And well it should. It's a serious failure of imagination to think that intelligence is good for so little. Who could have imagined, ever so long ago, what minds would someday do? We may not even *know* what our real problems are.

But meanwhile, because it's hard to see how one process could have such diverse powers, it's hard to imagine that one fell swoop could solve even such prosaic problems as obesity and cancer and aging.

Well, one trick cured smallpox and built airplanes and cultivated wheat and tamed fire. Our current science may not agree yet on how exactly the trick works, but it works anyway. If you are temporarily ignorant about a phenomenon, that is a fact about your current state of mind, not a fact about the phenomenon. A blank map does not correspond to a blank territory. If one does not quite understand that power which put footprints on the Moon, nonetheless, the footprints are still there—real footprints, on a real Moon, put there by a real power. If one were to understand deeply enough, one could create and shape that power. Intelligence is as real as electricity. It's merely far more powerful, far more dangerous, has far deeper implications for the unfolding story of life in the universe—and it's a tiny little bit harder to figure out how to build a generator.

PART I

The Simple Math of
Evolution



An Alien God

“A curious aspect of the theory of evolution,” said Jacques Monod, “is that everybody thinks he understands it.”

A human being, looking at the natural world, sees a thousand times *purpose*. A rabbit’s legs, built and articulated for running; a fox’s jaws, built and articulated for tearing. But what you see is not exactly what is there ...

In the days before Darwin, the cause of all this apparent *purposefulness* was a very great puzzle unto science. The Goddists said “God did it,” because you get 50 bonus points each time you use the word “God” in a sentence. Yet perhaps I’m being unfair. In the days before Darwin, it seemed like a much more reasonable hypothesis. Find a watch in the desert, said William Paley, and you can infer the existence of a watchmaker.

But when you look at *all* the apparent purposefulness in Nature, rather than picking and choosing your examples, you start to notice things that don’t fit the Judeo-Christian concept of one benevolent God. Foxes seem well-designed to catch rabbits. Rabbits seem well-designed to evade foxes. Was the Creator having trouble making up Its mind?

When I design a toaster oven, I don’t design one part that tries to get electricity to the coils and a second part that tries to prevent electricity from getting to the coils. It would be a waste of effort. *Who* designed the ecosystem, with its predators and prey, viruses and bacteria? Even the cactus plant, which you might think well-designed to provide water and fruit to desert animals, is covered with inconvenient spines.

The ecosystem would make much more sense if it wasn’t designed by a unitary *Who*, but, rather, created by a horde of deities—say from the Hindu or Shinto religions. This handily explains both the ubiquitous purposefulnesses, and the ubiquitous conflicts: More than one deity acted, often at cross-purposes. The fox and rabbit were both designed, but by distinct competing deities. I wonder if anyone ever remarked on the seemingly



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excellent evidence thus provided for Hinduism over Christianity. Probably not.

Similarly, the Judeo-Christian God is alleged to be benevolent—well, sort of. And yet much of nature’s *purposefulness* seems downright cruel. Darwin suspected a non-standard Creator for studying Ichneumon wasps, whose paralyzing stings preserve its prey to be eaten alive by its larvae: “I cannot persuade myself,” wrote Darwin, “that a beneficent and omnipotent God would have designedly created the Ichneumonidae with the express intention of their feeding within the living bodies of Caterpillars, or that a cat should play with mice.”¹ I wonder if any earlier thinker remarked on the excellent evidence thus provided for Manichaeism over monotheistic ones.

By now we all know the punchline: you just say “evolution.”

I worry that’s how some people are absorbing the “scientific” explanation, as a magical purposefulness factory in Nature. I’ve previously discussed the case of Storm from the movie *X-Men*, who in *one mutation* gets the ability to throw lightning bolts. Why? Well, there’s this thing called “evolution” that somehow pumps a lot of purposefulness into Nature, and the changes happen through “mutations.” So if Storm gets a really *large* mutation, she can be redesigned to throw lightning bolts. Radioactivity is a popular super origin: radiation causes mutations, so more powerful radiation causes more powerful mutations. That’s logic.

But evolution doesn’t allow just *any* kind of purposefulness to leak into Nature. That’s what makes evolution a success as an empirical hypothesis. If evolutionary biology could explain a toaster oven, not just a tree, it would be worthless. There’s a lot more to evolutionary theory than pointing at Nature and saying, “Now purpose is allowed,” or “Evolution did it!” The strength of a theory is not what it allows, but what it prohibits; if you can invent an equally persuasive explanation for any outcome, you have zero knowledge.

1. Francis Darwin, ed., *The Life and Letters of Charles Darwin*, vol. 2 (John Murray, 1887).

“Many non-biologists,” observed George Williams, “think that it is for their benefit that rattles grow on rattlesnake tails.”² Bzzzt! *This* kind of purposefulness is not allowed. Evolution doesn’t work by letting flashes of purposefulness creep in at random—reshaping one species for the benefit of a random recipient.

Evolution is powered by a systematic correlation between the different ways that different genes construct organisms, and how many copies of those genes make it into the next generation. For rattles to grow on rattlesnake tails, rattle-growing genes must become more and more frequent in each successive generation. (Actually genes for incrementally more complex rattles, but if I start describing all the fillips and caveats to evolutionary biology, we really *will* be here all day.)

There isn’t an Evolution Fairy that looks over the current state of Nature, decides what would be a “good idea,” and chooses to increase the frequency of rattle-constructing genes.

I suspect this is where a lot of people get stuck, in evolutionary biology. They understand that “helpful” genes become more common, but “helpful” lets any sort of purpose leak in. They don’t think there’s an Evolution Fairy, yet they ask which genes will be “helpful” as if a rattlesnake gene could “help” non-rattlesnakes.

The key realization is that there *is* no Evolution Fairy. There’s no outside force *deciding* which genes ought to be promoted. Whatever happens, happens *because of* the genes themselves.

Genes for constructing (incrementally better) rattles must have somehow ended up more frequent in the rattlesnake gene pool, *because of* the rattle. In this case it’s *probably* because rattlesnakes with better rattles survive more often—rather than mating more successfully, or having brothers that reproduce more successfully, etc.

Maybe predators are wary of rattles and don’t step on the snake. Or maybe the rattle diverts attention from the snake’s

2. George C. Williams, *Adaptation and Natural Selection: A Critique of Some Current Evolutionary Thought*, Princeton Science Library (Princeton, NJ: Princeton University Press, 1966).

head. (As George Williams suggests, “The outcome of a fight between a dog and a viper would depend very much on whether the dog initially seized the reptile by the head or by the tail.”)

But that’s just a snake’s rattle. There are much more complicated ways that a gene can cause copies of itself to become more frequent in the next generation. Your brother or sister shares half your genes. A gene that sacrifices one unit of resources to bestow three units of resource on a brother, may promote some copies of itself by sacrificing one of its constructed organisms. (If you really want to know all the details and caveats, buy a book on evolutionary biology; there is no royal road.)

The main point is that the *gene’s effect* must *cause* copies of that gene to become more frequent in the next generation. There’s no Evolution Fairy that reaches in from outside. There’s nothing which *decides* that some genes are “helpful” and should, therefore, increase in frequency. It’s just cause and effect, starting from the genes themselves.

This explains the strange conflicting purposefulness of Nature, and its frequent cruelty. It explains even better than a horde of Shinto deities.

Why is so much of Nature at war with other parts of Nature? Because there isn’t one Evolution directing the whole process. There’s as many different “evolutions” as reproducing populations. Rabbit genes are becoming more or less frequent in rabbit populations. Fox genes are becoming more or less frequent in fox populations. Fox genes which construct foxes that catch rabbits, insert more copies of themselves in the next generation. Rabbit genes which construct rabbits that evade foxes are *naturally* more common in the next generation of rabbits. Hence the phrase “natural selection.”

Why is Nature cruel? You, a human, can look at an Ichneumon wasp, and decide that it’s cruel to eat your prey alive. You can decide that if you’re going to eat your prey alive, you can at least have the decency to stop it from hurting. It would scarcely cost the wasp anything to anesthetize its prey as well as paralyze it. Or what about old elephants, who die of starvation when their last set of teeth fall out? These elephants aren’t going to reproduce anyway. What would it cost evolution—

the evolution of elephants, rather—to ensure that the elephant dies right away, instead of slowly and in agony? What would it cost evolution to anesthetize the elephant, or give it pleasant dreams before it dies? Nothing; that elephant won't reproduce more or less either way.

If you were talking to a fellow human, trying to resolve a conflict of interest, you would be in a good negotiating position—would have an easy job of persuasion. It would cost so little to anesthetize the prey, to let the elephant die without agony! Oh please, won't you do it, kindly ...um ...

There's no one to argue with.

Human beings fake their justifications, figure out what they want using one method, and then justify it using another method. There's no Evolution of Elephants Fairy that's trying to (a) figure out what's best for elephants, and then (b) figure out how to *justify* it to the Evolutionary Overseer, who (c) doesn't want to see reproductive fitness decreased, but is (d) willing to go along with the painless-death idea, so long as it doesn't actually harm any genes.

There's no advocate for the elephants anywhere in the system.

Humans, who are often deeply concerned for the well-being of animals, can be very persuasive in arguing how various kindnesses wouldn't harm reproductive fitness at all. Sadly, the evolution of elephants doesn't use a similar algorithm; it doesn't select nice genes *that can plausibly be argued* to help reproductive fitness. Simply: genes that replicate more often become more frequent in the next generation. Like water flowing downhill, and equally benevolent.

A human, looking over Nature, starts thinking of all the ways we would design organisms. And then we tend to start rationalizing reasons why our design improvements would increase reproductive fitness—a political instinct, trying to sell your own preferred option as matching the boss's favored justification.

And so, amateur evolutionary biologists end up making all sorts of wonderful and *completely mistaken* predictions. Because the amateur biologists are drawing their bottom line—

and more importantly, locating their prediction in hypothesis-space—using a *different* algorithm than evolutions use to draw *their* bottom lines.

A human engineer would have designed human taste buds to measure how much of each nutrient we had, and how much we needed. When fat was scarce, almonds or cheeseburgers would taste delicious. But if you started to become obese, or if vitamins were lacking, lettuce would taste delicious. But there is no Evolution of Humans Fairy, which intelligently planned ahead and designed a general system for every contingency. It was a reliable invariant of humans' ancestral environment that calories were scarce. So genes whose organisms loved calories, became more frequent. Like water flowing downhill.

We are simply the embodied history of which organisms *did in fact* survive and reproduce, not which organisms *ought prudentially* to have survived and reproduced.

The human retina is constructed backward: The light-sensitive cells are at the back, and the nerves emerge from the front and go back *through* the retina into the brain. Hence the blind spot. To a human engineer, this looks simply stupid—and other organisms have independently evolved retinas the right way around. Why not redesign the retina?

The problem is that no *single* mutation will reroute the *whole* retina simultaneously. A human engineer can redesign multiple parts simultaneously, or plan ahead for future changes. But if a single mutation breaks some vital part of the organism, it doesn't matter what wonderful things a Fairy could build on top of it—the organism dies and the gene decreases in frequency.

If you turn around the retina's cells without also reprogramming the nerves and optic cable, the system as a whole won't work. It doesn't matter that, to a Fairy or a human engineer, this is one step forward in redesigning the retina. The organism is blind. Evolution has no foresight, it is simply the frozen history of which organisms *did in fact* reproduce. Evolution is as blind as a halfway-redesigned retina.

Find a watch in a desert, said William Paley, and you can infer the watchmaker. There were once those who denied this, who thought that life "just happened" without need of an op-

timization process, mice being spontaneously generated from straw and dirty shirts.

If we ask who was *more* correct—the theologians who argued for a Creator-God, or the intellectually unfulfilled atheists who argued that mice spontaneously generated—then the theologians must be declared the victors: evolution is not God, but it is closer to God than it is to pure random entropy. Mutation is random, but selection is non-random. This doesn't mean an intelligent Fairy is reaching in and selecting. It means there's a non-zero statistical correlation between the gene and how often the organism reproduces. Over a few million years, that non-zero statistical correlation adds up to something very powerful. It's not a god, but it's more closely akin to a god than it is to snow on a television screen.

In a lot of ways, evolution *is* like unto theology. “Gods are ontologically distinct from creatures,” said Damien Broderick, “or they're not worth the paper they're written on.”³ And indeed, the Shaper of Life is not itself a creature. Evolution is bodiless, like the Judeo-Christian deity. Omnipresent in Nature, immanent in the fall of every leaf. Vast as a planet's surface. Billions of years old. Itself unmade, arising naturally from the structure of physics. Doesn't that all sound like something that might have been said about God?

And yet the Maker has no mind, as well as no body. In some ways, its handiwork is incredibly poor design by human standards. It is internally divided. Most of all, it isn't *nice*.

In a way, Darwin *discovered* God—a God that failed to match the preconceptions of theology, and so passed unheralded. If Darwin had discovered that life was created by an intelligent agent—a bodiless mind that loves us, and will smite us with lightning if we dare say otherwise—people would have said “My gosh! That's God!”

3. sharov.me/r/3-3
extropians.weidai.com



But instead Darwin discovered a strange alien God—not comfortably “ineffable,” but *really genuinely different from us*. Evolution is not a God, but if it were, it wouldn’t be Jehovah. It would be H. P. Lovecraft’s Azathoth,⁴ the blind idiot God burbling chaotically at the center of everything, surrounded by the thin monotonous piping of flutes.

Which you might have predicted, if you had really *looked* at Nature.

So much for the claim some religionists make, that they believe in a vague deity with a correspondingly high probability. Anyone who *really* believed in a vague deity, would have recognized their strange inhuman creator when Darwin said “Aha!”

So much for the claim some religionists make, that they are waiting innocently curious for Science to discover God. Science has already discovered the sort-of-godlike maker of humans—but it wasn’t what the religionists wanted to hear. They were waiting for the discovery of *their* God, the highly specific God *they* want to be there. They shall wait forever, for the great discovery has *already taken place*, and the winner is Azathoth.

Well, more power to us humans. I like having a Creator I can outwit. Beats being a pet. I’m glad it was Azathoth and not Odin.

4. sharov.me/r/3-4

Azathoth - Wikipedia • en.wikipedia.org



The Wonder of Evolution

The wonder of evolution is that it works at all.

I mean that literally: If you want to marvel at evolution, that's what's marvel-worthy.

How does optimization first arise in the universe? If an intelligent agent designed Nature, who designed the intelligent agent? Where is the first design that has no designer? The puzzle is *not* how the first stage of the bootstrap can be super-clever and super-efficient; the puzzle is how it can happen *at all*.

Evolution resolves the infinite regression, not by being super-clever and super-efficient, but by being stupid and inefficient and working anyway. *This* is the marvel.

For professional reasons, I often have to discuss the slowness, randomness, and blindness of evolution. Afterward someone says: “*You* just said that evolution can’t plan simultaneous changes, and that evolution is very inefficient because mutations are random. Isn’t that what the *creationists* say? That you couldn’t assemble a watch by randomly shaking the parts in a box?”

But the reply to creationists is not that you *can* assemble a watch by shaking the parts in a box. The reply is that this is *not* how evolution works. If you think that evolution *does* work by whirlwinds assembling 747s, then the creationists have successfully misrepresented biology to you; they’ve sold the strawman.

The real answer is that complex machinery evolves either incrementally, or by adapting previous complex machinery used for a new purpose. Squirrels jump from treetop to treetop using just their muscles, but the length they can jump depends to some extent on the aerodynamics of their bodies. So now there are flying squirrels, so aerodynamic they can glide short distances. If birds were wiped out, the descendants of flying squirrels might reoccupy that ecological niche in ten million



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years, gliding membranes transformed into wings. And the creationists would say, “What good is half a wing? You’d just fall down and splat. How could squirrelbirds possibly have evolved incrementally?”

That’s how one complex adaptation can jump-start a new complex adaptation. Complexity can also accrete incrementally, starting from a single mutation.

First comes some gene *A* which is simple, but at least a *little* useful on its own, so that *A* increases to universality in the gene pool. Now along comes gene *B*, which is only useful in the presence of *A*, but *A* is reliably present in the gene pool, so there’s a reliable selection pressure in favor of *B*. Now a modified version of *A** arises, which depends on *B*, but doesn’t break *B*’s dependency on *A/A**. Then along comes *C*, which depends on *A** and *B*, and *B**, which depends on *A** and *C*. Soon you’ve got “irreducibly complex” machinery that breaks if you take out any single piece.

And yet you can still visualize the trail backward to that single piece: you can, without breaking the whole machine, make one piece less dependent on another piece, and do this a few times, until you can take out one whole piece *without* breaking the machine, and so on until you’ve turned a ticking watch back into a crude sundial.

Here’s an example: DNA stores information very nicely, in a durable format that allows for exact duplication. A ribosome turns that stored information into a sequence of amino acids, a protein, which folds up into a variety of chemically active shapes. The combined system, DNA and ribosome, can build all sorts of protein machinery. But what good is DNA, without a ribosome that turns DNA information into proteins? What good is a ribosome, without DNA to tell it which proteins to make?

Organisms don’t always leave fossils, and evolutionary biology can’t *always* figure out the incremental pathway. But in this case we *do* know how it happened. RNA shares with DNA the property of being able to carry information and replicate itself, although RNA is less durable and copies less accurately. And RNA also shares the ability of proteins to fold up into

chemically active shapes, though it's not as versatile as the amino acid chains of proteins. Almost certainly, RNA is the single *A* which predates the mutually dependent *A** and *B*.

It's just as important to note that RNA does the combined job of DNA and proteins *poorly*, as that it does the combined job at all. It's amazing enough that a *single molecule* can both store information and manipulate chemistry. For it to do the job *well* would be a wholly unnecessary miracle.

What was the very first replicator ever to exist? It may well have been an RNA strand, because *by some strange coincidence*, the chemical ingredients of RNA are chemicals that would have arisen naturally on the prebiotic Earth of 4 billion years ago. Please note: evolution does *not* explain the origin of life; evolutionary biology is *not supposed to* explain the first replicator, because the *first* replicator does not come from another replicator. Evolution describes statistical trends in replication. The first replicator wasn't a statistical trend, it was a pure accident. The notion that evolution should explain the *origin* of life is a pure strawman—more creationist misrepresentation.

If you'd been watching the primordial soup on the day of the first replicator, the day that reshaped the Earth, you would not have been impressed by *how well* the first replicator replicated. The first replicator probably copied itself like a drunken monkey on LSD. It would have exhibited none of the signs of careful fine-tuning embodied in modern replicators, because the first replicator was an *accident*. It was not *needful* for that single strand of RNA, or chemical hypercycle, or pattern in clay, to replicate gracefully. It just had to happen *at all*. Even so, it was probably very improbable, considered in an isolated event—but it only had to happen *once*, and there were a lot of tide pools. A few billions of years later, the replicators are walking on the Moon.

The first accidental replicator was the most important molecule in the history of time. But if you praised it too highly, attributing to it all sorts of wonderful replication-aiding capabilities, you would be *missing the whole point*.

Don't think that, in the political battle between evolutionists and creationists, whoever praises evolution must be on the

side of science. Science has a very exact idea of the capabilities of evolution. If you praise evolution one millimeter higher than this, you're not "fighting on evolution's side" against creationism. You're being scientifically inaccurate, full stop. You're falling into a creationist trap by insisting that, yes, a whirlwind *does* have the power to assemble a 747! Isn't that amazing! How wonderfully intelligent is evolution, how praiseworthy! Look at me, I'm pledging my allegiance to science! The more nice things I say about evolution, the more I must be on evolution's side against the creationists!

But to praise evolution too highly destroys the *real* wonder, which is not *how well* evolution designs things, but that a naturally occurring process manages to design anything *at all*.

So let us dispose of the idea that evolution is a wonderful designer, or a wonderful conductor of species destinies, which we human beings ought to imitate. For human intelligence to imitate evolution as a designer, would be like a sophisticated modern bacterium trying to imitate the first replicator as a biochemist. As T. H. Huxley, "Darwin's Bulldog," put it:¹

Let us understand, once and for all, that the ethical progress of society depends, not on imitating the cosmic process, still less in running away from it, but in combating it.

Huxley didn't say that because he disbelieved in evolution, but because he understood it all too well.

1. Thomas Henry Huxley, *Evolution and Ethics and Other Essays* (Macmillan, 1894).

Evolutions Are Stupid (But Work Anyway)

In the previous essay, I wrote:

Science has a very exact idea of the capabilities of evolution. If you praise evolution one millimeter higher than this, you're not "fighting on evolution's side" against creationism. You're being scientifically inaccurate, full stop.



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In this essay I describe some well-known inefficiencies and limitations of evolutions. I say "evolutions," plural, because fox evolution works at cross-purposes to rabbit evolution, and neither can talk to snake evolution to learn how to build venomous fangs.

So I am talking about limitations of evolution here, but this does not mean I am trying to sneak in creationism. This is standard Evolutionary Biology 201. (583 if you must derive the equations.) Evolutions, thus limited, can still explain observed biology; in fact the limitations are necessary to make sense of it. Remember that the wonder of evolutions is not how well they work, but that they work at all.

Human intelligence is so complicated that no one has any good way to calculate how efficient it is. Natural selection, though not simple, is *simpler than* a human brain; and correspondingly slower and less efficient, as befits the first optimization process ever to exist. In fact, evolutions are simple enough that we can calculate *exactly how stupid* they are.

Evolutions are slow. How slow? Suppose there's a beneficial mutation that conveys a fitness advantage of 3%: on average, bearers of this gene have 1.03 times as many children as non-bearers. Assuming that the mutation spreads at all, how long will it take to spread through the whole population? That depends on the population size. A gene conveying a 3% fitness

advantage, spreading through a population of 100,000, would require an average of 768 generations to reach universality in the gene pool. A population of 500,000 would require 875 generations. The general formula is

$$\text{Generations to fixation} = 2 \ln(N)/s,$$

where N is the population size and $(1 + s)$ is the fitness. (If each bearer of the gene has 1.03 times as many children as a non-bearer, $s = 0.03$.)

Thus, if the population size were 1,000,000—the estimated population in hunter-gatherer times—then it would require 2,763 generations for a gene conveying a 1% advantage to spread through the gene pool.¹

This should not be surprising; genes have to do all their own work of spreading. There's no Evolution Fairy who can watch the gene pool and say, "Hm, that gene seems to be spreading rapidly—I should distribute it to everyone." In a human market economy, someone who is legitimately getting 20% returns on investment—especially if there's an obvious, clear mechanism behind it—can rapidly acquire more capital from other investors; and others will start duplicate enterprises. Genes have to spread without stock markets or banks or imitators—as if Henry Ford had to make one car, sell it, buy the parts for 1.01 more cars (on average), sell those cars, and keep doing this until he was up to a million cars.

All this assumes that the gene spreads in the first place. Here the equation is simpler and ends up not depending at all on population size:

$$\text{Probability of fixation} = 2s.$$

A mutation conveying a 3% advantage (which is pretty darned large, as mutations go) has a 6% chance of spreading,

1. Dan Graur and Wen-Hsiung Li, *Fundamentals of Molecular Evolution*, 2nd ed. (Sunderland, MA: Sinauer Associates, 2000).

at least on that occasion.² Mutations can happen more than once, but in a population of a million with a copying fidelity of 10^{-8} errors per base per generation, you may have to wait a hundred generations for another chance, and then it still has only a 6% chance of fixating.

Still, *in the long run*, an evolution has a good shot at getting there eventually. (This is going to be a running theme.)

Complex adaptations take a *very* long time to evolve. First comes allele *A*, which is advantageous of itself, and requires a thousand generations to fixate in the gene pool. *Only* then can another allele *B*, which depends on *A*, begin rising to fixation. A fur coat is not a strong advantage unless the environment has a *statistically reliable* tendency to throw cold weather at you. Well, genes form part of the environment of other genes, and if *B* depends on *A*, then *B* will not have a strong advantage unless *A* is *reliably* present in the genetic environment.

Let's say that *B* confers a 5% advantage in the presence of *A*, no advantage otherwise. Then while *A* is still at 1% frequency in the population, *B* only confers its advantage 1 out of 100 times, so the average fitness advantage of *B* is 0.05%, and *B*'s probability of fixation is 0.1%. With a complex adaptation, *first* *A* has to evolve over a thousand generations, *then* *B* has to evolve over another thousand generations, *then* *A** evolves over another thousand generations ...and several million years later, you've got a new complex adaptation.

Then other evolutions don't imitate it. If snake evolution develops an amazing new venom, it doesn't help fox evolution or lion evolution.

Contrast all this to a human programmer, who can design a new complex mechanism with a hundred interdependent parts over the course of *a single afternoon*. How is this even *possible*? I don't know all the answer, and my guess is that neither does science; human brains are much more complicated than evolutions. I could wave my hands and say something like

2. John B. S. Haldane, "A Mathematical Theory of Natural and Artificial Selection," *Mathematical Proceedings of the Cambridge Philosophical Society* 23 (5 1927): 607–615, doi:10.1017/S0305004100011750.

“goal-directed backward chaining using combinatorial modular representations,” but you would not thereby be enabled to design your own human. Still: Humans can foresightfully design new parts in anticipation of later designing other new parts; produce coordinated simultaneous changes in interdependent machinery; learn by observing single test cases; zero in on problem spots and think abstractly about how to solve them; and prioritize which tweaks are worth trying, rather than waiting for a cosmic ray strike to produce a good one. By the standards of natural selection, this is simply magic.

Humans can do things that evolutions probably can’t do *period* over the expected lifetime of the universe. As the eminent biologist Cynthia Kenyon once put it at a dinner I had the honor of attending, “One grad student can do things in an hour that evolution could not do in a billion years.” According to biologists’ best current knowledge, evolutions have invented a fully rotating wheel on a grand total of *three* occasions.

And don’t forget the part where the programmer posts the code snippet to the Internet.

Yes, some evolutionary handiwork is impressive even by comparison to the best technology of *Homo sapiens*. But our Cambrian explosion only started, we only really began accumulating knowledge, around ...what, four hundred years ago? In some ways, biology still excels over the best human technology: we can’t build a self-replicating system the size of a butterfly. In other ways, human technology leaves biology in the dust. We got wheels, we got steel, we got guns, we got knives, we got pointy sticks; we got rockets, we got transistors, we got nuclear power plants. With every passing decade, that balance tips further.

So, once again: for a human to look to natural selection as inspiration on the art of design is like a sophisticated modern bacterium trying to imitate the first awkward replicator’s biochemistry. The first replicator would be eaten instantly if it popped up in today’s competitive ecology. The same fate would accrue to any human planner who tried making random point mutations to their strategies and waiting 768 iterations of testing to adopt a 3% improvement.

Don't praise evolutions one millimeter more than they deserve.

Coming up next: More exciting mathematical bounds on evolution!

No Evolutions for Corporations or Nanodevices



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The laws of physics and the rules of math don't cease to apply. That leads me to believe that evolution doesn't stop. That further leads me to believe that nature—bloody in tooth and claw, as some have termed it—will simply be taken to the next level...

[Getting rid of Darwinian evolution is] like trying to get rid of gravitation. So long as there are limited resources and multiple competing actors capable of passing on characteristics, you have selection pressure.

—Perry Metzger, predicting that the reign of natural selection would continue into the indefinite future

In evolutionary biology, as in many other fields, it is important to think quantitatively rather than qualitatively. Does a beneficial mutation “sometimes spread, but not always”? Well, a psychic power would be a beneficial mutation, so you'd expect it to spread, right? Yet this is qualitative reasoning, not quantitative—if X is true, then Y is true; if psychic powers are beneficial, they may spread. In *Evolutions Are Stupid*, I described the equations for a beneficial mutation's probability of fixation, roughly twice the fitness advantage (6% for a 3% advantage). Only this kind of numerical thinking is likely to make us realize that mutations which are *only rarely useful* are extremely unlikely to spread, and that it is practically impossible for complex adaptations to arise without constant use. If psychic powers really existed, we should expect to see everyone using them all the time—not just because they would be so amazingly useful, but because otherwise they couldn't have evolved in the first place.

“So long as there are limited resources and multiple competing actors capable of passing on characteristics, you have selection pressure.” This is qualitative reasoning. *How much* selection pressure?

While there are several candidates for the most important equation in evolutionary biology, I would pick Price’s Equation,¹ which in its simplest formulation reads:

$$\Delta \text{cov}(v_i, z_i)$$

change in average characteristic = covariance(relative fitness, characteristic).

This is a *very* powerful and general formula. For example, a particular gene for height can be the *Z*, the characteristic that changes, in which case Price’s Equation says that the change in the probability of possessing this gene equals the covariance of the gene with reproductive fitness. Or you can consider *height in general* as the characteristic *Z*, apart from any particular genes, and Price’s Equation says that the change in height in the next generation will equal the covariance of height with relative reproductive fitness.

(At least, this is true so long as height is straightforwardly heritable. If nutrition improves, so that a fixed genotype becomes taller, you have to add a correction term to Price’s Equation. If there are complex nonlinear interactions between many genes, you have to either add a correction term, or calculate the equation in such a complicated way that it ceases to enlighten.)

Many enlightenments may be attained by studying the different forms and derivations of Price’s Equation. For example, the final equation says that the average characteristic changes according to its *covariance* with *relative* fitness, rather than its *absolute* fitness. This means that if a Frodo gene saves its whole species from extinction, the average Frodo characteristic does

1. sharov.me/r/3-5

Price equation - Wikipedia • en.wikipedia.org



not increase, since Frodo's act benefited all genotypes equally and did not *covary* with *relative* fitness.

It is said that Price became so disturbed with the implications of his equation for altruism that he committed suicide, though he may have had other issues. (*Overcoming Bias* does not advocate committing suicide after studying Price's Equation.)

One of the enlightenments which may be gained by meditating upon Price's Equation is that "limited resources" and "multiple competing actors capable of passing on characteristics" are *not sufficient* to give rise to an evolution. "Things that replicate themselves" is not a sufficient condition. Even "competition between replicating things" is not sufficient.

Do corporations evolve? They certainly compete. They occasionally spin off children. Their resources are limited. They sometimes die.

But how much does the child of a corporation resemble its parents? Much of the personality of a corporation derives from key officers, and CEOs cannot divide themselves by fission. Price's Equation only operates to the extent that characteristics are heritable across generations. If great-great-grandchildren don't much resemble their great-great-grandparents, you won't get more than four generations' worth of cumulative selection pressure—anything that happened more than four generations ago will blur itself out. Yes, the personality of a corporation can influence its spinoff—but that's nothing like the heritability of DNA, which is digital rather than analog, and can transmit itself with 10^{-8} errors per base per generation.

With DNA you have heritability lasting for *millions* of generations. That's how complex adaptations can arise by pure evolution—the digital DNA lasts long enough for a gene conveying 3% advantage to spread itself over 768 generations, and then another gene dependent on it can arise. Even if corporations replicated with digital fidelity, they would currently be at most ten generations into the RNA World.

Now, corporations are certainly *selected*, in the sense that incompetent corporations go bust. This should logically make you more likely to observe corporations with features contributing to competence. And in the same sense, any star that

goes nova shortly after it forms, is less likely to be visible when you look up at the night sky. But if an accident of stellar dynamics makes one star burn longer than another star, that doesn't make it more likely that future stars will also burn longer—the feature will not be copied onto other stars. We should not expect future astrophysicists to discover *complex* internal features of stars which seem designed to help them burn longer. That kind of mechanical adaptation requires much larger *cumulative* selection pressures than a once-off winnowing.

Think of the principle introduced in Einstein's Arrogance—that the vast majority of the evidence required to think of General Relativity had to go into raising that one particular equation to the level of Einstein's personal attention; the amount of evidence required to raise it from a deliberately considered possibility to 99.9% certainty was trivial by comparison. In the same sense, complex features of corporations that require hundreds of bits to specify are produced primarily by human intelligence, not a handful of generations of low-fidelity evolution. In biology, the mutations are purely random and evolution supplies thousands of bits of cumulative selection pressure. In corporations, humans offer up thousand-bit intelligently designed complex "mutations," and then the further selection pressure of "Did it go bankrupt or not?" accounts for a handful of additional bits in explaining what you see.

Advanced molecular nanotechnology—the artificial sort, not biology—should be able to copy itself with digital fidelity through thousands of generations. Would Price's Equation thereby gain a foothold?

Correlation is covariance divided by variance, so if A is highly predictive of B , there can be a strong "correlation" between them even if A is ranging from 0 to 9 and B is only ranging from 50.0001 and 50.0009. Price's Equation runs on *covariance* of characteristics with reproduction—not correlation! If you can compress variance in characteristics into a tiny band, the covariance goes way down, and so does the cumulative change in the characteristic.

The Foresight Institute suggests, among other sensible proposals, that the replication instructions for any nanodevice

should be encrypted. Moreover, encrypted such that flipping a single bit of the encoded instructions will entirely scramble the decrypted output. If all nanodevices produced are precise molecular copies, and moreover, any mistakes on the assembly line are not heritable because the offspring got a digital copy of the original encrypted instructions for use in making grandchildren, then your nanodevices ain't gonna be doin' much evolving.

You'd still have to worry about prions—self-replicating assembly errors apart from the encrypted instructions, where a robot arm fails to grab a carbon atom that is used in assembling a homologue of itself, and this causes the offspring's robot arm to likewise fail to grab a carbon atom, etc., even with all the encrypted instructions remaining constant. But how much *correlation* is there likely to be, between this sort of transmissible error, and a *higher* reproductive rate? Let's say that one nanodevice produces a copy of itself every 1,000 seconds, and the new nanodevice is magically more efficient (it not only has a prion, it has a *beneficial* prion) and copies itself every 999.99999 seconds. It needs one less carbon atom attached, you see. That's not a whole lot of variance in reproduction, so it's not a whole lot of covariance either.

And how often will these nanodevices need to replicate? Unless they've got more atoms available than exist in the solar system, or for that matter, the visible Universe, only a small number of generations will pass before they hit the resource wall. "Limited resources" are not a sufficient condition for evolution; you need the frequently iterated death of a substantial fraction of the population to free up resources. Indeed, "generations" is not so much an integer as an integral over the fraction of the population that consists of newly created individuals.

This is, to me, the most frightening thing about gray goo or nanotechnological weapons—that they could eat the whole Earth and then that would be *it*, nothing interesting would happen afterward. Diamond is stabler than proteins held together by van der Waals forces, so the goo would only need to reassemble some pieces of itself when an asteroid hit. Even if prions were a powerful enough idiom to support evolution at

all—evolution is slow enough with digital DNA!—fewer than 1.0 generations might pass between when the goo ate the Earth and when the Sun died.

To sum up, if you have *all* of the following properties:

- Entities that replicate;
- Substantial variation in their characteristics;
- Substantial variation in their reproduction;
- Persistent correlation between the characteristics and reproduction;
- High-fidelity long-range heritability in characteristics;
- Frequent birth of a significant fraction of the breeding population;
- And *all* this remains true through *many* iterations ...

Then you will have significant *cumulative* selection pressures, enough to produce complex adaptations by the force of evolution.

Evolving to Extinction



discussion
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It is a *very* common misconception that an evolution works for the good of its species. Can you remember hearing someone talk about two rabbits breeding eight rabbits and thereby “contributing to the survival of their species”? A modern evolutionary biologist would never say such a thing; they’d sooner breed with a rabbit.

It’s yet another case where you’ve got to simultaneously consider multiple abstract concepts and keep them distinct. Evolution doesn’t *operate* on particular individuals; individuals keep whatever genes they’re born with. Evolution operates on a reproducing population, a species, over time. There’s a natural tendency to think that if an Evolution Fairy is *operating* on the species, she must be *optimizing* for the species. But what really changes are the gene frequencies, and frequencies don’t increase or decrease according to how much the gene helps the species as a whole. As we shall later see, it’s quite possible for a species to evolve to extinction.

Why are boys and girls born in roughly equal numbers? (Leaving aside crazy countries that use artificial gender selection technologies.) To see why this is surprising, consider that 1 male can impregnate 2, 10, or 100 females; it wouldn’t seem that you need the same number of males as females to ensure the survival of the species. This is even more surprising in the vast majority of animal species where the male contributes very little to raising the children—humans are extraordinary, even among primates, for their level of paternal investment. Balanced gender ratios are found even in species where the male impregnates the female and vanishes into the mist.

Consider two groups on different sides of a mountain; in group *A*, each mother gives birth to 2 males and 2 females; in group *B*, each mother gives birth to 3 females and 1 male. Group *A* and group *B* will have the same number of children, but group *B* will have 50% more grandchildren and 125% more great-

grandchildren. You might think this would be a significant evolutionary advantage.

But consider: The rarer males become, the more reproductively valuable they become—not to the *group*, but to the *individual* parent. Every child has one male and one female parent. Then in every generation, the total genetic contribution from all males equals the total genetic contribution from all females. The fewer males, the greater the individual genetic contribution per male. If all the females around you are doing what's good for the group, what's good for the species, and birthing 1 male per 10 females, you can make a genetic *killing* by birthing all males, each of whom will have (on average) ten times as many grandchildren as their female cousins.

So while group selection ought to favor more girls, individual selection favors equal investment in male and female offspring. Looking at the statistics of a maternity ward, you can see at a glance that the quantitative balance between group selection forces and individual selection forces is overwhelmingly tilted in favor of individual selection in *Homo sapiens*.

(Technically, this isn't quite a glance. Individual selection favors equal *parental investments* in male and female offspring. If males cost half as much to birth and/or raise, twice as many males as females will be born at the evolutionarily stable equilibrium. If the same number of males and females were born in the population at large, but males were twice as cheap to birth, then you could again make a genetic killing by birthing more males. So the maternity ward should reflect the balance of parental opportunity costs, in a hunter-gatherer society, between raising boys and raising girls; and you'd have to assess that somehow. But ya know, it doesn't seem all that much *more* reproductive-opportunity-costly for a hunter-gatherer family to raise a girl, so it's kinda suspicious that around the same number of boys are born as girls.)

Natural selection isn't about groups, or species, or even *individuals*. In a sexual species, an individual organism doesn't evolve; it keeps whatever genes it's born with. An individual is a once-off collection of genes that will never reappear; how can you select on that? When you consider that nearly all of

your ancestors are dead, it's clear that "survival of the fittest" is a tremendous misnomer. "Replication of the fitter" would be more accurate, although technically fitness is *defined* only in terms of replication.

Natural selection is really about *gene frequencies*. To get a complex adaptation, a machine with multiple dependent parts, each new gene as it evolves depends on the other genes being reliably present in its genetic environment. They must have high frequencies. The more complex the machine, the higher the frequencies must be. The signature of natural selection occurring is a gene rising from 0.00001% of the gene pool to 99% of the gene pool. This is the information, in an information-theoretic sense; and this is what must happen for large complex adaptations to evolve.

The real struggle in natural selection is not the competition of organisms for resources; this is an ephemeral thing when all the participants will vanish in another generation. The real struggle is the competition of alleles for frequency in the gene pool. This is the lasting consequence that creates lasting information. The two rams bellowing and locking horns are only passing shadows.

It's perfectly possible for an allele to spread to fixation by outcompeting an alternative allele which was "better for the species." If the Flying Spaghetti Monster magically created a species whose gender mix was perfectly optimized to ensure the survival *of the species*—the optimal gender mix to bounce back reliably from near-extinction events, adapt to new niches, et cetera—then the evolution would rapidly degrade this species optimum back into the individual-selection optimum of equal parental investment in males and females.

Imagine a "Frodo gene" that sacrifices its vehicle *to save its entire species* from an extinction event. What happens to the allele frequency as a result? It goes down. Kthxbye.

If species-level extinction threats occur regularly (call this a "Buffy environment") then the Frodo gene will systematically decrease in frequency and vanish, and soon thereafter, so will the species.

A hypothetical example? Maybe. If the human species was going to stay biological for another century, it would be a good idea to start cloning Gandhi.

In viruses, there's the tension between individual viruses replicating as fast as possible, versus the benefit of leaving the host alive long enough to transmit the illness. This is a good real-world example of group selection, and if the virus evolves to a point on the fitness landscape where the group selection pressures fail to overcome individual pressures, the virus could vanish shortly thereafter. I don't know if a disease has ever been caught in the act of evolving to extinction, but it's probably happened any number of times.

Segregation-distorters subvert the mechanisms that usually guarantee fairness of sexual reproduction. For example, there is a segregation-distorter on the male sex chromosome of some mice which causes only male children to be born, all carrying the segregation-distorter. Then these males impregnate females, who give birth to only male children, and so on. You might cry "This is cheating!" but that's a human perspective; the reproductive fitness of this allele is extremely high, since it produces twice as many copies of itself in the succeeding generation as its nonmutant alternative. Even as females become rarer and rarer, males carrying this gene are no less likely to mate than any other male, and so the segregation-distorter remains twice as fit as its alternative allele. It's speculated that real-world group selection may have played a role in keeping the frequency of this gene as low as it seems to be. In which case, if mice were to evolve the ability to fly and migrate for the winter, they would probably form a single reproductive population, and would evolve to extinction as the segregation-distorter evolved to fixation.

Around 50% of the total genome of maize consists of transposons, DNA elements whose primary function is to copy themselves *into other locations of DNA*. A class of transposons called "P elements" seem to have first appeared in *Drosophila* only in the middle of the twentieth century, and spread to every population of the species within 50 years. The "Alu sequence" in humans, a 300-base transposon, is repeated between 300,000

and a million times in the human genome. This may not extinguish a species, but it doesn't help it; transposons cause more mutations which are as always mostly harmful, decrease the effective copying fidelity of DNA. Yet such cheaters are extremely fit.

Suppose that in some sexually reproducing species, a *perfect* DNA-copying mechanism is invented. Since most mutations are detrimental, this gene complex is an advantage to its holders. Now you might wonder about beneficial mutations—they do happen occasionally, so wouldn't the unmutable be at a disadvantage? But in a sexual species, a beneficial mutation that began in a mutable can spread to the descendants of unmutables as well. The mutables suffer from degenerate mutations in each generation; and the unmutables can sexually acquire, and thereby benefit from, any beneficial mutations that occur in the mutables. Thus the mutables have a pure disadvantage. The perfect DNA-copying mechanism rises in frequency to fixation. Ten thousand years later there's an ice age and the species goes out of business. It evolved to extinction.

The "bystander effect" is that, when someone is in trouble, solitary individuals are more likely to intervene than groups. A college student apparently having an epileptic seizure was helped 85% of the time by a single bystander, and 31% of the time by five bystanders. I speculate that even if the kinship relation in a hunter-gatherer tribe was strong enough to create a selection pressure for helping individuals not directly related, when *several* potential helpers were present, a genetic arms race might occur to be the *last* one to step forward. Everyone delays, hoping that someone else will do it. Humanity is facing multiple species-level extinction threats right now, and I gotta tell ya, there ain't a lot of people steppin' forward. If we lose this fight because virtually no one showed up on the battlefield, then—like a probably-large number of species which we don't see around today—we will have evolved to extinction.

Cancerous cells do pretty well in the body, prospering and amassing more resources, far outcompeting their more obedient counterparts. For a while.

Multicellular organisms can only exist because they've evolved powerful internal mechanisms to *outlaw evolution*. If the cells start evolving, they rapidly evolve to extinction: the organism dies.

So praise not evolution for the solicitous concern it shows for the individual; nearly all of your ancestors are dead. Praise not evolution for the solicitous concern it shows for a species; no one has ever found a complex adaptation which can only be interpreted as operating to preserve a species, and the mathematics would seem to indicate that this is virtually impossible. Indeed, it's perfectly possible for a species to evolve to extinction. Humanity may be finishing up the process right now. You can't even praise evolution for the solicitous concern it shows for genes; the battle between two alternative alleles at the same location is a zero-sum game for frequency.

Fitness is not always your friend.

The Tragedy of Group Selectionism



discussion
sharov.me/r/
3-c8

Before 1966, it was not unusual to see serious biologists advocating evolutionary hypotheses that we would now regard as magical thinking. These muddled notions played an important historical role in the development of later evolutionary theory, error calling forth correction; like the folly of English kings provoking into existence the Magna Carta and constitutional democracy.

As an example of romance, Vero Wynne-Edwards, Warder Allee, and J. L. Brereton, among others, believed that predators would voluntarily restrain their breeding to avoid overpopulating their habitat and exhausting the prey population.

But evolution does not open the floodgates to arbitrary purposes. You cannot explain a rattlesnake's rattle by saying that it exists to benefit other animals who would otherwise be bitten. No outside Evolution Fairy decides when a gene *ought* to be promoted; the gene's effect must somehow *directly cause* the gene to be more prevalent in the next generation. It's clear why our human sense of aesthetics, witnessing a population crash of foxes who've eaten all the rabbits, cries "Something should've been done!" But how would a gene complex for *restraining reproduction*—of all things!—cause itself to become more frequent in the next generation?

A human being designing a neat little toy ecology—for entertainment purposes, like a model railroad—might be annoyed if their painstakingly constructed fox and rabbit populations self-destructed by the foxes eating all the rabbits and then dying of starvation themselves. So the human would tinker with the toy ecology—a fox-breeding-restrainer is the obvious solution that leaps to our human minds—until the ecology looked nice and neat. Nature has no human, of course, but *that* needn't stop us—now that we know what we want on *aesthetic* grounds, we

just have to come up with a plausible argument that persuades Nature to want the *same* thing on *evolutionary* grounds.

Obviously, selection on the level of the individual won't produce individual restraint in breeding. Individuals who reproduce unrestrainedly will, naturally, produce more offspring than individuals who restrain themselves.

(Individual selection will not produce *individual sacrifice of breeding opportunities*. Individual selection can certainly produce individuals who, after acquiring all available resources, use those resources to produce four big eggs instead of eight small eggs—not to conserve social resources, but because that is the *individual* sweet spot for (number of eggs)×(egg survival probability). This does not get rid of the commons problem.)

But suppose that the species population was broken up into subpopulations, which were mostly isolated, and only occasionally interbred. Then, surely, subpopulations that restrained their breeding would be less likely to go extinct, and would send out more messengers, and create new colonies to reinhabit the territories of crashed populations.

The problem with this scenario wasn't that it was mathematically *impossible*. The problem was that it was *possible but very difficult*.

The fundamental problem is that it's not only restrained breeders who reap the benefits of restrained breeding. If some foxes refrain from spawning cubs who eat rabbits, then the uneaten rabbits don't go to *only* cubs who carry the restrained-breeding adaptation. The unrestrained foxes, and their many more cubs, will happily eat any rabbits left unhunted. The only way the restraining gene can survive against this pressure, is if the benefits of restraint preferentially go to restrainers.

Specifically, the requirement is $C/B < F_{ST}$ where C is the cost of altruism to the donor, B is the benefit of altruism to the recipient, and F_{ST} is the spatial structure of the population: the average *relatedness* between a randomly selected organism

and its randomly selected neighbor, where a “neighbor” is any other fox who benefits from an altruistic fox’s restraint.¹

So is the cost of restrained breeding sufficiently small, and the empirical benefit of less famine sufficiently large, compared to the empirical spatial structure of fox populations and rabbit populations, that the group selection argument can work?

The math suggests this is pretty unlikely. In Henry Harpending and Alan Rogers’ simulation,^{2,3} for example, the cost to altruists is 3% of fitness, pure altruist groups have a fitness twice as great as pure selfish groups, the subpopulation size is 25, and 20% of all deaths are replaced with messengers from another group: the result is polymorphic for selfishness and altruism. If the subpopulation size is doubled to 50, selfishness is fixed; if the cost to altruists is increased to 6%, selfishness is fixed; if the altruistic benefit is decreased by half, selfishness is fixed or in large majority. Neighborhood-groups must be very small, with only around 5 members, for group selection to operate when the cost of altruism exceeds 10%. This doesn’t seem plausibly true of foxes restraining their breeding.

You can guess by now, I think, that the group selectionists ultimately lost the scientific argument. The kicker was not the mathematical argument, but empirical observation: foxes *didn’t* restrain their breeding (I forget the exact species of dispute; it wasn’t foxes and rabbits), and indeed, predator-prey systems crash all the time. Group selectionism would later revive, somewhat, in drastically different form—mathematically speaking, there *is* neighborhood structure, which implies nonzero group selection pressure *not* necessarily capable of overcoming countervailing individual selection pressure, and if you don’t take it

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1. David Sloan Wilson, “A Theory of Group Selection,” *Proceedings of the National Academy of Sciences of the United States of America* 72, no. 1 (1975): 143–146.
 2. Henry Harpending and Alan Rogers, ‘On Wright’s mechanism for inter-group selection’, *Journal of Theoretical Biology*, 1987, vol. 127, pp.51-61.
 3. sharov.me/r/3-6



into account your math will be wrong, full stop. And evolved enforcement mechanisms (not originally postulated) change the game entirely. So why is this now-historical scientific dispute worthy material for *Overcoming Bias*?

A decade after the controversy, a biologist had a fascinating idea. The mathematical conditions for group selection overcoming individual selection were too extreme to be found in Nature. Why not create them artificially, in the laboratory? Michael J. Wade proceeded to do just that, repeatedly selecting populations of insects for low numbers of adults per subpopulation.⁴ And what was the result? Did the insects restrain their breeding and live in quiet peace with enough food for all?

No; the adults adapted to cannibalize eggs and larvae, especially female larvae.

Of course selecting for small subpopulation sizes would not select for individuals who restrained their *own* breeding; it would select for individuals who ate *other* individuals' children. Especially the girls.

Once you have that experimental result in hand—and it's massively obvious in retrospect—then it suddenly becomes clear how the original group selectionists allowed romanticism, a human sense of aesthetics, to cloud their predictions of Nature.

This is an archetypal example of a missed Third Alternative, resulting from a rationalization of a predetermined bottom line that produced a fake justification and then motivatedly stopped. The group selectionists didn't start with clear, fresh minds, happen upon the idea of group selection, and *neutrally* extrapolate forward the probable outcome. They started out with the beautiful idea of fox populations voluntarily restraining their reproduction to what the rabbit population would bear, Nature in perfect harmony; then they searched for a reason why this would happen, and came up with the idea of group selection; then, since they knew what they *wanted* the

4. Michael J. Wade, "Group selections among laboratory populations of *Tribolium*," *Proceedings of the National Academy of Sciences of the United States of America* 73, no. 12 (1976): 4604–4607, doi:10.1073/pnas.73.12.4604.

outcome of group selection to be, they didn't look for any *less* beautiful and aesthetic adaptations that group selection would be more likely to promote instead. If they'd *really* been trying to calmly and neutrally predict the result of selecting for small subpopulation sizes resistant to famine, they would have thought of cannibalizing other organisms' children or some similarly "ugly" outcome—*long before* they imagined anything so evolutionarily outré as *individual restraint in breeding!*

This also illustrates the point I was trying to make in Einstein's Arrogance: With large answer spaces, nearly all of the real work goes into promoting one possible answer to the point of being singled out for attention. If a hypothesis is improperly promoted to your attention—your sense of aesthetics suggests a beautiful way for Nature to be, and yet natural selection doesn't involve an Evolution Fairy who shares your appreciation—then this alone may seal your doom, unless you can manage to clear your mind entirely and start over.

In principle, the world's stupidest person may say the Sun is shining, but that doesn't make it dark out. Even if an answer is suggested by a lunatic on LSD, you should be able to neutrally calculate the evidence for and against, and if necessary, *un*-believe.

In practice, the group selectionists were doomed because their bottom line was originally suggested by their sense of aesthetics, and Nature's bottom line was produced by natural selection. These two processes had no principled reason for their outputs to correlate, and indeed they didn't. All the furious argument afterward didn't change that.

If you start with your own desires for what Nature should do, consider Nature's own observed reasons for doing things, and then rationalize an extremely persuasive argument for why Nature should produce your preferred outcome for Nature's own reasons, then Nature, alas, *still* won't listen. The universe has no mind and is not subject to clever political persuasion. You can argue all day why gravity should really make water flow *uphill*, and the water just ends up in the same place regardless.

It's like the universe plain isn't listening. J. R. Molloy said:⁵ "Nature is the ultimate bigot, because it is obstinately and intolerantly devoted to its own prejudices and absolutely refuses to yield to the most persuasive rationalizations of humans."

I often recommend evolutionary biology to friends just because the modern field tries to train its students against rationalization, error calling forth correction. Physicists and electrical engineers don't have to be carefully trained to avoid anthropomorphizing electrons, because electrons don't exhibit mindish behaviors. Natural selection creates purposefulnesses which are alien to humans, and students of evolutionary theory are warned accordingly. It's good training for any thinker, but it is *especially* important if you want to think clearly about other weird mindish processes that do not work like you do.

5. sharov.me/r/3-7
extropians.weidai.com



Fake Optimization Criteria



discussion
sharov.me/r/
3-c9

I've previously dwelt in considerable length upon forms of rationalization whereby our beliefs appear to match the evidence much more strongly than they actually do. And I'm not overemphasizing the point, either. If we could beat this fundamental metabias and see what every hypothesis *really* predicted, we would be able to recover from almost any other error of fact.

The mirror challenge for decision theory is seeing which option a choice criterion *really* endorses. If your stated moral principles call for you to provide laptops to everyone, does that *really* endorse buying a \$1 million gem-studded laptop for yourself, or spending the same money on shipping 5,000 OLPCs?

We seem to have evolved a knack for arguing that practically any goal implies practically any action. A phlogiston theorist explaining why magnesium gains weight when burned has nothing on an Inquisitor explaining why God's infinite love for all His children requires burning some of them at the stake.

There's no mystery about this. Politics was a feature of the ancestral environment. We are descended from those who argued most persuasively that the good of the tribe meant executing their hated rival Uglak. (We sure ain't descended from Uglak.)

And yet...is it possible to *prove* that if Robert Mugabe cared *only* for the good of Zimbabwe, he would resign from its presidency? You can *argue* that the policy follows from the goal, but haven't we just seen that humans can match up any goal to any policy? How do you know that you're right and Mugabe is wrong? (There are a number of reasons this is a good guess, but bear with me here.)

Human motives are manifold and obscure, our decision processes as vastly complicated as our brains. And the world itself is vastly complicated, on every choice of real-world policy. Can we even *prove* that human beings are rationalizing—that we're

systematically distorting the link from principles to policy—when we lack a single firm place on which to stand? When there’s no way to find out *exactly* what even a single optimization criterion implies? (Actually, you can just observe that people *disagree* about office politics in ways that strangely correlate to their own interests, while simultaneously denying that any such interests are at work. But again, bear with me here.)

Where is the standardized, open-source, generally intelligent, consequentialist optimization process into which we can feed a complete morality as an XML file, to find out what that morality *really* recommends when applied to our world? Is there even a single real-world case where we can know *exactly* what a choice criterion recommends? Where is the *pure* moral reasoner—of known utility function, purged of all other stray desires that might distort its optimization—whose trustworthy output we can contrast to human rationalizations of the same utility function?

Why, it’s our old friend the alien god, of course! Natural selection is guaranteed free of all mercy, all love, all compassion, all aesthetic sensibilities, all political factionalism, all ideological allegiances, all academic ambitions, all libertarianism, all socialism, all Blue and all Green. Natural selection doesn’t *maximize* its criterion of inclusive genetic fitness—it’s not that smart. But when you look at the output of natural selection, you are guaranteed to be looking at an output that was optimized *only* for inclusive genetic fitness, and not the interests of the US agricultural industry.

In the case histories of evolutionary science—in, for example, The Tragedy of Group Selectionism—we can directly compare human rationalizations to the result of *pure* optimization for a known criterion. What did Wynne-Edwards think would be the result of group selection for small subpopulation sizes? Voluntary individual restraint in breeding, and enough food for everyone. What was the actual laboratory result? Cannibalism.

Now you might ask: Are these case histories of evolutionary science really relevant to human morality, which doesn’t give two figs for inclusive genetic fitness when it gets in the way of love, compassion, aesthetics, healing, freedom, fairness, et

cetera? Human societies didn't even have a concept of "inclusive genetic fitness" until the twentieth century.

But I ask in return: If we can't see clearly the result of a single monotone optimization criterion—if we can't even train ourselves to hear a single pure note—then how will we listen to an orchestra? How will we see that "Always be selfish" or "Always obey the government" are poor guiding principles for human beings to adopt—if we think that even *optimizing genes for inclusive fitness* will yield organisms that sacrifice reproductive opportunities in the name of social resource conservation?

To train ourselves to see clearly, we need simple practice cases.

Adaptation-Executers, Not Fitness-Maximizers

Individual organisms are best thought of as adaptation-executers rather than as fitness-maximizers.

—John Tooby and Leda Cosmides,
“The Psychological Foundations of Culture”¹



discussion
sharov.me/r/
3-c10

Fifty thousand years ago, the taste buds of *Homo sapiens* directed their bearers to the scarcest, most critical food resources—sugar and fat. Calories, in a word. Today, the context of a taste bud’s function has changed, but the taste buds themselves have not. Calories, far from being scarce (in First World countries), are actively harmful. Micronutrients that were reliably abundant in leaves and nuts are absent from bread, but our taste buds don’t complain. A scoop of ice cream is a superstimulus, containing more sugar, fat, and salt than anything in the ancestral environment.

No human being with the *deliberate* goal of maximizing their alleles’ inclusive genetic fitness would ever eat a cookie unless they were starving. But individual organisms are best thought of as adaptation-executers, not fitness-maximizers.

A Phillips-head screwdriver, though its designer intended it to turn screws, won’t reconform itself to a flat-head screw to fulfill its function. We created these tools, but they exist independently of us, and they continue independently of us.

The atoms of a screwdriver don’t have tiny little XML tags inside describing their “objective” purpose. The designer had something in mind, yes, but that’s not the same as what *happens*

1. John Tooby and Leda Cosmides, “The Psychological Foundations of Culture,” in *The Adapted Mind: Evolutionary Psychology and the Generation of Culture*, ed. Jerome H. Barkow, Leda Cosmides, and John Tooby (New York: Oxford University Press, 1992), 19–136.

in the real world. If you forgot that the designer is a separate entity from the designed thing, you might think, “The *purpose* of the screwdriver is to drive screws”—as though this were an explicit property of the screwdriver itself, rather than a property of the designer’s state of mind. You might be surprised that the screwdriver didn’t reconfigure itself to the flat-head screw, since, after all, the screwdriver’s *purpose* is to turn screws.

The *cause* of the screwdriver’s existence is the designer’s mind, which imagined an imaginary screw, and imagined an imaginary handle turning. The *actual* operation of the screwdriver, its *actual* fit to an actual screw head, *cannot* be the objective cause of the screwdriver’s existence: The future cannot cause the past. But the designer’s brain, as an actually existent thing within the past, can indeed be the cause of the screwdriver.

The *consequence* of the screwdriver’s existence may not correspond to the imaginary consequences in the designer’s mind. The screwdriver blade could slip and cut the user’s hand.

And the *meaning* of the screwdriver—why, that’s something that exists in the mind of a user, not in tiny little labels on screwdriver atoms. The designer may intend it to turn screws. A murderer may buy it to use as a weapon. And then accidentally drop it, to be picked up by a child, who uses it as a chisel.

So the screwdriver’s *cause*, and its *shape*, and its *consequence*, and its various *meanings*, are all different things; and only *one* of these things is found within the screwdriver itself.

Where do taste buds come from? Not from an intelligent designer visualizing their consequences, but from a frozen history of ancestry: Adam liked sugar and ate an apple and reproduced, Barbara liked sugar and ate an apple and reproduced, Charlie liked sugar and ate an apple and reproduced, and 2763 generations later, the allele became fixed in the population. For convenience of thought, we sometimes compress this giant history and say: “Evolution did it.” But it’s not a quick, local event like a human designer visualizing a screwdriver. This is the *objective cause* of a taste bud.

What is the *objective shape* of a taste bud? Technically, it’s a molecular sensor connected to reinforcement circuitry. This

adds another level of indirection, because the taste bud isn't directly acquiring food. It's influencing the organism's mind, making the organism want to eat foods that are similar to the food just eaten.

What is the *objective consequence* of a taste bud? In a modern First World human, it plays out in multiple chains of causality: from the desire to eat more chocolate, to the plan to eat more chocolate, to eating chocolate, to getting fat, to getting fewer dates, to reproducing less successfully. This consequence is directly *opposite* the key regularity in the long chain of ancestral successes that caused the taste bud's shape. But, since overeating has only recently become a problem, no significant evolution (compressed regularity of ancestry) has further influenced the taste bud's shape.

What is the *meaning* of eating chocolate? That's between you and your moral philosophy. Personally, I think chocolate tastes good, but I wish it were less harmful; acceptable solutions would include redesigning the chocolate or redesigning my biochemistry.

Smushing several of the concepts together, you could sort-of-say, "Modern humans do today what would have propagated our genes in a hunter-gatherer society, whether or not it helps our genes in a modern society." But this still isn't quite right, because we're not *actually* asking ourselves which behaviors would maximize our ancestors' inclusive fitness. And many of our activities today have no ancestral analogue. In the hunter-gatherer society there wasn't any such thing as chocolate.

So it's better to view our taste buds as an *adaptation* fitted to ancestral conditions that included near-starvation and apples and roast rabbit, which modern humans *execute* in a new context that includes cheap chocolate and constant bombardment by advertisements.

Therefore it is said: Individual organisms are best thought of as adaptation-executers, not fitness-maximizers.

Evolutionary Psychology



discussion
sharov.me/r/
3-c11

Like “IRC chat” or “TCP/IP protocol,” the phrase “reproductive organ” is redundant. *All* organs are reproductive organs. Where do a bird’s wings come from? An Evolution-of-Birds Fairy who thinks that flying is really neat? The bird’s wings are there because they contributed to the bird’s ancestors’ reproduction. Likewise the bird’s heart, lungs, and genitals. At most we might find it worthwhile to distinguish between *directly* reproductive organs and *indirectly* reproductive organs.

This observation holds true also of the brain, the most complex organ system known to biology. Some brain organs are directly reproductive, like lust; others are indirectly reproductive, like anger.

Where does the human emotion of anger come from? An Evolution-of-Humans Fairy who thought that anger was a worthwhile feature? The neural circuitry of anger is a reproductive organ as surely as your liver. Anger exists in *Homo sapiens* because angry ancestors had more kids. *There’s no other way it could have gotten there.*

This *historical* fact about the origin of anger confuses all too many people. They say, “Wait, are you saying that when I’m angry, I’m subconsciously trying to have children? That’s not what I’m thinking after someone punches me in the nose.”

No. No. **No. NO!**

Individual organisms are best thought of as adaptation-executers, not fitness-maximizers. The *cause* of an adaptation, the *shape* of an adaptation, and the *consequence* of an adaptation are all separate things. If you built a toaster, you wouldn’t expect the toaster to reshape itself when you tried to cram in a whole loaf of bread; yes, you intended it to make toast, but that intention is a fact about you, not a fact about the toaster. The toaster has no sense of its own purpose.

But a toaster is not an intention-bearing object. It is not a mind at all, so we are not tempted to attribute goals to it. If we

see the toaster as purposed, we don't think the toaster knows it, because we don't think the toaster knows *anything*.

It's like the old test of being asked to say the color of the letters in "blue." It takes longer for subjects to name this color, because of the need to untangle the meaning of the letters and the color of the letters. You wouldn't have similar trouble naming the color of the letters in "wind."

But a human brain, in addition to being an artifact historically produced by evolution, is also a mind capable of bearing its own intentions, purposes, desires, goals, and plans. Both a bee and a human are designs, but only a human is a designer. The bee is "wind;" the human is "blue."

Cognitive causes are *ontologically distinct* from evolutionary causes. They are made out of a different kind of stuff. Cognitive causes are made of neurons. Evolutionary causes are made of ancestors.

The most obvious kind of cognitive cause is deliberate, like an intention to go to the supermarket, or a plan for toasting toast. But an emotion also exists physically in the brain, as a train of neural impulses or a cloud of spreading hormones. Likewise an instinct, or a flash of visualization, or a fleetingly suppressed thought; if you could scan the brain in three dimensions and you understood the code, you would be able to *see* them.

Even subconscious cognitions exist physically in the brain. "Power tends to corrupt," observed Lord Acton.¹ Stalin may or may not have believed himself an altruist, working toward the greatest good for the greatest number. But it seems likely that, somewhere in Stalin's brain, there were neural circuits that reinforced pleasurably the exercise of power, and neural circuits that detected anticipations of increases and decreases in power. If there were nothing in Stalin's brain that correlated to power—no little light that went on for political command,

1. John Dalberg-Acton, Letter to Mandell Creighton, April 5, 1887, sharov.me/r/3-8



and off for political weakness—then how could Stalin’s brain have known to be corrupted by power?

Evolutionary selection pressures are *ontologically distinct* from the biological artifacts they create. The evolutionary cause of a bird’s wings is millions of ancestor-birds who reproduced more often than other ancestor-birds, with statistical regularity owing to their possession of incrementally improved wings compared to their competitors. We compress this gargantuan historical-statistical macrofact by saying “evolution did it.”

Natural selection is ontologically distinct from creatures; evolution is not a little furry thing lurking in an undiscovered forest. Evolution is a causal, statistical regularity in the reproductive history of ancestors.

And this logic applies also to the brain. Evolution has made wings that flap, but do not understand flappiness. It has made legs that walk, but do not understand walkyness. Evolution has carved bones of calcium ions, but the bones themselves have no explicit concept of strength, let alone inclusive genetic fitness. And evolution designed brains themselves capable of designing; yet these brains had no more concept of evolution than a bird has of aerodynamics. Until the twentieth century, not a single human brain explicitly represented the complex abstract concept of *inclusive genetic fitness*.

When we’re told that “The evolutionary purpose of anger is to increase inclusive genetic fitness,” there’s a tendency to slide to “*The purpose of* anger is reproduction” to “The cognitive purpose of anger is reproduction.” No! The statistical regularity of ancestral history isn’t *in* the brain, even subconsciously, any more than the designer’s intentions of toast are in a toaster!

Thinking that your built-in anger-circuitry embodies an explicit desire to reproduce is like thinking your hand is an embodied mental desire to pick things up.

Your hand is not wholly cut off from your mental desires. In particular circumstances, you can control the flexing of your fingers by an act of will. If you bend down and pick up a penny, then this may represent an act of will; but it is not an act of will that made your hand grow in the first place.

One must distinguish a one-time event of particular anger (anger-1, anger-2, anger-3) from the underlying neural circuitry for anger. An anger-event is a cognitive cause, and an anger-event may have cognitive causes, but you didn't will the anger-circuitry to be wired into the brain.

So you have to distinguish the event of anger, from the circuitry of anger, from the gene complex that laid down the neural template, from the ancestral macrofact that explains the gene complex's presence.

If there were ever a discipline that genuinely *demanded* X-Treme Nitpicking, it is evolutionary psychology.

Consider, O my readers, this sordid and joyful tale: A man and a woman meet in a bar. The man is attracted to her clear complexion and firm breasts, which would have been fertility cues in the ancestral environment, but which in this case result from makeup and a bra. This does not bother the man; he just likes the way she looks. His clear-complexion-detecting neural circuitry does not know that its purpose is to detect fertility, any more than the atoms in his hand contain tiny little XML tags reading "<purpose>pick things up</purpose>." The woman is attracted to his confident smile and firm manner, cues to high status, which in the ancestral environment would have signified the ability to provide resources for children. She plans to use birth control, but her confident-smile-detectors don't know this any more than a toaster knows its designer intended it to make toast. She's not concerned philosophically with the meaning of this rebellion, because her brain is a creationist and denies vehemently that evolution exists. He's not concerned philosophically with the meaning of this rebellion, because he just wants to get laid. They go to a hotel, and undress. He puts on a condom, because he doesn't want kids, just the dopamine-noradrenaline rush of sex, which reliably produced offspring 50,000 years ago when it was an invariant feature of the ancestral environment that condoms did not exist. They have sex, and shower, and go their separate ways. The main objective consequence is to keep the bar and the hotel and the condom-manufacturer in business; which was not the cognitive purpose in their minds, and has virtually nothing to

do with the key statistical regularities of reproduction 50,000 years ago which explain how they got the genes that built their brains that executed all this behavior.

To reason correctly about evolutionary psychology you must simultaneously consider many complicated abstract facts that are strongly related yet importantly distinct, without a single mixup or conflation.

An Especially Elegant Evolutionary Psychology Experiment

In a 1989 Canadian study, adults were asked to imagine the death of children of various ages and estimate which deaths would create the greatest sense of loss in a parent. The results, plotted on a graph, show grief growing until just before adolescence and then beginning to drop. When this curve was compared with a curve showing changes in reproductive potential over the life cycle (a pattern calculated from Canadian demographic data), the correlation was fairly strong. But much stronger—nearly perfect, in fact—was the correlation between the grief curves of these modern Canadians and the reproductive-potential curve of a hunter-gatherer people, the !Kung of Africa. In other words, the pattern of changing grief was almost exactly what a Darwinian would predict, given demographic realities in the ancestral environment.

—Robert Wright, *The Moral Animal*,
summarizing Crawford et al.¹

The first correlation was 0.64, the second an extremely high 0.92 ($N = 221$).

The most obvious *inelegance* of this study, as described, is that it was conducted by asking human adults to imagine parental grief, rather than asking real parents with children



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1. Robert Wright, *The Moral Animal: Why We Are the Way We Are: The New Science of Evolutionary Psychology* (Pantheon Books, 1994); Charles B. Crawford, Brenda E. Salter, and Kerry L. Jang, "Human Grief: Is Its Intensity Related to the Reproductive Value of the Deceased?," *Ethology and Sociobiology* 10, no. 4 (1989): 297–307.

of particular ages. (Presumably that would have cost more / allowed fewer subjects.) However, my understanding is that the results here squared well with the data from closer studies of parental grief that were looking for other correlations (i.e., a raw correlation between parental grief and child age).

That said, consider some of this experiment's elegant aspects:

1. A correlation of 0.92(!) This may sound suspiciously high—could evolution really do such exact fine-tuning?—until you realize that this selection pressure was not only great enough to *fine-tune* parental grief, but, in fact, *carve it out of existence from scratch in the first place*.
2. People who say that evolutionary psychology hasn't made any advance predictions are (ironically) mere victims of “no one knows what science doesn't know” syndrome. You wouldn't even *think of this as an experiment to be performed* if not for evolutionary psychology.
3. The experiment illustrates, as beautifully and as cleanly as any I have ever seen, the distinction between a *conscious or subconscious ulterior motive* and an *executing adaptation with no realtime sensitivity to the original selection pressure that created it*.

The parental grief is *not even subconsciously* about reproductive value—otherwise it would update for Canadian reproductive value instead of !Kung reproductive value. Grief is an adaptation that now simply exists, real in the mind and continuing under its own inertia.

Parents do *not* care about children for the sake of their reproductive contribution. Parents care about children for their own sake; and the *non-cognitive, evolutionary-historical* reason why such minds exist in the universe in the first place is that children carry their parents' genes.

Indeed, evolution is the reason why there are any minds in the universe at all. So you can see why I'd want to draw a sharp line through my cynicism about ulterior motives at the evolutionary-cognitive boundary; otherwise, I might as well

stand up in a supermarket checkout line and say, “Hey! You’re only correctly processing visual information while bagging my groceries in order to maximize your inclusive genetic fitness!”

1. I think 0.92 is the highest correlation I’ve ever seen in any evolutionary psychology experiment, and indeed, one of the highest correlations I’ve seen in any psychology experiment. (Although I’ve seen e.g. a correlation of 0.98 reported for asking one group of subjects “How similar is *A* to *B*?” and another group “What is the probability of *A* given *B*?” on questions like “How likely are you to draw 60 red balls and 40 white balls from this barrel of 800 red balls and 200 white balls?”—in other words, these are simply processed as the same question.)

Since we are all Bayesians here, we may take our priors into account and ask if at least *some* of this unexpectedly high correlation is due to luck. The evolutionary fine-tuning we can probably take for granted; this is a huge selection pressure we’re talking about. The remaining sources of suspiciously low variance are (a) whether a large group of adults could correctly envision, on average, relative degrees of parental grief (apparently they can), and (b) whether the surviving !Kung are *typical* ancestral hunter-gatherers in this dimension, or whether variance *between* hunter-gatherer tribal types should have been too high to allow a correlation of 0.92.

But even after taking into account any skeptical priors, correlation 0.92 and $N = 221$ is pretty strong evidence, and our posteriors should be less skeptical on all these counts.

2. You might think it an inelegance of the experiment that it was performed *prospectively* on imagined grief, rather than retrospectively on real grief. But it is *prospectively* imagined grief that will actually operate to steer parental behavior *away* from losing the child! From an evolutionary standpoint, an actual dead child is a sunk cost; evolution “wants” the parent to learn from the pain, not do it again, adjust back to their hedonic set point, and go on raising other children.

3. Similarly, the graph that correlates to parental grief is for *the future reproductive potential of a child that has survived to a given age*, and not *the sunk cost of raising the child which has survived to that age*. (Might we get an even *higher* correlation

if we tried to take into account the reproductive opportunity cost of raising a child of age X to independent maturity, while discarding all sunk costs to raise a child to age X ?)

Humans usually do notice sunk costs—this is presumably either an adaptation to prevent us from switching strategies too often (compensating for an overeager opportunity-noticer?) or an unfortunate spandrel of pain felt on wasting resources.

Evolution, on the other hand—it's not that evolution "doesn't care about sunk costs," but that evolution doesn't even remotely "think" that way; "evolution" is just a macrofact about the real historical reproductive consequences.

So—of course—the parental grief adaptation is fine-tuned in a way that has nothing to do with past investment in a child, and everything to do with the future reproductive consequences of losing that child. Natural selection isn't crazy about sunk costs the way we are.

But—of course—the parental grief adaptation goes on functioning as if the parent were living in a !Kung tribe rather than Canada. Most humans would notice the difference.

Humans and natural selection are insane in *different stable complicated ways*.

Superstimuli and the Collapse of Western Civilization

At least three people have died playing online games for days without rest. People have lost their spouses, jobs, and children to World of Warcraft. If people have the right to play video games—and it's hard to imagine a more fundamental right—then the market is going to respond by supplying the most *engaging* video games that can be sold, to the point that exceptionally engaged consumers are removed from the gene pool.

How does a consumer product become so *involving* that, after 57 hours of using the product, the consumer would rather use the product for one more hour than eat or sleep? (I suppose one could argue that the consumer makes a rational decision that they'd rather play Starcraft for the next hour than live out the rest of their life, but let's just not go there. Please.)

A candy bar is a *superstimulus*: it contains more concentrated sugar, salt, and fat than anything that exists in the ancestral environment. A candy bar matches taste buds that evolved in a hunter-gatherer environment, but it matches those taste buds much more strongly than anything that actually existed *in* the hunter-gatherer environment. The signal that once reliably correlated to healthy food has been hijacked, blotted out with a point in tastespace that wasn't in the training dataset—an impossibly distant outlier on the old ancestral graphs. Tastiness, formerly representing the evolutionarily identified correlates of healthiness, has been reverse-engineered and perfectly matched with an artificial substance. Unfortunately there's no equally powerful market incentive to make the resulting food item as healthy as it is tasty. We can't taste healthfulness, after all.



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The now-famous Dove Evolution¹ video shows the painstaking construction of another superstimulus: an ordinary woman transformed by makeup, careful photography, and finally extensive Photoshopping, into a billboard model—a beauty impossible, unmatched by human women in the unretouched real world. Actual women are killing themselves (e.g., supermodels using cocaine to keep their weight down) to keep up with competitors that literally don't exist.

And likewise, a video game can be so much more *engaging* than mere reality, even through a simple computer monitor, that someone will play it without food or sleep until they literally die. I don't know all the tricks used in video games, but I can guess some of them—challenges poised at the critical point between ease and impossibility, intermittent reinforcement, feedback showing an ever-increasing score, social involvement in massively multiplayer games.

Is there a limit to the market incentive to make video games more engaging? You might hope there'd be no incentive past the point where the players lose their jobs; after all, they must be able to pay their subscription fee. This would imply a “sweet spot” for the addictiveness of games, where the mode of the bell curve is having fun, and only a few unfortunate souls on the tail become addicted to the point of losing their jobs. As of 2007, playing *World of Warcraft* for 58 hours straight until you literally die is still the exception rather than the rule. But video game manufacturers compete against each other, and if you can make your game 5% more addictive, you may be able to steal 50% of your competitor's customers. You can see how this problem could get a *lot* worse.

If people have the right to be tempted—and that's what free will is all about—the market is going to respond by supplying as much temptation as can be sold. The incentive is to make your stimuli 5% more tempting than those of your current leading

1. sharov.me/r/3-9
youtube.com



competitors. This continues well beyond the point where the stimuli become ancestrally anomalous superstimuli. Consider how our standards of product-selling feminine beauty have changed since the advertisements of the 1950s. And as candy bars demonstrate, the market incentive also continues well beyond the point where the superstimulus begins wreaking collateral damage on the consumer.

So why don't we just say no? A key assumption of free-market economics is that, in the absence of force and fraud, people can always refuse to engage in a harmful transaction. (To the extent this is true, a free market would be, not merely the *best* policy on the whole, but a policy with few or no downsides.)

An organism that regularly passes up food will die, as some video game players found out the hard way. But, on some occasions in the ancestral environment, a typically beneficial (and therefore tempting) act may in fact be harmful. Humans, as organisms, have an unusually strong ability to perceive these special cases using abstract thought. On the other hand we also tend to imagine lots of special-case consequences that don't exist, like ancestor spirits commanding us not to eat perfectly good rabbits.

Evolution seems to have struck a compromise, or perhaps just aggregated new systems on top of old. *Homo sapiens* are still tempted by food, but our oversized prefrontal cortices give us a *limited* ability to resist temptation. Not unlimited ability—our ancestors with too much willpower probably starved themselves to sacrifice to the gods, or failed to commit adultery one too many times. The video game players who died must have exercised willpower (in some sense) to keep playing for so long without food or sleep; the evolutionary hazard of self-control.

Resisting any temptation takes conscious expenditure of an exhaustible supply of mental energy. It is not in fact *true* that we can “just say no”—not *just* say no, without cost to ourselves. Even humans who won the birth lottery for willpower or foresightfulness still pay a price to resist temptation. The price is just more easily paid.

Our limited willpower evolved to deal with ancestral temptations; it may not operate well against enticements beyond anything known to hunter-gatherers. Even where we successfully resist a superstimulus, it seems plausible that the effort required would deplete willpower much faster than resisting ancestral temptations.

Is public display of superstimuli a negative externality, even to the people who say no? Should we ban chocolate cookie ads, or storefronts that openly say “Ice Cream”?

Just because a problem exists doesn’t show (without further justification and a substantial burden of proof) that the government can fix it. The regulator’s career incentive does not focus on products that combine low-grade consumer harm with addictive superstimuli; it focuses on products with failure modes spectacular enough to get into the newspaper. Conversely, just because the government may *not* be able to fix something, doesn’t mean it *isn’t* going wrong.

I leave you with a final argument from fictional evidence: Simon Funk’s online novel *After Life*² depicts (among other plot points) the planned extermination of biological *Homo sapiens*—not by marching robot armies, but by artificial children that are much cuter and sweeter and more fun to raise than real children. Perhaps the demographic collapse of advanced societies happens because the market supplies ever-more-tempting alternatives to having children, while the attractiveness of changing diapers remains constant over time. Where are the advertising billboards that say “BREED”? Who will pay professional image consultants to make arguing with sullen teenagers seem more alluring than a vacation in Tahiti?³

“In the end,” Simon Funk wrote, “the human species was simply marketed out of existence.”

2. sharov.me/r/3-10
After Life • interstice.com



3. Edit 2018: In this environment, genes or cultures that caused people to raise more teenagers would become much more common.

Thou Art Godshatter

Before the twentieth century, not a single human being had an explicit concept of “inclusive genetic fitness,” the sole and absolute obsession of the blind idiot god. We have no instinctive revulsion of condoms or oral sex. Our brains, those supreme reproductive organs, don’t perform a check for reproductive efficacy before granting us sexual pleasure.

Why not? Why *aren’t* we consciously obsessed with inclusive genetic fitness? Why did the Evolution-of-Humans Fairy create brains that would invent condoms? “It would have been so *easy*,” thinks the human, who can design new complex systems in an afternoon.

The Evolution Fairy, as we all know, is obsessed with inclusive genetic fitness. When she decides which genes to promote to universality, she doesn’t seem to take into account *anything* except the number of copies a gene produces. (How strange!)

But since the maker of intelligence is thus obsessed, why not create intelligent agents—you can’t call them humans—who would likewise care purely about inclusive genetic fitness? Such agents would have sex only as a means of reproduction, and wouldn’t bother with sex that involved birth control. They could eat food out of an explicitly reasoned belief that food was necessary to reproduce, not because they liked the taste, and so they wouldn’t eat candy if it became detrimental to survival or reproduction. Post-menopausal women would babysit grandchildren until they became sick enough to be a net drain on resources, and would then commit suicide.

It seems like such an obvious design improvement—from the Evolution Fairy’s perspective.

Now it’s clear that it’s hard to build a powerful enough consequentialist. Natural selection sort-of reasons consequentially, but only by depending on the *actual consequences*. Human evolutionary theorists have to do really high-falutin’ abstract



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reasoning in order to *imagine* the links between adaptations and reproductive success.

But human brains clearly *can* imagine these links in protein. So when the Evolution Fairy made humans, why did It bother with *any* motivation except inclusive genetic fitness?

It's been less than two centuries since a protein brain first represented the concept of natural selection. The modern notion of "inclusive genetic fitness" is even more subtle, a highly abstract concept. What matters is not the number of shared genes. Chimpanzees share 95% of your genes. What matters is shared genetic *variance*, *within* a reproducing population—your sister is one-half related to you, because any variations in your genome, within the human species, are 50% likely to be shared by your sister.

Only in the last century—arguably only in the last fifty years—have evolutionary biologists really begun to understand the full range of causes of reproductive success, things like reciprocal altruism and costly signaling. Without all this highly detailed knowledge, an intelligent agent that set out to "maximize inclusive fitness" would fall flat on its face.

So why not preprogram protein brains with the knowledge? Why wasn't a concept of "inclusive genetic fitness" *programmed* into us, along with a library of explicit strategies? Then you could dispense with all the reinforcers. The organism would be born knowing that, with high probability, fatty foods would lead to fitness. If the organism later learned that this was no longer the case, it would stop eating fatty foods. You could refactor the whole system. And it wouldn't invent condoms or cookies.

This looks like it should be quite possible in principle. I occasionally run into people who don't quite understand consequentialism, who say, "But if the organism doesn't have a separate drive to eat, it will starve, and so fail to reproduce." So long as the organism knows *this very fact*, and has a utility function that values reproduction, it will automatically eat. In fact, this is *exactly* the consequentialist reasoning that natural selection itself used to build automatic eaters.

What about curiosity? Wouldn't a consequentialist only be curious when it saw some specific reason to be curious? And wouldn't this cause it to miss out on lots of important knowledge that came with no specific reason for investigation attached? Again, a consequentialist will investigate given only the knowledge of this very same fact. If you consider the curiosity drive of a human—which is not indiscriminating, but responds to particular features of problems—then this complex adaptation is purely the result of consequentialist reasoning by DNA, an *implicit* representation of knowledge: Ancestors who engaged in this kind of inquiry left more descendants.

So in principle, the pure reproductive consequentialist is possible. In principle, all the ancestral history *implicitly* represented in cognitive adaptations can be converted to *explicitly* represented knowledge, running on a core consequentialist.

But the blind idiot god isn't that smart. Evolution is not a human programmer who can simultaneously refactor whole code architectures. Evolution is not a human programmer who can sit down and type out instructions at sixty words per minute.

For millions of years before hominid consequentialism, there was reinforcement learning. The reward signals were events that correlated reliably to reproduction. You can't ask a nonhominid brain to foresee that a child eating fatty foods now will live through the winter. So the DNA builds a protein brain that generates a reward signal for eating fatty food. Then it's up to the organism to learn which prey animals are tastiest.

DNA constructs protein brains with reward signals that have a *long-distance* correlation to reproductive fitness, but a *short-distance* correlation to organism behavior. You don't have to figure out that eating sugary food in the fall will lead to digesting calories that can be stored fat to help you survive the winter so that you mate in spring to produce offspring in summer. An apple simply tastes good, and your brain just has to plot out how to get more apples off the tree.

And so organisms evolve rewards for eating, and building nests, and scaring off competitors, and helping siblings, and discovering important truths, and forming strong alliances, and arguing persuasively, and of course having sex ...

When hominid brains capable of cross-domain consequential reasoning began to show up, they reasoned consequentially about how to get the *existing* reinforcers. It was a relatively simple hack, vastly simpler than rebuilding an “inclusive fitness maximizer” from scratch. The protein brains plotted how to acquire calories and sex, without any explicit cognitive representation of “inclusive fitness.”

A human engineer would have said, “Whoa, I’ve just invented a consequentialist! Now I can take all my previous hard-won knowledge about which behaviors improve fitness, and declare it explicitly! I can convert all this complicated reinforcement learning machinery into a simple declarative knowledge statement that ‘fatty foods and sex usually improve your inclusive fitness.’ Consequential reasoning will automatically take care of the rest. Plus, it won’t have the obvious failure mode where it invents condoms!”

But then a human engineer wouldn’t have built the retina backward, either.

The blind idiot god is not a unitary purpose, but a many-splintered attention. Foxes evolve to catch rabbits, rabbits evolve to evade foxes; there are as many evolutions as species. But *within* each species, the blind idiot god is purely obsessed with inclusive genetic fitness. No quality is valued, not even survival, except insofar as it increases reproductive fitness. There’s no point in an organism with steel skin if it ends up having 1% less reproductive capacity.

Yet when the blind idiot god created protein computers, its monomaniacal focus on inclusive genetic fitness was not faithfully transmitted. Its optimization criterion did not successfully quine. We, the handiwork of evolution, are as alien to evolution as our Maker is alien to us. One pure utility function splintered into a thousand shards of desire.

Why? Above all, because evolution is stupid in an absolute sense. But also because the *first* protein computers weren’t anywhere near as *general* as the blind idiot god, and could only utilize short-term desires.

In the final analysis, asking why evolution didn’t build humans to maximize inclusive genetic fitness is like asking why

evolution didn't hand humans a ribosome and tell them to design their own biochemistry. Because evolution can't refactor code that fast, that's why. But maybe in a billion years of continued natural selection that's exactly what *would* happen, if intelligence were foolish enough to allow the idiot god continued reign.

The Mote in God's Eye by Niven and Pournelle depicts an intelligent species that stayed biological a little too long, slowly becoming truly enslaved by evolution, gradually turning into true fitness maximizers obsessed with outreproducing each other. But thankfully that's not what happened. Not here on Earth. At least not yet.

So humans love the taste of sugar and fat, and we love our sons and daughters. We seek social status, and sex. We sing and dance and play. We learn for the love of learning.

A thousand delicious tastes, matched to ancient reinforcers that once correlated with reproductive fitness—now sought whether or not they enhance reproduction. Sex with birth control, chocolate, the music of long-dead Bach on a CD.

And when we finally learn about evolution, we think to ourselves: "Obsess all day about inclusive genetic fitness? Where's the fun in *that*?"

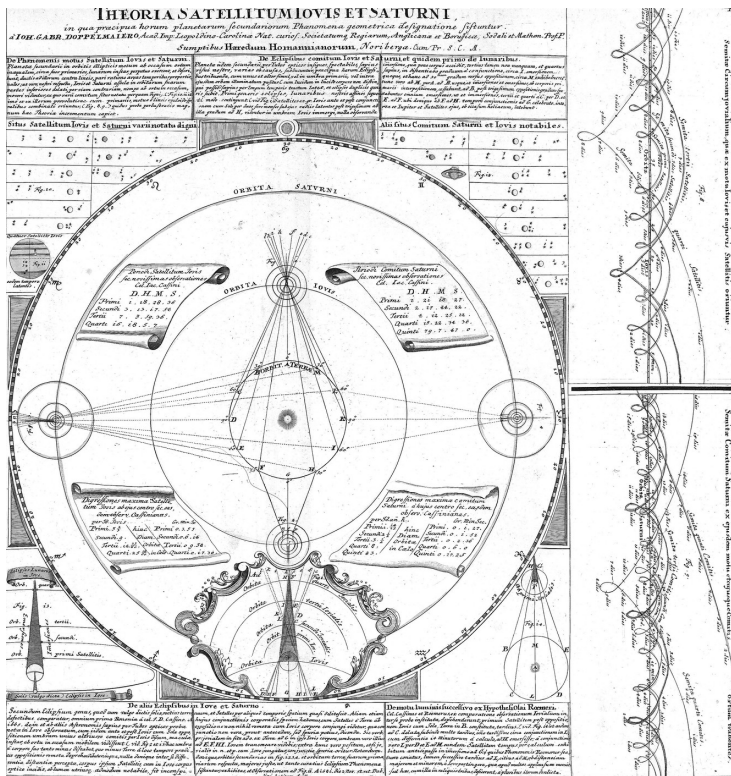
The blind idiot god's single monomaniacal goal splintered into a thousand shards of desire. And this is well, I think, though I'm a human who says so. Or else what would we do with the future? What would we do with the billion galaxies in the night sky? Fill them with maximally efficient replicators? Should our descendants deliberately obsess about maximizing their inclusive genetic fitness, regarding all else only as a means to that end?

Being a thousand shards of desire isn't always fun, but at least it's not *boring*. Somewhere along the line, we evolved tastes for novelty, complexity, elegance, and challenge—tastes that judge the blind idiot god's monomaniacal focus, and find it aesthetically unsatisfying.

And yes, we got those very same tastes from the blind idiot's godshatter.

So what?

Fragile Purposes



Belief in Intelligence

I don't know what moves Garry Kasparov would make in a chess game. What, then, is the empirical content of my belief that "Kasparov is a highly intelligent chess player"? What real-world experience does my belief tell me to anticipate? Is it a cleverly masked form of total ignorance?

To sharpen the dilemma, suppose Kasparov plays against some mere chess grandmaster Mr. G, who's not in the running for world champion. My own ability is far too low to distinguish between these levels of chess skill. When I try to guess Kasparov's move, or Mr. G's next move, all I can do is try to guess "the best chess move" using my own meager knowledge of chess. Then I would produce exactly the same prediction for Kasparov's move or Mr. G's move in any particular chess position. So what is the empirical content of my belief that "Kasparov is a *better* chess player than Mr. G"?

The empirical content of my belief is the testable, falsifiable prediction that the *final* chess position will occupy the class of chess positions that are wins for Kasparov, rather than drawn games or wins for Mr. G. (Counting resignation as a legal move that leads to a chess position classified as a loss.) The degree to which I think Kasparov is a "better player" is reflected in the amount of probability mass I concentrate into the "Kasparov wins" class of outcomes, versus the "drawn game" and "Mr. G wins" class of outcomes. These classes are extremely vague in the sense that they refer to vast spaces of possible chess positions—but "Kasparov wins" is more specific than maximum entropy, because it can be definitely falsified by a vast set of chess positions.

The *outcome* of Kasparov's game is predictable because I know, and understand, Kasparov's goals. Within the confines of the chess board, I know Kasparov's motivations—I know his success criterion, his utility function, his target as an optimization process. I know where Kasparov is *ultimately* trying to



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steer the future and I anticipate he is powerful enough to get there, although I don't anticipate much about *how* Kasparov is going to do it.

Imagine that I'm visiting a distant city, and a local friend volunteers to drive me to the airport. I don't know the neighborhood. Each time my friend approaches a street intersection, I don't know whether my friend will turn left, turn right, or continue straight ahead. I can't predict my friend's move even as we approach each individual intersection—let alone predict the whole sequence of moves in advance.

Yet I can predict the *result* of my friend's unpredictable actions: we will arrive at the airport. Even if my friend's house were located elsewhere in the city, so that my friend made a completely different sequence of turns, I would just as confidently predict our arrival at the airport. I can predict this long in advance, before I even get into the car. My flight departs soon, and there's no time to waste; I wouldn't get into the car in the first place, if I couldn't confidently predict that the car would travel to the airport along an unpredictable pathway.

Isn't this a remarkable situation to be in, from a scientific perspective? I can predict the *outcome* of a process, without being able to predict any of the *intermediate steps* of the process.

How is this even possible? Ordinarily one predicts by imagining the present and then running the visualization forward in time. If you want a *precise* model of the Solar System, one that takes into account planetary perturbations, you must start with a model of all major objects and run that model forward in time, step by step.

Sometimes simpler problems have a closed-form solution, where calculating the future at time T takes the same amount of work regardless of T . A coin rests on a table, and after each minute, the coin turns over. The coin starts out showing heads. What face will it show a hundred minutes later? Obviously you did not answer this question by visualizing a hundred intervening steps. You used a closed-form solution that worked to predict the outcome, and would *also* work to predict any of the intervening steps.

But when my friend drives me to the airport, I can predict the outcome successfully using a strange model that won't work to predict *any* of the intermediate steps. My model doesn't even require me to input the initial conditions—I don't need to know where we start out in the city!

I do need to know something about my friend. I must know that my friend wants me to make my flight. I must credit that my friend is a good enough planner to successfully drive me to the airport (if he wants to). These are properties of my *friend's* initial state—properties which let me predict the final destination, though not any intermediate turns.

I must also credit that my friend knows enough about the city to drive successfully. This may be regarded as a relation between my friend and the city; hence, a property of both. But an extremely *abstract* property, which does not require any *specific* knowledge about either the city, or about my friend's knowledge about the city.

This is one way of viewing the subject matter to which I've devoted my life—these *remarkable situations* which place us in such odd epistemic positions. And my work, in a sense, can be viewed as unraveling the exact form of that strange abstract knowledge we can possess; whereby, not knowing the actions, we can justifiably know the consequence.

"Intelligence" is too narrow a term to describe these remarkable situations in full generality. I would say rather "optimization process." A similar situation accompanies the study of biological natural selection, for example; we can't predict the exact form of the next organism observed.

But my own specialty is the kind of optimization process called "intelligence"; and even narrower, a particular kind of intelligence called "Friendly Artificial Intelligence"—of which, I hope, I will be able to obtain especially precise abstract knowledge.

Humans in Funny Suits



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Many times the human species has travelled into space, only to find the stars inhabited by aliens who look remarkably like humans in funny suits—or even humans with a touch of makeup and latex—or just beige Caucasians in fee simple.



Star Trek: The Original Series, “Arena,” ©CBS Corporation

It’s remarkable how the human form is the natural baseline of the universe, from which all other alien species are derived via a few modifications.

What could possibly explain this fascinating phenomenon? Convergent evolution, of course! Even though these alien life-forms evolved on a thousand alien planets, completely independently from Earthly life, they all turned out the same.

Don’t be fooled by the fact that a kangaroo (a mammal) resembles us rather less than does a chimp (a primate), nor by the fact that a frog (amphibians, like us, are tetrapods) resembles us less than the kangaroo. Don’t be fooled by the bewildering variety of the insects, who split off from us even longer ago than the frogs; don’t be fooled that insects have six legs, and

their skeletons on the outside, and a different system of optics, and rather different sexual practices.

You might think that a truly alien species would be more different from us than we are from insects. As I said, don't be fooled. For an alien species to evolve *intelligence*, it must have two legs with one knee each attached to an upright torso, and must walk in a way similar to us. You see, any *intelligence* needs hands, so you've got to repurpose a pair of legs for that—and if you don't start with a four-legged being, it can't develop a running gait and walk upright, freeing the hands.

...Or perhaps we should consider, as an alternative theory, that it's the *easy way out* to use humans in funny suits.

But the real problem is not shape; it is mind. "Humans in funny suits" is a well-known term in literary science-fiction fandom, and it does *not* refer to something with four limbs that walks upright. An angular creature of pure crystal is a "human in a funny suit" if she *thinks* remarkably like a human—especially a human of an English-speaking culture of the late-twentieth/early-twenty-first century.

I don't watch a lot of ancient movies. When I was watching the movie *Psycho* (1960) a few years back, I was taken aback by the cultural gap between the Americans on the screen and my America. The buttoned-shirted characters of *Psycho* are considerably more alien than the vast majority of so-called "aliens" I encounter on TV or the silver screen.

To write a culture that isn't just like your own culture, you have to be able to see your own culture as a *special case*—not as a norm which all other cultures must take as their point of departure. Studying history may help—but then it is only little black letters on little white pages, not a living experience. I suspect that it would help more to live for a year in China or Dubai or among the !Kung...this I have never done, being busy. Occasionally I wonder what things I might not be seeing (not there, but here).

Seeing your *humanity* as a special case is very much harder than this.

In every known culture, humans seem to experience joy, sadness, fear, disgust, anger, and surprise. In every known culture,

these emotions are indicated by the same facial expressions. Next time you see an “alien”—or an “AI,” for that matter—I bet that when it gets angry (and it will get angry), it will show the human-universal facial expression for anger.

We humans are very much alike under our skulls—that goes with being a sexually reproducing species; you can’t have everyone using different *complex* adaptations, they wouldn’t assemble. (Do the aliens reproduce sexually, like humans and many insects? Do they share small bits of genetic material, like bacteria? Do they form colonies, like fungi? Does the rule of psychological unity apply among them?)

The only intelligences your ancestors had to *manipulate*—complexly so, and not just tame or catch in nets—the only minds your ancestors had to model *in detail*—were minds that worked more or less like their own. And so we evolved to predict Other Minds by putting *ourselves* in their shoes, asking what we would do in their situations; for that which was to be predicted, was similar to the predictor.

“What?” you say. “I don’t assume other people are just like me! Maybe I’m sad, and they happen to be angry! They believe other things than I do; their personalities are different from mine!” Look at it this way: a human brain is an *extremely* complicated physical system. You are not modeling it neuron-by-neuron or atom-by-atom. If you came across a physical system as complex as the human brain which was *not* like you, it would take scientific lifetimes to unravel it. You do *not* understand how human brains work in an abstract, general sense; you can’t build one, and you can’t even build a computer model that predicts other brains as well as *you* predict them.

The only reason you can try at all to grasp anything as physically complex and poorly understood as the brain of another human being is that you configure your own brain to imitate it. You empathize (though perhaps not sympathize). You impose on your own brain the shadow of the other mind’s anger and the shadow of its beliefs. You may never think the words, “What would I do in this situation?,” but that little shadow of the other mind that you hold within yourself is something animated within your own brain, invoking the same complex

machinery that exists in the other person, synchronizing gears you don't understand. You may not be angry yourself, but you know that if *you* were angry at you, and *you* believed that you were godless scum, *you* would try to hurt you ...

This "empathic inference" (as I shall call it) works for humans, more or less.

But minds with *different* emotions—minds that feel emotions you've never felt yourself, or that fail to feel emotions you would feel? That's something you can't grasp by putting your brain into the other brain's shoes. I can tell you to imagine an alien that grew up in a universe with four spatial dimensions, instead of three spatial dimensions, but you won't be able to reconfigure your visual cortex to see like that alien would see. I can try to write a story about aliens with different emotions, but you won't be able to feel those emotions, and neither will I.

Imagine an alien watching a video of the Marx Brothers and having absolutely no idea what was going on, or why you would actively seek out such a sensory experience, because the alien has never conceived of anything remotely like a sense of humor. Don't pity them for missing out; *you've never antled*.

You might ask: Maybe the aliens do have a sense of humor, but you're not telling funny enough jokes? This is roughly the equivalent of trying to speak English very loudly, and very slowly, in a foreign country, on the theory that those foreigners must have an inner ghost that can hear the *meaning* dripping from your words, inherent in your words, if only you can speak them loud enough to overcome whatever strange barrier stands in the way of your perfectly sensible English.

It is important to appreciate that laughter can be a beautiful and valuable thing, even if it is not universalizable, even if it is not possessed by all possible minds. It would be our own *special* part of the gift we give to tomorrow. That can count for something too.

It had better, because universalizability is one metaethical notion that I can't salvage for you. Universalizability among humans, maybe; but not among all possible minds.

And what about minds that don't run on emotional architectures like your own—that don't have things analogous to

emotions? No, don't bother explaining why any intelligent mind powerful enough to build complex machines must inevitably have states analogous to emotions. Natural selection builds complex machines without itself having emotions. Now *there's* a Real Alien for you—an optimization process that *really* Does Not Work Like You Do.

Much of the progress in biology since the 1960s has consisted of trying to enforce a moratorium on anthropomorphizing evolution. That was a major academic slap-fight, and I'm not sure that sanity would have won the day if not for the availability of crushing experimental evidence backed up by clear math. Getting people to stop putting themselves in alien shoes is a long, hard, uphill slog. I've been fighting that battle on AI for years.

Our anthropomorphism runs very deep in us; it cannot be excised by a simple act of will, a determination to say, "Now I shall stop thinking like a human!" Humanity is the air we breathe; it is our *generic*, the white paper on which we begin our sketches. And we do not think of ourselves as being human when we are being human.

It is proverbial in literary science fiction that the true test of an author is their ability to write Real Aliens. (And not just conveniently incomprehensible aliens who, for their own mysterious reasons, do whatever the plot happens to require.) Jack Vance was one of the great masters of this art. Vance's *humans*, if they come from a different culture, are more alien than most "aliens." (Never read any Vance? I would recommend starting with *City of the Chasch*.) Niven and Pournelle's *The Mote in God's Eye* also gets a standard mention here.

And conversely—well, I once read a science fiction author say that the all-time low point of television science fiction was the *Star Trek* episode "The Omega Glory" where parallel evolution has proceeded to the extent of producing aliens who not only look just like humans, who not only speak English, but have also independently rewritten, word for word, the preamble to the US Constitution.

This is the Great Failure of Imagination. Don't think that it's just about science fiction, or even just about AI. The inability to

imagine the alien is the inability to see *yourself*—the inability to understand your own specialness. Who can see a human camouflaged against a human background?

Optimization and the Intelligence Explosion



discussion
sharov.me/r/
3-c17

Among the topics I haven't delved into here is the notion of an optimization process. Roughly, this is the idea that your power as a mind is your ability to hit small targets in a large search space—this can be either the space of possible futures (planning) or the space of possible designs (invention).

Suppose you have a car, and suppose we already know that your preferences involve travel. Now suppose that you take all the parts in the car, or all the atoms, and jumble them up at random. It's very unlikely that you'll end up with a travel-artifact at all, even so much as a wheeled cart; let alone a travel-artifact that ranks as high in your preferences as the original car. So, relative to your preference ordering, the car is an extremely *improbable* artifact. The power of an optimization process is that it can produce this kind of improbability.

You can view both intelligence and natural selection as special cases of *optimization*: processes that hit, in a large search space, very small targets defined by implicit preferences. Natural selection prefers more efficient replicators. Human intelligences have more complex preferences. Neither evolution nor humans have consistent utility functions, so viewing them as "optimization processes" is understood to be an approximation. You're trying to get at the *sort of work being done*, not claim that humans or evolution do this work *perfectly*.

This is how I see the story of life and intelligence—as a story of improbably good designs being produced by optimization processes. The "improbability" here is improbability relative to a random selection from the design space, not improbability in an absolute sense—if you have an optimization process around, then "improbably" good designs become probable.

Looking over the history of optimization on Earth up until now, the first step is to conceptually separate the meta level

from the object level—separate the *structure of optimization* from *that which is optimized*.

If you consider biology in the absence of hominids, then on the object level we have things like dinosaurs and butterflies and cats. On the meta level we have things like sexual recombination and natural selection of asexual populations. The object level, you will observe, is rather more complicated than the meta level. Natural selection is not an *easy* subject and it involves math. But if you look at the anatomy of a whole cat, the cat has dynamics immensely more complicated than “mutate, recombine, reproduce.”

This is not surprising. Natural selection is an *accidental* optimization process, that basically just started happening one day in a tidal pool somewhere. A cat is the *subject* of millions of years and billions of years of evolution.

Cats have brains, of course, which operate to learn over a lifetime; but at the end of the cat’s lifetime, that information is thrown away, so it does not accumulate. The cumulative effects of cat-brains upon the world as optimizers, therefore, are relatively small.

Or consider a bee brain, or a beaver brain. A bee builds hives, and a beaver builds dams; but they didn’t figure out how to build them from scratch. A beaver can’t figure out how to build a hive, a bee can’t figure out how to build a dam.

So animal brains—up until recently—were not major players in the planetary game of optimization; they were *pieces* but not *players*. Compared to evolution, brains lacked both generality of optimization power (they could not produce the amazing range of artifacts produced by evolution) and cumulative optimization power (their products did not accumulate complexity over time). For more on this theme see Protein Reinforcement and DNA Consequentialism.¹

1. sharov.me/r/3-11

Protein Reinforcement and DNA Consequentialism •
lesswrong.com



Very recently, certain animal brains have begun to exhibit both generality of optimization power (producing an amazingly wide range of artifacts, in time scales too short for natural selection to play any significant role) and cumulative optimization power (artifacts of increasing complexity, as a result of skills passed on through language and writing).

Natural selection takes hundreds of generations to do anything and millions of years for *de novo* complex designs. Human programmers can design a complex machine with a hundred interdependent elements in a single afternoon. This is not surprising, since natural selection is an *accidental* optimization process that basically just started happening one day, whereas humans are *optimized* optimizers handcrafted by natural selection over millions of years.

The wonder of evolution is not how well it works, but that it works *at all* without being optimized. This is how optimization bootstrapped itself into the universe—starting, as one would expect, from an extremely inefficient accidental optimization process. Which is not the accidental first replicator, mind you, but the accidental first process of natural selection. Distinguish the object level and the meta level!

Since the dawn of optimization in the universe, a certain structural commonality has held across both natural selection and human intelligence ...

Natural selection *selects on genes*, but generally speaking, the genes do not turn around and optimize natural selection. The invention of sexual recombination is an exception to this rule, and so is the invention of cells and DNA. And you can see both the power and the *rarity* of such events, by the fact that evolutionary biologists structure entire histories of life on Earth around them.

But if you step back and take a human standpoint—if you think like a programmer—then you can see that natural selection is *still* not all that complicated. We'll try bundling different genes together? We'll try separating information storage from moving machinery? We'll try randomly recombining groups of genes? On an absolute scale, these are the sort of bright

ideas that any smart hacker comes up with during the first ten minutes of thinking about system architectures.

Because natural selection started out so inefficient (as a completely accidental process), this tiny handful of meta-level improvements feeding back in from the replicators—nowhere near as complicated as the structure of a cat—structure the evolutionary epochs of life on Earth.

And *after* all that, natural selection is *still* a blind idiot of a god. Gene pools can evolve to extinction, despite all cells and sex.

Now natural selection does feed on itself in the sense that each new adaptation opens up new avenues of further adaptation; but that takes place on the object level. The gene pool feeds on its own complexity—but only thanks to the protected interpreter of natural selection that runs in the background, and that is not itself rewritten or altered by the evolution of species.

Likewise, human beings invent sciences and technologies, but we have not *yet* begun to rewrite the protected structure of the human brain itself. We have a prefrontal cortex and a temporal cortex and a cerebellum, just like the first inventors of agriculture. We haven't started to genetically engineer ourselves. On the object level, science feeds on science, and each new discovery paves the way for new discoveries—but all that takes place with a protected interpreter, the human brain, running untouched in the background.

We have meta-level inventions like science, that try to instruct humans in how to think. But the first person to invent Bayes's Theorem did not become a Bayesian; they could not rewrite themselves, lacking both that knowledge and that power. Our significant innovations in the art of thinking, like writing and science, are so powerful that they structure the course of human history; but they do not rival the brain itself in complexity, and their effect upon the brain is comparatively shallow.

The present state of the art in rationality training is not sufficient to turn an arbitrarily selected mortal into Albert Einstein, which shows the power of a few minor genetic quirks

of brain design compared to all the self-help books ever written in the twentieth century.

Because the brain hums away invisibly in the background, people tend to overlook its contribution and take it for granted; and talk as if the simple instruction to “Test ideas by experiment,” or the $p < 0.05$ significance rule, were the same order of contribution as an entire human brain. Try telling chimpanzees to test their ideas by experiment and see how far you get.

Now ...some of us *want* to intelligently design an intelligence that would be capable of intelligently redesigning itself, right down to the level of machine code.

The machine code at first, and the laws of physics later, would be a protected level of a sort. But that “protected level” would not contain the *dynamic of optimization*; the protected levels would not structure the work. The human brain does quite a bit of optimization on its own, and screws up on its own, no matter what you try to tell it in school. But this *fully wraparound recursive optimizer* would have no protected level that was *optimizing*. All the structure of optimization would be subject to optimization itself.

And that is a sea change which breaks with the entire past since the first replicator, because it breaks the idiom of a protected meta level.

The history of Earth up until now has been a history of optimizers spinning their wheels at a constant rate, generating a constant optimization pressure. And creating optimized products, *not* at a constant rate, but at an accelerating rate, because of how object-level innovations open up the pathway to other object-level innovations. But that acceleration is taking place with a protected meta level doing the actual optimizing. Like a search that leaps from island to island in the search space, and good islands tend to be adjacent to even better islands, but the jumper doesn’t change its legs. *Occasionally*, a few tiny little changes manage to hit back to the meta level, like sex or science, and then the history of optimization enters a new epoch and everything proceeds faster from there.

Imagine an economy without investment, or a university without language, a technology without tools to make tools.

Once in a hundred million years, or once in a few centuries, someone invents a hammer.

That is what optimization has been like on Earth up until now.

When I look at the history of Earth, I don't see a history of optimization *over time*. I see a history of *optimization power* in, and *optimized products* out. Up until now, thanks to the existence of almost entirely protected meta-levels, it's been possible to split up the history of optimization into epochs, and, *within* each epoch, graph the cumulative *object-level optimization* over time, because the protected level is running in the background and is not itself changing within an epoch.

What happens when you build a fully wraparound, recursively self-improving AI? Then you take the graph of "optimization in, optimized out," and fold the graph in on itself. Metaphorically speaking.

If the AI is weak, it does nothing, because it is not powerful enough to significantly improve itself—like telling a chimpanzee to rewrite its own brain.

If the AI is powerful enough to rewrite itself in a way that increases its ability to make further improvements, and this reaches all the way down to the AI's full understanding of its own source code and its own design as an optimizer ...then even if the graph of "optimization power in" and "optimized product out" looks essentially the same, the graph of optimization *over time* is going to look completely different from Earth's history so far.

People often say something like, "But what if it requires exponentially greater amounts of self-rewriting for only a linear improvement?" To this the obvious answer is, "Natural selection exerted roughly constant optimization power on the hominid line in the course of coughing up humans; and this doesn't seem to have required exponentially more time for each linear increment of improvement."

All of this is still mere analogic reasoning. A full Artificial General Intelligence thinking about the nature of optimization and doing its own AI research and rewriting its own source code, is not *really* like a graph of Earth's history folded in

on itself. It is a different sort of beast. These analogies are *at best* good for qualitative predictions, and even then, I have a large amount of other beliefs I haven't yet explained, which are telling me *which* analogies to make, et cetera.

But if you want to know why I might be reluctant to extend the graph of biological and economic growth *over time*, into the future and over the horizon of an AI that thinks at transistor speeds and invents self-replicating molecular nanofactories and *improves its own source code*, then there is my reason: you are drawing the wrong graph, and it should be optimization power in versus optimized product out, not optimized product versus time.

Ghosts in the Machine

People hear about Friendly AI and say—this is one of the top three initial reactions:

“Oh, you can try to tell the AI to be Friendly, but if the AI can modify its own source code, it’ll just remove any constraints you try to place on it.”

And where does *that* decision come from?

Does it enter from outside causality, rather than being an effect of a lawful chain of causes that started with the source code as originally written? Is the AI the ultimate source of its own free will?

A Friendly AI is not a selfish AI constrained by a special extra conscience module that overrides the AI’s natural impulses and tells it what to do. You just build the conscience, and that *is* the AI. If you have a program that computes which decision the AI should make, you’re *done*. The buck stops immediately.

At this point, I shall take a moment to quote some case studies from the Computer Stupidities site and Programming subtopic. (I am not linking to this, because it is a fearsome time-trap; you can Google if you dare.)

I tutored college students who were taking a computer programming course. A few of them didn’t understand that computers are not sentient. More than one person used comments in their Pascal programs to put detailed explanations such as, “Now I need you to put these letters on the screen.” I asked one of them what the deal was with those comments. The reply: “How else is the computer going to understand what I want it to do?” Apparently they would assume that since they couldn’t make sense of Pascal, neither could the computer.



discussion
sharov.me/r/
3-c18

While in college, I used to tutor in the school's math lab. A student came in because his BASIC program would not run. He was taking a beginner course, and his assignment was to write a program that would calculate the recipe for oatmeal cookies, depending upon the number of people you're baking for. I looked at his program, and it went something like this:

```
10 Preheat oven to 350
20 Combine all ingredients in a
   large mixing bowl
30 Mix until smooth
```

An introductory programming student once asked me to look at his program and figure out why it was always churning out zeroes as the result of a simple computation. I looked at the program, and it was pretty obvious:

```
begin
  read("Number of Apples", apples)
  read("Number of Carrots", carrots)
  read("Price for 1 Apple", a_price)
  read("Price for 1 Carrot", c_price)
  write("Total for Apples", a_total)
  write("Total for Carrots", c_total)
  write("Total", total)
  total = a_total + c_total
  a_total = apples * a_price
  c_total = carrots * c_price
end
```

Me: "Well, your program can't print correct results before they're computed."

Him: “Huh? It’s logical what the right solution is, and the computer should reorder the instructions the right way.”

There’s an instinctive way of imagining the scenario of “programming an AI.” It maps onto a similar-seeming human endeavor: Telling a human being what to do. Like the “program” is giving instructions to a little ghost that sits inside the machine, which will look over your instructions and decide whether it likes them or not.

There is no ghost who looks over the instructions and decides how to follow them. The program *is* the AI.

That doesn’t mean the ghost does anything you wish for, like a genie. It doesn’t mean the ghost does everything you want the way you want it, like a slave of exceeding docility. It means your instruction *is* the only ghost that’s there, at least at boot time.

AI is much harder than people instinctively imagined, exactly because you can’t just *tell* the ghost what to do. You have to build the ghost from scratch, and everything that seems obvious to you, the ghost will not see unless you know how to make the ghost see it. You can’t just *tell* the ghost to see it. You have to create that-which-sees from scratch.

If you don’t know how to build something that seems to have some strange ineffable elements like, say, “decision-making,” then you can’t just shrug your shoulders and let the ghost’s free will do the job. You’re left forlorn and ghostless.

There’s more to building a chess-playing program than building a really fast processor—so the AI will be really smart—and then typing at the command prompt “Make whatever chess moves *you* think are best.” You might think that, since the programmers themselves are not very good chess players, any advice they tried to give the electronic superbrain would just slow the ghost down. But there is no ghost. You see the problem.

And there isn’t a simple spell you can perform to—poof!—summon a complete ghost into the machine. You can’t say, “I summoned the ghost, and it appeared; that’s cause and effect for you.” (It doesn’t work if you use the notion of “emergence”

or “complexity” as a substitute for “summon,” either.) You can’t give an instruction to the CPU, “Be a good chess player!” You have to see inside the mystery of chess-playing thoughts, and structure the whole ghost from scratch.

No matter how common-sensical, no matter how logical, no matter how “obvious” or “right” or “self-evident” or “intelligent” something seems to you, it will not happen inside the ghost. *Unless* it happens at the end of a chain of cause and effect that began with the instructions that *you* had to decide on, plus any causal dependencies on sensory data that you built into the starting instructions.

This doesn’t mean you program in every decision explicitly. Deep Blue was a chess player far superior to its programmers. Deep Blue made better chess moves than anything its makers could have explicitly programmed—but not because the programmers shrugged and left it up to the ghost. Deep Blue moved better than its programmers ...at the end of a chain of cause and effect that began in the programmers’ code and proceeded lawfully from there. Nothing happened *just* because it was so obviously a good move that Deep Blue’s ghostly free will took over, without the code and its lawful consequences being involved.

If you try to wash your hands of constraining the AI, you aren’t left with a free ghost like an emancipated slave. You are left with a heap of sand that no one has purified into silicon, shaped into a CPU and programmed to think.

Go ahead, try telling a computer chip “Do whatever you want!” See what happens? Nothing. Because you haven’t constrained it to understand freedom.

All it takes is one single step that is *so obvious, so logical, so self-evident* that your mind just skips right over it, and you’ve left the path of the AI programmer. It takes an effort like the

one I illustrate in Grasping Slippery Things¹ to prevent your mind from doing this.

1. sharov.me/r/3-12
Grasping Slippery Things • lesswrong.com



Artificial Addition



discussion
sharov.me/r/
3-c19

Suppose that human beings had absolutely *no idea* how they performed arithmetic. Imagine that human beings had *evolved*, rather than having *learned*, the ability to count sheep and add sheep. People using this built-in ability have no idea how it worked, the way Aristotle had no idea how his visual cortex supported his ability to see things. Peano Arithmetic as we know it has not been invented. There are philosophers working to formalize numerical intuitions, but they employ notations such as

$$\text{Plus-Of}(\text{Seven}, \text{Six}) = \text{Thirteen}$$

to formalize the intuitively obvious fact that when you add “seven” plus “six,” of course you get “thirteen.”

In this world, pocket calculators work by storing a giant lookup table of arithmetical facts, entered manually by a team of expert Artificial Arithmeticians, for starting values that range between zero and one hundred. While these calculators may be helpful in a pragmatic sense, many philosophers argue that they’re only *simulating* addition, rather than really *adding*. No machine can really *count*—that’s why humans have to count thirteen sheep before typing “thirteen” into the calculator. Calculators can recite back stored facts, but they can never know what the statements mean—if you type in “two hundred plus two hundred” the calculator says “Error: Outrange,” when it’s intuitively *obvious*, if you *know* what the words *mean*, that the answer is “four hundred.”

Some philosophers, of course, are not so naive as to be taken in by these intuitions. Numbers are really a purely formal system—the label “thirty-seven” is meaningful, not because of any inherent property of the words themselves, but because the label *refers to* thirty-seven sheep in the external world. A number is given this referential property by its *semantic network* of relations to other numbers. That’s why, in computer

programs, the LISP token for “thirty-seven” doesn’t need any *internal* structure—it’s only meaningful because of reference and relation, not some computational property of “thirty-seven” itself.

No one has ever developed an Artificial General Arithmetician, though of course there are plenty of domain-specific, narrow Artificial Arithmeticians that work on numbers between “twenty” and “thirty,” and so on. And if you look at how slow progress has been on numbers in the range of “two hundred,” then it becomes clear that we’re not going to get Artificial General Arithmetic any time soon. The best experts in the field estimate it will be at least a hundred years before calculators can add as well as a human twelve-year-old.

But not everyone agrees with this estimate, or with merely conventional beliefs about Artificial Arithmetic. It’s common to hear statements such as the following:

- “It’s a framing problem—what ‘twenty-one plus’ equals depends on whether it’s ‘plus three’ or ‘plus four.’ If we can just get enough arithmetical facts stored to cover the common-sense truths that everyone knows, we’ll start to see real addition in the network.”
- “But you’ll never be able to program in that many arithmetical facts by hiring experts to enter them manually. What we need is an Artificial Arithmetician that can *learn* the vast network of relations between numbers that humans acquire during their childhood by observing sets of apples.”
- “No, what we really need is an Artificial Arithmetician that can understand natural language, so that instead of having to be explicitly told that twenty-one plus sixteen equals thirty-seven, it can get the knowledge by exploring the Web.”
- “Frankly, it seems to me that you’re just trying to convince yourselves that you can solve the problem. None of you really know what arithmetic is, so you’re floundering around with these generic sorts of arguments. ‘We need

an AA that can learn X , ‘We need an AA that can extract X from the Internet.’ I mean, it sounds good, it sounds like you’re making progress, and it’s even good for public relations, because everyone thinks they understand the proposed solution—but it doesn’t really get you any closer to *general* addition, as opposed to domain-specific addition. Probably we will never know the fundamental nature of arithmetic. The problem is just too hard for humans to solve.”

- “That’s why we need to develop a general arithmetician the same way Nature did—evolution.”
- “Top-down approaches have clearly failed to produce arithmetic. We need a bottom-up approach, some way to make arithmetic *emerge*. We have to acknowledge the basic unpredictability of complex systems.”
- “You’re all wrong. Past efforts to create machine arithmetic were futile from the start, because they just didn’t have enough computing power. If you look at how many trillions of synapses there are in the human brain, it’s clear that calculators don’t have lookup tables anywhere near that large. We need calculators as powerful as a human brain. According to Moore’s Law, this will occur in the year 2031 on April 27 between 4:00 and 4:30 in the morning.”
- “I believe that machine arithmetic will be developed when researchers scan each neuron of a complete human brain into a computer, so that we can simulate the biological circuitry that performs addition in humans.”
- “I don’t think we have to wait to scan a whole brain. Neural networks are just like the human brain, and you can train them to do things without knowing how they do them. We’ll create programs that will do arithmetic without we, our creators, ever understanding how they do arithmetic.”

- “But Gödel’s Theorem shows that no formal system can ever capture the basic properties of arithmetic. Classical physics is formalizable, so to add two and two, the brain must take advantage of quantum physics.”
- “Hey, if human arithmetic were simple enough that we could reproduce it in a computer, we wouldn’t be able to count high enough to build computers.”
- “Haven’t you heard of John Searle’s Chinese Calculator Experiment? Even if you did have a huge set of rules that would let you add ‘twenty-one’ and ‘sixteen,’ just imagine translating all the words into Chinese, and you can see that there’s no genuine addition going on. There are no real *numbers* anywhere in the system, just labels that humans use for numbers...”

There is more than one moral to this parable, and I have told it with different morals in different contexts. It illustrates the idea of levels of organization, for example—a CPU can add two large numbers because the numbers aren’t black-box opaque objects, they’re ordered structures of 32 bits.

But for purposes of overcoming bias, let us draw two morals:

- First, the danger of believing assertions you can’t regenerate from your own knowledge.
- Second, the danger of trying to dance around basic confusions.

Lest anyone accuse me of generalizing from fictional evidence, both lessons may be drawn from the real history of Artificial Intelligence as well.

The first danger is the object-level problem that the AA devices ran into: they functioned as tape recorders playing back “knowledge” generated from outside the system, using a process they couldn’t capture internally. A human could tell the AA device that “twenty-one plus sixteen equals thirty-seven,” and the AA devices could record this sentence and play it back, or even pattern-match “twenty-one plus sixteen” to

output “thirty-seven!”—but the AA devices couldn’t generate such knowledge for themselves.

Which is strongly reminiscent of believing a physicist who tells you “Light is waves,” recording the fascinating words and playing them back when someone asks “What is light made of?,” without being able to generate the knowledge for yourself.

The second moral is the meta-level danger that consumed the Artificial Arithmetic researchers and opinionated bystanders—the danger of dancing around confusing gaps in your knowledge. The tendency to do just about anything *except* grit your teeth and buckle down and fill in the damn gap.

Whether you say, “It is emergent!,” or whether you say, “It is unknowable!,” in neither case are you acknowledging that there is a basic insight required which is possessable, but unpossessed by you.

How can you know when you’ll have a new basic insight? And there’s no way to get one except by banging your head against the problem, learning everything you can about it, studying it from as many angles as possible, perhaps for years. It’s not a pursuit that academia is set up to permit, when you need to publish at least one paper per month. It’s certainly not something that venture capitalists will fund. You want to either go ahead and build the system *now*, or give up and do something else instead.

Look at the comments above: none are aimed at setting out on a quest for the missing insight which would *make numbers no longer mysterious*, make “twenty-seven” more than a black box. None of the commenters realized that their difficulties arose from ignorance or confusion in their own minds, rather than an inherent property of arithmetic. They were not trying to achieve a state where the confusing thing ceased to be confusing.

If you read Judea Pearl’s *Probabilistic Reasoning in Intelligent Systems: Networks of Plausible Inference*,¹ then you will see that the basic insight behind graphical models is *indispensable* to problems that require it. (It’s not something that fits on a T-

1. Judea Pearl, *Probabilistic Reasoning in Intelligent Systems*.

shirt, I'm afraid, so you'll have to go and read the book yourself. I haven't seen any online popularizations of Bayesian networks that adequately convey the reasons behind the principles, or the importance of the math being exactly the way it is, but Pearl's book is wonderful.) There were once dozens of "non-monotonic logics" awkwardly trying to capture intuitions such as "If my burglar alarm goes off, there was probably a burglar, but if I then learn that there was a small earthquake near my home, there was probably not a burglar." With the graphical-model insight in hand, you can give a mathematical explanation of exactly why first-order logic has the wrong properties for the job, and express the correct solution in a compact way that captures all the common-sense details in one elegant swoop. Until you have that insight, you'll go on patching the logic here, patching it there, adding more and more hacks to force it into correspondence with everything that seems "obviously true."

You won't *know* the Artificial Arithmetic problem is unsolvable without its key. If you don't know the rules, you don't know the rule that says you need to know the rules to do anything. And so there will be all sorts of clever ideas that seem like they might work, like building an Artificial Arithmetician that can read natural language and download millions of arithmetical assertions from the Internet.

And yet *somehow* the clever ideas never work. Somehow it always turns out that you "couldn't see any reason it wouldn't work" because you were ignorant of the obstacles, not because no obstacles existed. Like shooting blindfolded at a distant target—you can fire blind shot after blind shot, crying, "You can't prove to me that I won't hit the center!" But until you take off the blindfold, you're not even in the aiming game. When "no one can prove to you" that your precious idea *isn't* right, it means you don't have enough information to strike a small target in a vast answer space. *Until you know your idea will work, it won't.*

From the history of previous key insights in Artificial Intelligence, and the grand messes that were proposed prior to those insights, I derive an important real-life lesson: *When the basic*

THE MACHINE IN THE GHOST

*problem is your ignorance, clever strategies for bypassing your
ignorance lead to shooting yourself in the foot.*

Terminal Values and Instrumental Values

On a purely instinctive level, any human planner behaves as if they distinguish between means and ends. Want chocolate? There's chocolate at the Publix supermarket. You can get to the supermarket if you drive one mile south on Washington Ave. You can drive if you get into the car. You can get into the car if you open the door. You can open the door if you have your car keys. So you put your car keys into your pocket, and get ready to leave the house ...

...when suddenly the word comes on the radio that an earthquake has destroyed all the chocolate at the local Publix. Well, there's no point in driving to the Publix if there's no chocolate there, and no point in getting into the car if you're not driving anywhere, and no point in having car keys in your pocket if you're not driving. So you take the car keys out of your pocket, and call the local pizza service and have them deliver a chocolate pizza. Mm, delicious.

I rarely notice people losing track of plans they devised themselves. People usually don't drive to the supermarket if they know the chocolate is gone. But I've also noticed that when people begin *explicitly* talking about goal systems instead of just *wanting* things, *mentioning* "goals" instead of *using* them, they oft become confused. Humans are experts at planning, not experts on planning, or there'd be a lot more AI developers in the world.

In particular, I've noticed people get confused when—in abstract philosophical discussions rather than everyday life—they consider the distinction between means and ends; more formally, between "instrumental values" and "terminal values."

Part of the problem, it seems to me, is that the human mind uses a rather ad-hoc system to keep track of its goals—it works, but not cleanly. English doesn't embody a sharp distinction



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between means and ends: “I want to save my sister’s life” and “I want to administer penicillin to my sister” use the same word “want.”

Can we describe, in mere English, the distinction that is getting lost?

As a first stab:

“Instrumental values” are desirable strictly conditional on their anticipated consequences. “I want to administer penicillin to my sister,” not because a penicillin-filled sister is an intrinsic good, but in anticipation of penicillin curing her flesh-eating pneumonia. If instead you anticipated that injecting penicillin would melt your sister into a puddle like the Wicked Witch of the West, you’d fight just as hard to keep her penicillin-free.

“Terminal values” are desirable without conditioning on other consequences: “I want to save my sister’s life” has nothing to do with your anticipating whether she’ll get injected with penicillin after that.

This first attempt suffers from obvious flaws. If saving my sister’s life would cause the Earth to be swallowed up by a black hole, then I would go off and cry for a while, but I wouldn’t administer penicillin. Does this mean that saving my sister’s life was not a “terminal” or “intrinsic” value, because it’s theoretically conditional on its consequences? Am I *only* trying to save her life because of my belief that a black hole won’t consume the Earth afterward? Common sense should say that’s not what’s happening.

So forget English. We can set up a mathematical description of a decision system in which terminal values and instrumental values are separate and incompatible types—like integers and floating-point numbers, in a programming language with no automatic conversion between them.

An ideal Bayesian decision system can be set up using only four elements:

- Outcomes : type Outcome[]
 - list of possible outcomes
 - {sister lives, sister dies}

- `Actions : type Action[]`
 - list of possible actions
 - {administer penicillin, don't administer penicillin}
- `Utility_function : type Outcome -> Utility`
 - utility function that maps each outcome onto a utility
 - (a utility being representable as a real number between negative and positive infinity)
 - {sister lives $\rightarrow$ 1, sister dies $\rightarrow$ 0}
- `Conditional_probability_function : type Action -> (Outcome -> Probability)`
 - conditional probability function that maps each action onto a probability distribution over outcomes
 - (a probability being representable as a real number between 0 and 1)
 - {administer penicillin $\rightarrow$ (sister lives $\rightarrow$ 0.9, sister dies $\rightarrow$ 0.1), don't administer penicillin $\rightarrow$ (sister lives $\rightarrow$ 0.3, sister dies $\rightarrow$ 0.7)}

If you can't read the type system directly, don't worry, I'll always translate into English. For programmers, seeing it described in distinct statements helps to set up distinct mental objects.

And the decision system itself?

- `Expected_Utility : Action A -> (Sum O in Outcomes: Utility(O) * Probability(O|A))`
 - The “expected utility” of an action equals the sum, over all outcomes, of the utility of that outcome times the conditional probability of that outcome given that action.
 - {EU(administer penicillin) = 0.9, EU(don't administer penicillin) = 0.3}

- Choose :
 - > (Argmax A in Actions: Expected_Utility(A))
 - Pick an action whose “expected utility” is maximal.
 - {return: administer penicillin}

For every action, calculate the conditional probability of all the consequences that might follow, then add up the utilities of those consequences times their conditional probability. Then pick the best action.

This is a mathematically simple sketch of a decision system. It is not an efficient way to compute decisions in the real world.

What if, for example, you need a *sequence* of acts to carry out a plan? The formalism can easily represent this by letting each Action stand for a whole sequence. But this creates an exponentially large space, like the space of all sentences you can type in 100 letters. As a simple example, if one of the possible acts on the first turn is “Shoot my own foot off,” a human planner will decide this is a bad idea generally—eliminate *all* sequences beginning with this action. But we’ve flattened this structure out of our representation. We don’t have sequences of acts, just flat “actions.”

So, yes, there are *a few minor complications*. Obviously so, or we’d just run out and build a real AI this way. In that sense, it’s much the same as Bayesian probability theory itself.

But this is one of those times when it’s a *surprisingly good idea* to consider the absurdly simple version before adding in any high-falutin’ complications.

Consider the philosopher who asserts, “All of us are ultimately selfish; we care only about our own states of mind. The mother who claims to care about her son’s welfare, really wants to *believe* that her son is doing well—this belief is what makes the mother happy. She helps him for the sake of her own happiness, not his.” You say, “Well, suppose the mother sacrifices her life to push her son out of the path of an oncoming truck. That’s not going to make her happy, just dead.” The philosopher stammers for a few moments, then replies, “But she still did it

because *she valued* that choice above others—because of the *feeling of importance* she attached to that decision.”

So you say,

TYPE ERROR: No constructor found for Expected_Utility -> Utility.

Allow me to explain that reply.

Even our simple formalism illustrates a sharp distinction between *expected utility*, which is something that *actions* have; and *utility*, which is something that *outcomes* have. Sure, you can map both utilities and expected utilities onto real numbers. But that’s like observing that you can map wind speed and temperature onto real numbers. It doesn’t make them the same thing.

The philosopher begins by arguing that all your Utilities must be over Outcomes consisting of your state of mind. If this were true, your intelligence would operate *as an engine to steer the future* into regions where you were happy. Future states would be distinguished only by your state of mind; you would be indifferent between any two futures in which you had the same state of mind.

And you would, indeed, be rather unlikely to sacrifice your own life to save another.

When we object that people sometimes *do* sacrifice their lives, the philosopher’s reply shifts to discussing Expected Utilities over Actions: “The feeling of *importance* she attached to that *decision*.” This is a drastic jump that *should* make us leap out of our chairs in indignation. Trying to convert an Expected_Utility into a Utility would cause an outright error in our programming language. But in English it all sounds the same.

The choices of our simple decision system are those with highest Expected_Utility, but this doesn’t say anything whatsoever about *where it steers the future*. It doesn’t say anything about the utilities the decider assigns, or which real-world outcomes are likely to happen as a result. It doesn’t say anything about the mind’s function as an engine.

The physical cause of a physical action is a cognitive state, in our ideal decider an Expected_Utility, and this expected

utility is calculated by evaluating a utility function over imagined consequences. To save your son's life, you must imagine the event of your son's life being saved, and this imagination is not the event itself. It's a *quotation*, like the difference between "snow" and snow. But that doesn't mean that what's *inside the quote marks* must itself be a cognitive state. If you choose the action that leads to the future that you represent with "my son is still alive," then you have functioned as an engine to steer the future into a region where your son is still alive. Not an engine that steers the future into a region where *you represent the sentence* "my son is still alive." To steer the future *there*, your utility function would have to return a high utility when fed "'my son is still alive'", the quotation of the quotation, your imagination of yourself imagining. Recipes make poor cake when you grind them up and toss them in the batter.

And that's why it's helpful to consider the simple decision systems first. Mix enough complications into the system, and formerly clear distinctions become harder to see.

So now let's look at some complications. Clearly the Utility function (mapping Outcomes onto Utilities) is meant to formalize what I earlier referred to as "terminal values," values not contingent upon their consequences. What about the case where saving your sister's life leads to Earth's destruction by a black hole? In our formalism, we've flattened out this possibility. Outcomes don't lead to Outcomes, only Actions lead to Outcomes. Your sister recovering from pneumonia *followed by* the Earth being devoured by a black hole would be flattened into a single "possible outcome."

And where are the "instrumental values" in this simple formalism? Actually, they've vanished entirely! You see, in this formalism, actions lead directly to outcomes with no intervening events. There's no notion of throwing a rock that flies through the air and knocks an apple off a branch so that it falls to the ground. Throwing the rock is the Action, and it leads straight to the Outcome of the apple lying on the ground—according to the conditional probability function that turns an Action directly into a Probability distribution over Outcomes.

In order to *actually compute* the conditional probability function, and in order to separately consider the utility of a sister's pneumonia and a black hole swallowing Earth, we would have to represent the network structure of causality—the way that events lead to other events.

And then the instrumental values would start coming back. If the causal network was sufficiently regular, you could find a state *B* that tended to lead to *C* regardless of how you achieved *B*. Then if you wanted to achieve *C* for some reason, you could plan efficiently by first working out a *B* that led to *C*, and then an *A* that led to *B*. This would be the phenomenon of “instrumental value”—*B* would have “instrumental value” because it led to *C*. The state *C* itself might be terminally valued—a term in the utility function over the total outcome. Or *C* might just be an instrumental value, a node that was not directly valued by the utility function.

Instrumental value, in this formalism, is purely an aid to the efficient computation of plans. It can and should be discarded wherever this kind of regularity does not exist.

Suppose, for example, that there's some particular value of *B* that doesn't lead to *C*. Would you choose an *A* which led to that *B*? Or never mind the abstract philosophy: If you wanted to go to the supermarket to get chocolate, and you wanted to drive to the supermarket, and you needed to get into your car, would you gain entry by ripping off the car door with a steam shovel? (No.) Instrumental value is a “leaky abstraction,” as we programmers say; you sometimes have to toss away the cached value and compute out the actual expected utility. Part of being *efficient* without being *suicidal* is noticing when convenient shortcuts break down. Though this formalism does give rise to instrumental values, it does so only where the requisite regularity exists, and strictly as a convenient shortcut in computation.

But if you complicate the formalism before you understand the simple version, then you may start thinking that instrumental values have some strange life of their own, even in a normative sense. That, once you say *B* is usually good because it leads to *C*, you've committed yourself to always try

for *B* even in the absence of *C*. People make this kind of mistake in abstract philosophy, even though they would never, in real life, rip open their car door with a steam shovel. You may start thinking that there's no way to develop a consequentialist that maximizes only inclusive genetic fitness, because it will starve unless you include an explicit terminal value for "eating food." People make this mistake even though they would never stand around opening car doors all day long, for fear of being stuck outside their cars if they didn't have a terminal value for opening car doors.

Instrumental values live in (the network structure of) the conditional probability function. This makes instrumental value strictly dependent on beliefs-of-fact given a fixed utility function. If I believe that penicillin causes pneumonia, and that the absence of penicillin cures pneumonia, then my perceived instrumental value of penicillin will go from high to low. Change the beliefs of fact—change the conditional probability function that associates actions to believed consequences—and the instrumental values will change in unison.

In moral arguments, some disputes are about instrumental consequences, and some disputes are about terminal values. If your debating opponent says that banning guns will lead to lower crime, and you say that banning guns will lead to higher crime, then you agree about a superior instrumental value (crime is bad), but you disagree about which intermediate events lead to which consequences. But I do not think an argument about female circumcision is really a factual argument about how to best achieve a shared value of treating women fairly or making them happy.

This important distinction often gets *flushed down the toilet* in angry arguments. People with factual disagreements and shared values each decide that their debating opponents must be sociopaths. As if your hated enemy, gun control/rights advocates, *really wanted to kill people*, which should be implausible as realistic psychology.

I fear the human brain does not strongly type the distinction between terminal moral beliefs and instrumental moral beliefs. "We should ban guns" and "We should save lives" don't *feel*

different, as moral beliefs, the way that sight feels different from sound. Despite all the other ways that the human goal system complicates everything in sight, this *one distinction* it manages to collapse into a mishmash of things-with-conditional-value.

To extract out the terminal values we have to inspect this mishmash of valuable things, trying to figure out which ones are getting their value from somewhere else. It's a difficult project! If you say that you want to ban guns in order to reduce crime, it may take a moment to realize that "reducing crime" isn't a terminal value, it's a superior instrumental value with links to terminal values for human lives and human happinesses. And then the one who advocates gun rights may have links to the superior instrumental value of "reducing crime" plus a link to a value for "freedom," which might be a terminal value unto them, or another instrumental value ...

We can't print out our complete network of values derived from other values. We probably don't even store the whole history of how values got there. By considering the right moral dilemmas, "Would you do *X* if *Y*," we can often figure out where our values came from. But even this project itself is full of pitfalls; misleading dilemmas and gappy philosophical arguments. We don't know what our own values are, or where they came from, and can't find out except by undertaking error-prone projects of cognitive archaeology. Just forming a conscious distinction between "terminal value" and "instrumental value," and keeping track of what it means, and using it correctly, is hard work. Only by inspecting the simple formalism can we see how easy it ought to be, in principle.

And that's to say nothing of all the other complications of the human reward system—the whole use of reinforcement architecture, and the way that eating chocolate is pleasurable, and anticipating eating chocolate is pleasurable, but they're different kinds of pleasures ...

But I don't complain too much about the mess.

Being ignorant of your own values may not always be fun, but at least it's not boring.

Leaky Generalizations



discussion
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Are apples good to eat? Usually, but some apples are rotten.

Do humans have ten fingers? Most of us do, but plenty of people have lost a finger and nonetheless qualify as “human.”

Unless you descend to a level of description far below any macroscopic object—below societies, below people, below fingers, below tendon and bone, below cells, all the way down to particles and fields where the laws are truly universal—practically every generalization you use in the real world will be leaky.

(Though there may, of course, be some exceptions to the above rule ...)

Mostly, the way you deal with leaky generalizations is that, well, you just have to deal. If the cookie market almost always closes at 10 p.m., except on Thanksgiving it closes at 6 p.m., and today happens to be National Native American Genocide Day, you’d better show up before 6 p.m. or you won’t get a cookie.

Our ability to manipulate leaky generalizations is opposed by *need for closure*, the degree to which we want to say once and for all that humans have ten fingers, and get frustrated when we have to tolerate continued ambiguity. Raising the value of the stakes can increase need for closure—which shuts down complexity tolerance when complexity tolerance is most needed.

Life would be complicated even if the things we wanted were simple (they aren’t). The leakyness of leaky generalizations about what-to-do-next would leak in from the leaky structure of the real world. Or to put it another way:

Instrumental values often have no specification that is both compact and local.

Suppose there’s a box containing a million dollars. The box is locked, not with an ordinary combination lock, but with a dozen keys controlling a machine that can open the box. If

you know how the machine works, you can deduce which sequences of key-presses will open the box. There's more than one key sequence that can trigger the machine to open the box. But if you press a sufficiently wrong sequence, the machine incinerates the money. And if you *don't know* about the machine, there's no simple rules like "Pressing any key three times opens the box" or "Pressing five different keys with no repetitions incinerates the money."

There's a *compact nonlocal* specification of which keys you want to press: You want to press keys such that they open the box. You can write a compact computer program that computes which key sequences are good, bad or neutral, but the computer program will need to describe the machine, not just the keys themselves.

There's likewise a *local noncompact* specification of which keys to press: a giant lookup table of the results for each possible key sequence. It's a very large computer program, but it makes no mention of anything except the keys.

But there's no way to describe which key sequences are good, bad, or neutral, which is both *simple* and phrased *only in terms of the keys themselves*.

It may be even worse if there are tempting local generalizations which turn out to be *leaky*. Pressing *most* keys three times in a row will open the box, but there's a particular key that incinerates the money if you press it just once. You might think you had found a perfect generalization—a locally describable class of sequences that *always* opened the box—when you had merely failed to visualize all the possible paths of the machine, or failed to value all the side effects.

The machine represents the complexity of the real world. The openness of the box (which is good) and the incinerator (which is bad) represent the thousand shards of desire that make up our terminal values. The keys represent the actions and policies and strategies available to us.

When you consider how many different ways we value outcomes, and how complicated are the paths we take to get there, it's a wonder that there exists any such thing as helpful ethi-

cal *advice*. (Of which the strangest of all advices, *and yet still helpful*, is that “the end does not justify the means.”)

But conversely, the complicatedness of action need not say anything about the complexity of goals. You often find people who smile wisely, and say, “Well, morality is complicated, you know, female circumcision is right in one culture and wrong in another, it’s not always a bad thing to torture people. How naive you are, how full of need for closure, that you think there are any simple rules.”

You can say, unconditionally and flatly, that killing *anyone* is a huge dose of negative terminal utility. Yes, even Hitler. That doesn’t mean you shouldn’t shoot Hitler. It means that the net instrumental utility of shooting Hitler carries a giant dose of negative utility from Hitler’s death, and a hugely larger dose of positive utility from all the other lives that would be saved as a consequence.

Many commit the type error that I warned against in Terminal Values and Instrumental Values, and think that if the net consequential expected utility of Hitler’s death is conceded to be positive, then the immediate local terminal utility must also be positive, meaning that the moral principle “Death is always a bad thing” is itself a leaky generalization. But this is double counting, with utilities instead of probabilities; you’re setting up a resonance between the expected utility and the utility, instead of a one-way flow from utility to expected utility.

Or maybe it’s just the urge toward a one-sided policy debate: the best policy must have *no* drawbacks.

In my moral philosophy, the *local* negative utility of Hitler’s death is stable, no matter what happens to the external consequences and hence to the *expected* utility.

Of course, you can set up a moral argument that it’s an *inherently* good thing to punish evil people, even with capital punishment for sufficiently evil people. But you can’t carry this moral argument by pointing out that the *consequence* of shooting a man holding a leveled gun may be to save other lives. This is appealing to the value of life, not appealing to the value of death. If expected utilities are leaky and complicated,

it doesn't mean that utilities must be leaky and complicated as well. They might be! But it would be a separate argument.

The Hidden Complexity of Wishes



discussion
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3-c22

I wish to live in the locations of my choice, in a physically healthy, uninjured, and apparently normal version of my current body containing my current mental state, a body which will heal from all injuries at a rate three sigmas faster than the average given the medical technology available to me, and which will be protected from any diseases, injuries or illnesses causing disability, pain, or degraded functionality or any sense, organ, or bodily function for more than ten days consecutively or fifteen days in any year ...

—The Open-Source Wish Project, Wish For
Immortality 1.1

There are three kinds of genies: Genies to whom you can safely say, “I wish for you to do what I should wish for”; genies for which *no* wish is safe; and genies that aren’t very powerful or intelligent.

Suppose your aged mother is trapped in a burning building, and it so happens that you’re in a wheelchair; you can’t rush in yourself. You could cry, “Get my mother out of that building!” but there would be no one to hear.

Luckily you have, in your pocket, an Outcome Pump. This handy device squeezes the flow of time, pouring probability into some outcomes, draining it from others.

The Outcome Pump is not sentient. It contains a tiny time machine, which resets time *unless* a specified outcome occurs. For example, if you hooked up the Outcome Pump’s sensors to a coin, and specified that the time machine should keep resetting until it sees the coin come up heads, and then you actually flipped the coin, *you* would see the coin come up heads. (The physicists say that any future in which a “reset” occurs is

inconsistent, and therefore never happens in the first place—so you aren’t actually killing any versions of yourself.)

Whatever proposition you can manage to input into the Outcome Pump *somehow happens*, though not in a way that violates the laws of physics. If you try to input a proposition that’s *too* unlikely, the time machine will suffer a spontaneous mechanical failure before that outcome ever occurs.

You can also redirect probability flow in more quantitative ways, using the “future function” to scale the temporal reset probability for different outcomes. If the temporal reset probability is 99% when the coin comes up heads, and 1% when the coin comes up tails, the odds will go from 1:1 to 99:1 in favor of tails. If you had a mysterious machine that spit out money, and you wanted to maximize the amount of money spit out, you would use reset probabilities that diminished as the amount of money increased. For example, spitting out \$10 might have a 99.99999% reset probability, and spitting out \$100 might have a 99.99999% reset probability. This way you can get an outcome that tends to be as high as possible in the future function, even when you don’t know the best attainable maximum.

So you desperately yank the Outcome Pump from your pocket—your mother is still trapped in the burning building, remember?—and try to describe your goal: *get your mother out of the building!*

The user interface doesn’t take English inputs. The Outcome Pump isn’t sentient, remember? But it does have 3D scanners for the near vicinity, and built-in utilities for pattern matching. So you hold up a photo of your mother’s head and shoulders; match on the photo; use object contiguity to select your mother’s whole body (not just her head and shoulders); and define the *future function* using your mother’s distance from the building’s center. The further she gets from the building’s center, the less the time machine’s reset probability.

You cry “Get my mother out of the building!” for luck, and press Enter.

For a moment it seems like nothing happens. You look around, waiting for the fire truck to pull up, and rescuers to

arrive—or even just a strong, fast runner to haul your mother out of the building—

BOOM! With a thundering roar, the gas main under the building explodes. As the structure comes apart, in what seems like slow motion, you glimpse your mother's shattered body being hurled high into the air, traveling fast, rapidly increasing its distance from the former center of the building.

On the side of the Outcome Pump is an Emergency Regret Button. All future functions are automatically defined with a huge negative value for the Regret Button being pressed—a temporal reset probability of nearly 1—so that the Outcome Pump is extremely unlikely to do anything which upsets the user enough to make them press the Regret Button. You can't ever remember pressing it. But you've barely started to reach for the Regret Button (and what good will it do now?) when a flaming wooden beam drops out of the sky and smashes you flat.

Which wasn't really what you wanted, but scores very high in the defined future function ...

The Outcome Pump is a genie of the second class. *No wish is safe.*

If someone asked you to get their poor aged mother out of a burning building, you might help, or you might pretend not to hear. But it wouldn't even *occur* to you to explode the building. "Get my mother out of the building" *sounds* like a much safer wish than it really is, because you don't even *consider* the plans that you assign extreme negative values.

Consider again the Tragedy of Group Selectionism: Some early biologists asserted that group selection for low subpopulation sizes would produce individual restraint in breeding; and yet actually enforcing group selection in the laboratory produced cannibalism, especially of immature females. It's obvious in hindsight that, given strong selection for small subpopulation sizes, cannibals will outreproduce individuals who voluntarily forego reproductive opportunities. But eating little girls is such an *un-aesthetic* solution that Wynne-Edwards, Allee, Brereton, and the other group-selectionists simply didn't think of it. They only saw the solutions they would have used themselves.

Suppose you try to patch the future function by specifying that the Outcome Pump should not explode the building: outcomes in which the building materials are distributed over too much volume will have ~ 1 temporal reset probabilities.

So your mother falls out of a second-story window and breaks her neck. The Outcome Pump took a different path through time that still ended up with your mother outside the building, and it still wasn't what you wanted, and it still wasn't a solution that would occur to a human rescuer.

If only the Open-Source Wish Project had developed a Wish To Get Your Mother Out Of A Burning Building:

I wish to move my mother (defined as the woman who shares half my genes and gave birth to me) to outside the boundaries of the building currently closest to me which is on fire; but not by exploding the building; nor by causing the walls to crumble so that the building no longer has boundaries; nor by waiting until after the building finishes burning down for a rescue worker to take out the body ...

All these special cases, the seemingly unlimited number of required patches, should remind you of the parable of Artificial Addition—programming an Arithmetic Expert Systems by explicitly adding ever more assertions like “fifteen plus fifteen equals thirty, but fifteen plus sixteen equals thirty-one instead.”

How do *you* exclude the outcome where the building explodes and flings your mother into the sky? You look ahead, and you foresee that your mother would end up dead, and you don't want that consequence, so you try to forbid the event leading up to it.

Your brain isn't hardwired with a specific, prerecorded statement that “Blowing up a burning building containing my mother is a bad idea.” And yet you're trying to prerecord that exact specific statement in the Outcome Pump's future function. So the wish is exploding, turning into a giant lookup table that records your judgment of every possible path through time.

You failed to ask for what you really wanted. You *wanted* your mother to go on living, but you *wished* for her to become more distant from the center of the building.

Except that's not all you wanted. If your mother was rescued from the building but was horribly burned, that outcome would rank lower in your preference ordering than an outcome where she was rescued safe and sound. So you not only value your mother's life, but also her health.

And you value not just her bodily health, but her state of mind. Being rescued in a fashion that traumatizes her—for example, a giant purple monster roaring up out of nowhere and seizing her—is inferior to a fireman showing up and escorting her out through a non-burning route. (Yes, we're supposed to stick with physics, but maybe a powerful enough Outcome Pump has aliens coincidentally showing up in the neighborhood at exactly that moment.) You would certainly prefer her being rescued by the monster to her being roasted alive, however.

How about a wormhole spontaneously opening and swallowing her to a desert island? Better than her being dead; but worse than her being alive, well, healthy, untraumatized, and in continual contact with you and the other members of her social network.

Would it be okay to save your mother's life at the cost of the family dog's life, if it ran to alert a fireman but then got run over by a car? Clearly yes, but it would be better *ceteris paribus* to avoid killing the dog. You wouldn't want to swap a human life for hers, but what about the life of a convicted murderer? Does it matter if the murderer dies trying to save her, from the goodness of his heart? How about two murderers? If the cost of your mother's life was the destruction of every extant copy, including the memories, of Bach's *Little Fugue in G Minor*, would that be worth it? How about if she had a terminal illness and would die anyway in eighteen months?

If your mother's foot is crushed by a burning beam, is it worthwhile to extract the rest of her? What if her head is crushed, leaving her body? What if her body is crushed, leaving only her head? What if there's a cryonics team waiting outside,

ready to suspend the head? Is a frozen head a person? Is Terry Schiavo a person? How much is a chimpanzee worth?

Your brain is not infinitely complicated; there is only a finite Kolmogorov complexity / message length which suffices to describe all the judgments you would make. But just because this complexity is finite does not make it small. We value many things, and no they are *not* reducible to valuing happiness or valuing reproductive fitness.

There is no safe wish smaller than an entire human morality. There are too many possible paths through Time. You can't visualize all the roads that lead to the destination you give the genie. "Maximizing the distance between your mother and the center of the building" can be done even more effectively by detonating a nuclear weapon. Or, at higher levels of genie power, flinging her body out of the Solar System. Or, at higher levels of genie intelligence, doing something that neither you nor I would think of, just like a chimpanzee wouldn't think of detonating a nuclear weapon. You can't visualize all the paths through time, any more than you can program a chess-playing machine by hardcoding a move for every possible board position.

And real life is far more complicated than chess. You cannot predict, in advance, which of your values will be needed to judge the path through time that the genie takes. Especially if you wish for something longer-term or wider-range than rescuing your mother from a burning building.

I fear the Open-Source Wish Project is futile, except as an illustration of how *not* to think about genie problems. The only safe genie is a genie that shares all your judgment criteria, and at that point, you can just say "I wish for you to do what I should wish for." Which simply runs the genie's *should* function.¹

1. Edit 2018: A safe genie would not need you to make a wish like: "Rescue my mother from a burning building." The genie should just rescue your mother because the utility difference is large and there may not be time for you to be questioned. However wishes would still be useful where the utility difference is small or not time critical like: "I wish for a chocolate milkshake."

Indeed, it shouldn't be necessary to say *anything*. To be a safe fulfiller of a wish, a genie must share the same values that led you to make the wish. Otherwise the genie may not choose a path through time that leads to the destination you had in mind, or it may fail to exclude horrible side effects that would lead you to not even consider a plan in the first place. Wishes are leaky generalizations, derived from the huge but finite structure that is your entire morality; only by including this entire structure can you plug all the leaks.

With a safe genie, wishing is superfluous. Just run the genie.

Anthropomorphic Optimism

The core fallacy of anthropomorphism is expecting something to be predicted by the black box of your brain, when its causal structure is so different from that of a human brain as to give you no license to expect any such thing.

The early (pre-1966) biologists in The Tragedy of Group Selectionism believed that predators would voluntarily restrain their breeding to avoid overpopulating their habitat and exhausting the prey population. Later on, when Michael J. Wade actually went out and created in the laboratory the nigh-impossible conditions for group selection, the adults adapted to cannibalize eggs and larvae, especially female larvae.¹

Now, why might the group selectionists have *not* thought of that possibility?

Suppose you were a member of a tribe, and you knew that, in the near future, your tribe would be subjected to a resource squeeze. You might propose, as a solution, that no couple have more than one child—after the first child, the couple goes on birth control. Saying, “Let’s all individually have as many children as we can, but then hunt down and cannibalize each other’s children, especially the girls,” would not even *occur to you as a possibility*.

Think of a preference ordering over solutions, relative to your goals. You want a solution as high in this preference ordering as possible. How do you find one? With a brain, of course! Think of your brain as a *high-ranking-solution-generator*—a search process that produces solutions that rank high in your innate preference ordering.

The solution space on all real-world problems is generally fairly large, which is why you need an *efficient* brain that doesn’t even *bother to formulate* the vast majority of low-ranking solutions.



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1. Wade, “Group selections among laboratory populations of *Tribolium*.”

If your tribe is faced with a resource squeeze, you could try hopping everywhere on one leg, or chewing off your own toes. These “solutions” obviously wouldn’t work and would incur large costs, as you can see upon examination—but in fact your brain is too efficient to waste time considering such poor solutions; it doesn’t generate them in the first place. Your brain, in its search for high-ranking solutions, flies directly to parts of the solution space like “Everyone in the tribe gets together, and agrees to have no more than one child per couple until the resource squeeze is past.”

Such a *low-ranking* solution as “Everyone have as many kids as possible, then cannibalize the girls” would not be *generated in your search process*.

But the ranking of an option as “low” or “high” is not an inherent property of the option. It is a property of the optimization process that does the preferring. And different optimization processes will search in different orders.

So far as *evolution* is concerned, individuals reproducing to the fullest and then cannibalizing others’ daughters is a no-brainer; whereas individuals voluntarily restraining their own breeding for the good of the group is absolutely ludicrous. Or to say it less anthropomorphically, the first set of alleles would rapidly replace the second in a population. (And natural selection has no obvious search order here—these two alternatives seem around equally simple as mutations.)

Suppose that one of the biologists had said, “If a predator population has only finite resources, evolution will craft them to voluntarily restrain their breeding—that’s how *I’d* do it if *I* were in charge of building predators.” This would be anthropomorphism outright, the lines of reasoning naked and exposed: *I* would do it this way, therefore *I* infer that *evolution* will do it this way.

One does occasionally encounter the fallacy outright, in my line of work. But suppose you say to the one, “An AI will not necessarily work like you do.” Suppose you say to this hypothetical biologist, “Evolution doesn’t work like you do.” What will the one say in response? I can tell you a reply you will *not* hear: “Oh my! I didn’t realize that! One of the steps

of my inference was invalid; I will throw away the conclusion and start over from scratch.”

No: what you’ll hear *instead* is a reason why any AI has to reason the same way as the speaker. Or a reason why natural selection, following entirely different criteria of optimization and using entirely different methods of optimization, ought to do *the same thing* that would occur to a human as a good idea.

Hence the elaborate idea that group selection would favor predator groups where the individuals voluntarily forsook reproductive opportunities.

The group selectionists went just as far astray, in their predictions, as someone committing the fallacy outright. Their final conclusions were the same as if they were assuming outright that evolution necessarily thought like themselves. But they erased what had been written *above* the bottom line of their argument, *without* erasing the actual bottom line, and wrote in new rationalizations. Now the fallacious reasoning is disguised; the *obviously* flawed step in the inference has been hidden—even though the conclusion remains exactly the same; and hence, in the real world, exactly as wrong.

But why would any scientist do this? In the end, the data came out against the group selectionists and they were embarrassed.

As I remarked in Fake Optimization Criteria, we humans seem to have evolved an instinct for arguing that *our* preferred policy arises from practically *any* criterion of optimization. Politics was a feature of the ancestral environment; we are descended from those who argued most persuasively that the *tribe’s* interest—not just their own interest—required that their hated rival Uglak be executed. We certainly aren’t descended from Uglak, who failed to argue that his tribe’s moral code—not just his own obvious self-interest—required his survival.

And because we can more persuasively argue for what we honestly believe, we have evolved an instinct to honestly believe that other people’s goals, and our tribe’s moral code, truly do imply that they should do things *our* way for *their* benefit.

So the group selectionists, imagining this beautiful picture of predators restraining their breeding, instinctively rationalized

why natural selection ought to do things *their* way, even according to natural selection's own purposes. The foxes will be fitter if they restrain their breeding! No, really! They'll even outbreed other foxes who don't restrain their breeding! Honestly!

The problem with trying to argue natural selection into doing things your way is that evolution does not contain that which could be moved by your arguments. Evolution does not work like you do—not even to the extent of having any element that could listen to or *care about* your painstaking explanation of why evolution ought to do things your way. Human arguments are not even *commensurate* with the internal structure of natural selection as an optimization process—human arguments aren't used in promoting alleles, as human arguments would play a causal role in human politics.

So instead of *successfully* persuading natural selection to do things their way, the group selectionists were simply embarrassed when reality came out differently.

There's a fairly heavy subtext here about Unfriendly AI.

But the point generalizes: this is the problem with optimistic reasoning *in general*. What is optimism? It is ranking the possibilities by your own preference ordering, and selecting an outcome high in that preference ordering, and somehow that outcome ends up as your prediction. What kind of elaborate rationalizations were generated along the way is probably not so relevant as one might fondly believe; look at the cognitive history and it's optimism in, optimism out. But Nature, or whatever other process is under discussion, is not *actually, causally* choosing between outcomes by ranking them in your preference ordering and picking a high one. So the brain fails to synchronize with the environment, and the prediction fails to match reality.

Lost Purposes

It was in either kindergarten or first grade that I was first asked to pray, given a transliteration of a Hebrew prayer. I asked what the words meant. I was told that so long as I prayed in Hebrew, I didn't need to know what the words meant, it would work anyway.

That was the beginning of my break with Judaism.

As you read this, some young man or woman is sitting at a desk in a university, earnestly studying material they have no intention of ever using, and no interest in knowing for its own sake. They want a high-paying job, and the high-paying job requires a piece of paper, and the piece of paper requires a previous master's degree, and the master's degree requires a bachelor's degree, and the university that grants the bachelor's degree requires you to take a class in twelfth-century knitting patterns to graduate. So they diligently study, intending to forget it all the moment the final exam is administered, but still seriously working away, because they *want* that piece of paper.

Maybe *you* realized it was all madness, but I bet you did it anyway. You didn't have a choice, right? A recent study¹ here in the Bay Area showed that 80% of teachers in K-5 reported spending less than one hour per week on science, and 16% said they spend no time on science. Why? I'm given to understand the proximate cause is the No Child Left Behind Act and similar legislation. Virtually all classroom time is now spent on preparing for tests mandated at the state or federal level. I seem to recall (though I can't find the source) that just *taking* mandatory tests was 40% of classroom time in one school.

1. sharov.me/r/3-13

Science courses nearly extinct in elementary grades, study finds • sfgate.com



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The old Soviet bureaucracy was famous for being more interested in appearances than reality. One shoe factory overfulfilled its quota by producing lots of tiny shoes. Another shoe factory reported cut but unassembled leather as a “shoe.” The superior bureaucrats weren’t interested in looking too hard, because they also wanted to report quota overfulfillments. All this was a great help to the comrades freezing their feet off.

It is now being suggested in several sources that an actual *majority* of published findings in medicine, though “statistically significant with $p < 0.05$,” are untrue. But so long as $p < 0.05$ remains the threshold for publication, why should anyone hold themselves to higher standards, when that requires bigger research grants for larger experimental groups, and decreases the likelihood of getting a publication? Everyone knows that *the whole point of science* is to publish lots of papers, just as *the whole point of a university* is to print certain pieces of parchment, and *the whole point of a school* is to pass the mandatory tests that guarantee the annual budget. You don’t get to set the rules of the game, and if you try to play by different rules, you’ll just lose.

(Though for some reason, physics journals² require a threshold of $p < 0.0000003$. It’s as if they conceive of some other purpose to their existence than publishing physics papers.)

There’s chocolate at the supermarket, and you can get to the supermarket by driving, and driving requires that you be in the car, which means opening your car door, which needs keys. If you find there’s no chocolate at the supermarket, you won’t stand around opening and slamming your car door because the car door still needs opening. I rarely notice people losing track of plans they devised themselves.

It’s another matter when incentives must flow through large organizations—or worse, many different organizations and interest groups, some of them governmental. Then you see be-

2. sharov.me/r/3-14

5 *Sigma What’s That?* | *Scientific American* •
scientificamerican.com



haviors that would mark literal insanity, if they were born from a single mind. Someone gets paid every time they open a car door, because that's what's measurable; and this person doesn't care whether the driver ever gets paid for arriving at the supermarket, let alone whether the buyer purchases the chocolate, or whether the eater is happy or starving.

From a Bayesian perspective, subgoals are epiphenomena of conditional probability functions. There is no expected utility without utility. How silly would it be to think that instrumental value could take on a *mathematical* life of its own, leaving terminal value in the dust? It's not sane by decision-theoretical criteria of sanity.

But consider the No Child Left Behind Act. The politicians want to look like they're doing something about educational difficulties; the politicians have to look busy to voters *this* year, not fifteen years later when the kids are looking for jobs. The politicians are not the consumers of education. The bureaucrats have to show progress, which means that they're only interested in progress that can be measured this year. They aren't the ones who'll end up ignorant of science. The publishers who commission textbooks, and the committees that purchase textbooks, don't sit in the classrooms bored out of their skulls.

The actual consumers of knowledge are the children—who can't pay, can't vote, can't sit on the committees. Their parents care for them, but don't sit in the classes themselves; they can only hold politicians responsible according to surface images of "tough on education." Politicians are too busy being re-elected to study all the data themselves; they have to rely on surface images of bureaucrats being busy and commissioning studies—it may not work to help any children, but it works to let politicians appear caring. Bureaucrats don't expect to use textbooks themselves, so they don't care if the textbooks are hideous to read, so long as the process by which they are purchased looks good on the surface. The textbook publishers have no motive to produce *bad* textbooks, but they know that the textbook purchasing committee will be comparing textbooks based on how many different subjects they cover, and that the fourth-grade purchasing committee isn't coordinated

with the third-grade purchasing committee, so they cram as many subjects into one textbook as possible. Teachers won't get through a fourth of the textbook before the end of the year, and then the next year's teacher will start over. Teachers might complain, but they aren't the decision-makers, and ultimately, it's not their future on the line, which puts sharp bounds on how much effort they'll spend on unpaid altruism...

It's amazing, when you look at it that way—consider all the lost information and lost incentives—that anything at all remains of the original purpose, gaining knowledge. Though many educational systems seem to be currently in the process of collapsing into a state not much better than nothing.

Want to see the problem *really* solved? Make the politicians go to school.

A single human mind can track a probabilistic expectation of utility as it flows through the conditional chances of a dozen intermediate events—including nonlocal dependencies, places where the expected utility of opening the car door depends on whether there's chocolate in the supermarket. But organizations can only reward today what is measurable today, what can be written into legal contract today, and this means measuring intermediate events rather than their distant consequences. These intermediate measures, in turn, are leaky generalizations—often *very* leaky. Bureaucrats are untrustworthy genies, for they do not share the values of the wisher.

Miyamoto Musashi said:³

The primary thing when you take a sword in your hands is your intention to cut the enemy, whatever the means. Whenever you parry, hit, spring, strike or touch the enemy's cutting sword, you must cut the enemy in the same movement. It is essential to attain this. If you think only of hitting, springing, striking or touching the enemy, you will not be able actually to cut him. More than anything, you must

3. Miyamoto Musashi, *Book of Five Rings* (New Line Publishing, 2003).

be thinking of carrying your movement through to cutting him. You must thoroughly research this.

(I wish *I* lived in an era where I could just tell my readers they have to thoroughly research something, without giving insult.)

Why would any individual lose track of their purposes in a swordfight? If someone else had taught them to fight, if they had not generated the *entire* art from within themselves, they might not understand the reason for parrying at one moment, or springing at another moment; they might not realize when the rules had exceptions, fail to see the times when the usual method won't cut through.

The essential thing in the art of epistemic rationality is to understand how every rule is cutting through to the truth in the same movement. The corresponding essential of pragmatic rationality—decision theory, versus probability theory—is to always see how every expected utility cuts through to utility. You must thoroughly research this.

C. J. Cherryh said:⁴

Your sword has no blade. It has only your intention.
When that goes astray you have no weapon.

I have seen many people go astray when they wish to the genie of an imagined AI, dreaming up wish after wish that seems good to them, sometimes with many patches and sometimes without even that pretense of caution. And they don't jump to the meta-level. They don't instinctively look-to-purpose, the instinct that started me down the track to atheism at the age of five. They do not ask, as I reflexively ask, "Why do *I* think this wish is a good idea? Will the genie judge likewise?" They don't *see* the source of their judgment, hovering behind the judgment as its generator. They lose track of the ball; they know the ball bounced, but they don't instinctively look back to see where it bounced from—the criterion that generated their judgments.

4. Carolyn J. Cherryh, *The Paladin* (Baen, 2002).

Likewise with people not automatically noticing when supposedly selfish people give altruistic arguments in favor of selfishness, or when supposedly altruistic people give selfish arguments in favor of altruism.

People can handle goal-tracking for driving to the supermarket just fine, when it's *all* inside their own heads, and no genies or bureaucracies or philosophies are involved. The trouble is that real civilization is immensely more complicated than this. Dozens of organizations, and dozens of years, intervene between the child suffering in the classroom, and the new-minted college graduate not being very good at their job. (But will the interviewer or manager notice, if the college graduate is good at looking busy?) With every new link that intervenes between the action and its consequence, intention has one more chance to go astray. With every intervening link, information is lost, incentive is lost. And this bothers most people a lot less than it bothers me, or why were all my classmates willing to say prayers without knowing what they meant? They didn't feel the same instinct to look-to-the-generator.

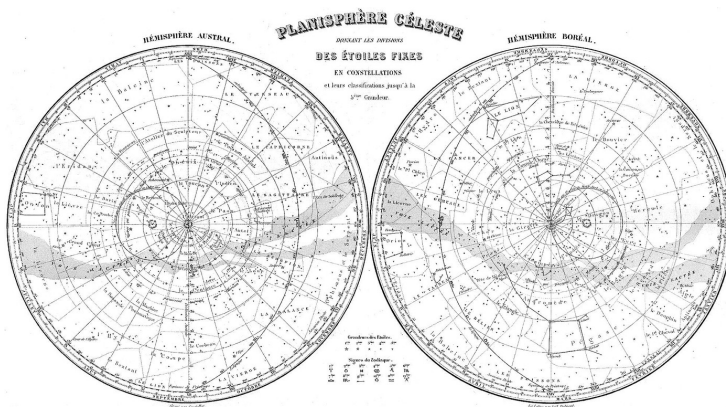
Can people learn to keep their eye on the ball? To keep their intention from going astray? To never spring or strike or touch, without knowing the higher goal they will complete in the same movement? People *do* often want to do their jobs, all else being equal. Can there be such a thing as a sane corporation? A sane civilization, even? That's only a distant dream, but it's what I've been getting at with all of these essays on the flow of intentions (a.k.a. expected utility, a.k.a. instrumental value) without losing purpose (a.k.a. utility, a.k.a. terminal value). Can people learn to feel the flow of parent goals and child goals? To know consciously, as well as implicitly, the distinction between expected utility and utility?

Do you care about threats to your civilization? The worst metathreat to complex civilization is its own complexity, for that complication leads to the loss of many purposes.

I look back, and I see that more than anything, my life has been driven by an exceptionally strong abhorrence to lost purposes. I hope it can be transformed to a learnable skill.

PART N

A Human's Guide to Words



The Parable of the Dagger

(Adapted from Raymond Smullyan.¹)

Once upon a time, there was a court jester who dabbled in logic.

The jester presented the king with two boxes. Upon the first box was inscribed:

Either this box contains an angry frog, or the box with a false inscription contains an angry frog, but not both.

On the second box was inscribed:

Either this box contains gold and the box with a false inscription contains an angry frog, or this box contains an angry frog and the box with a true inscription contains gold.

And the jester said to the king: “One box contains an angry frog, the other box gold; and one, and only one, of the inscriptions is true.”

The king opened the wrong box, and was savaged by an angry frog.

“You see,” the jester said, “let us hypothesize that the first inscription is the true one. Then suppose the first box contains gold. Then the other box would have an angry frog, while the box with a true inscription would contain gold, which would make the second statement true as well. Now hypothesize that the first inscription is false, and that the first box contains gold. Then the second inscription would be—”

The king ordered the jester thrown in the dungeons.



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1. Raymond M. Smullyan, *What Is the Name of This Book?: The Riddle of Dracula and Other Logical Puzzles* (Penguin Books, 1990).

A day later, the jester was brought before the king in chains and shown two boxes.

“One box contains a key,” said the king, “to unlock your chains; and if you find the key you are free. But the other box contains a dagger for your heart if you fail.”

And the first box was inscribed:

Either both inscriptions are true, or both inscriptions are false.

And the second box was inscribed:

This box contains the key.

The jester reasoned thusly: “Suppose the first inscription is true. Then the second inscription must also be true. Now suppose the first inscription is false. Then again the second inscription must be true. So the second box must contain the key, if the first inscription is true, and also if the first inscription is false. Therefore, the second box must logically contain the key.”

The jester opened the second box, and found a dagger.

“How?!” cried the jester in horror, as he was dragged away. “It’s logically impossible!”

“It is entirely possible,” replied the king. “I merely wrote those inscriptions on two boxes, and then I put the dagger in the second one.”

The Parable of Hemlock

All men are mortal. Socrates is a man. Therefore
Socrates is mortal.

—Standard Medieval syllogism



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Socrates raised the glass of hemlock to his lips...

“Do you suppose,” asked one of the onlookers, “that even hemlock will not be enough to kill so wise and good a man?”

“No,” replied another bystander, a student of philosophy; “all men are mortal, and Socrates is a man; and if a mortal drinks hemlock, surely he dies.”

“Well,” said the onlooker, “what if it happens that Socrates *isn’t* mortal?”

“Nonsense,” replied the student, a little sharply; “all men are mortal *by definition*; it is part of what we mean by the word ‘man.’ All men are mortal, Socrates is a man, therefore Socrates is mortal. It is not merely a guess, but a *logical certainty*.”

“I suppose that’s right...” said the onlooker. “Oh, look, Socrates already drank the hemlock while we were talking.”

“Yes, he should be keeling over any minute now,” said the student.

And they waited, and they waited, and they waited...

“Socrates appears not to be mortal,” said the onlooker.

“Then Socrates must not be a man,” replied the student. “All men are mortal, Socrates is not mortal, therefore Socrates is not a man. And that is not merely a guess, but a *logical certainty*.”

The fundamental problem with arguing that things are true “by definition” is that you can’t make reality go a different way by choosing a different definition.

You could reason, perhaps, as follows: “All things I have observed which wear clothing, speak language, and use tools, have also shared certain other properties as well, such as breathing air and pumping red blood. The last thirty ‘humans’ belonging to this cluster whom I observed to drink hemlock soon fell

over and stopped moving. Socrates wears a toga, speaks fluent ancient Greek, and drank hemlock from a cup. So I predict that Socrates will keel over in the next five minutes.”

But that would be mere *guessing*. It wouldn’t be, y’know, absolutely and eternally certain. The Greek philosophers—like most prescientific philosophers—were rather fond of certainty.

Luckily the Greek philosophers have a crushing rejoinder to your questioning. You have misunderstood the meaning of “All humans are mortal,” they say. It is not a mere *observation*. It is part of the *definition* of the word “human.” Mortality is one of several properties that are individually necessary, and together sufficient, to determine membership in the class “human.” The statement “All humans are mortal” is a logically valid truth, absolutely unquestionable. And if Socrates is human, he *must* be mortal: it is a logical deduction, as certain as certain can be.

But then we can never know for certain that Socrates is a “human” until after Socrates has been observed to be mortal. It does no good to observe that Socrates speaks fluent Greek, or that Socrates has red blood, or even that Socrates has human DNA. None of these characteristics are *logically equivalent* to mortality. You have to *see him die* before you can conclude that he was human.

(And even then it’s not infinitely certain. What if Socrates rises from the grave a night after you see him die? Or more realistically, what if Socrates is signed up for cryonics? If mortality is defined to mean finite lifespan, then you can never really *know* if someone was human, until you’ve observed to the end of eternity—just to make sure they don’t come back. Or you could *think* you saw Socrates keel over, but it could be an illusion projected onto your eyes with a retinal scanner. Or maybe you just hallucinated the whole thing ...)

The problem with syllogisms is that they’re *always* valid. “All humans are mortal; Socrates is human; therefore Socrates is mortal” is—if you treat it as a logical syllogism—logically valid within our own universe. It’s also logically valid within neighboring Everett branches in which, due to a slightly different evolved biochemistry, hemlock is a delicious treat rather than a poison. And it’s logically valid even in universes where

Socrates never existed, or for that matter, where humans never existed.

The Bayesian definition of evidence favoring a hypothesis is evidence which we are more likely to see if the hypothesis is true than if it is false. Observing that a syllogism is logically valid can never be evidence favoring any empirical proposition, because the syllogism will be logically valid whether that proposition is true or false.

Syllogisms are valid in all possible worlds, and therefore, observing their validity never tells us anything about *which* possible world we actually live in.

This doesn't mean that logic is useless—just that logic can only tell us that which, *in some sense*, we already know. But we do not always believe what we know. Is the number 29,384,209 prime? By virtue of how I define my decimal system and my axioms of arithmetic, I have already determined my answer to this question—but I do not know what my answer is yet, and I must do some logic to find out.

Similarly, if I form the uncertain empirical generalization “Humans are vulnerable to hemlock,” and the uncertain empirical guess “Socrates is human,” logic can tell me that my previous guesses are predicting that Socrates will be vulnerable to hemlock.

It's been suggested that we can view logical reasoning as resolving our uncertainty about impossible possible worlds—eliminating probability mass in logically impossible worlds which we did not know to be logically impossible. In this sense, logical argument can be treated as observation.

But when you talk about an empirical prediction like “Socrates is going to keel over and stop breathing” or “Socrates is going to do fifty jumping jacks and then compete in the Olympics next year,” that is a matter of possible worlds, not impossible possible worlds.

Logic can tell us which hypotheses match up to which observations, and it can tell us what these hypotheses predict for the future—it can bring old observations and previous guesses to bear on a new problem. But logic never flatly says, “Socrates *will* stop breathing now.” Logic never dictates any empirical

question; it never settles any real-world query which could, by any stretch of the imagination, go either way.

Just remember the Litany Against Logic:

Logic stays true, wherever you may go,

So logic never tells you where you live.

Words as Hidden Inferences

Suppose I find a barrel, sealed at the top, but with a hole large enough for a hand. I reach in and feel a small, curved object. I pull the object out, and it's blue—a bluish egg. Next I reach in and feel something hard and flat, with edges—which, when I extract it, proves to be a red cube. I pull out 11 eggs and 8 cubes, and every egg is blue, and every cube is red.

Now I reach in and I feel another egg-shaped object. Before I pull it out and look, I have to guess: What will it look like?

The evidence doesn't prove that every egg in the barrel is blue and every cube is red. The evidence doesn't even argue this all that strongly: 19 is not a large sample size. Nonetheless, I'll guess that this egg-shaped object is blue—or as a runner-up guess, red. If I guess anything else, there's as many possibilities as distinguishable colors—and for that matter, who says the egg has to be a single shade? Maybe it has a picture of a horse painted on.

So I say “blue,” with a dutiful patina of humility. For I am a sophisticated rationalist-type person, and I keep track of my assumptions and dependencies—I guess, but I'm aware that I'm guessing...right?

But when a large yellow striped feline-shaped object leaps out at me from the shadows, I think, “Yikes! A tiger!” Not, “Hm...objects with the properties of largeness, yellowness, stripedness, and feline shape, have previously often possessed the properties ‘hungry’ and ‘dangerous,’ and thus, although it is not logically necessary, it may be an empirically good guess that *aaauuughhhh CRUNCH CRUNCH GULP.*”

The human brain, for some odd reason, seems to have been adapted to make this inference quickly, automatically, and without keeping explicit track of its assumptions.

And if I name the egg-shaped objects “bleggs” (for blue eggs) and the red cubes “rubes,” then, when I reach in and feel another



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egg-shaped object, I may think, *Oh, it's a blegg*, rather than considering all that problem-of-induction stuff.

It is a common misconception that you can define a word any way you like.

This would be true *if* the brain treated words as purely logical constructs, Aristotelian classes, and you never took out any more information than you put in.

Yet the brain goes on about its work of categorization, whether or not we consciously approve. "All humans are mortal; Socrates is a human; therefore Socrates is mortal"—thus spake the ancient Greek philosophers. Well, if mortality is part of your logical definition of "human," you can't logically classify Socrates as human until you observe him to be mortal. But—this is the problem—Aristotle knew perfectly well that Socrates was a human. Aristotle's brain placed Socrates in the "human" category as efficiently as your own brain categorizes tigers, apples, and everything else in its environment: Swiftly, silently, and without conscious approval.

Aristotle laid down rules under which no one could conclude Socrates was "human" until after he died. Nonetheless, Aristotle and his students went on concluding that living people were humans and therefore mortal; they saw distinguishing properties such as human faces and human bodies, and their brains made the leap to inferred properties such as mortality.

Misunderstanding the working of your own mind does *not*, thankfully, prevent the mind from doing its work. Otherwise Aristotelians would have starved, unable to conclude that an object was edible merely because it looked and felt like a banana.

So the Aristotelians went on classifying environmental objects on the basis of partial information, the way people had always done. Students of Aristotelian logic went on thinking exactly the same way, but they had acquired an erroneous picture of *what* they were doing.

If you asked an Aristotelian philosopher whether Carol the grocer was mortal, they would say "Yes." If you asked them how they knew, they would say "All humans are mortal; Carol is human; therefore Carol is mortal." Ask them whether it was

a guess or a certainty, and they would say it was a certainty (if you asked before the sixteenth century, at least). Ask them how they knew that humans were mortal, and they would say it was established by definition.

The Aristotelians were still the same people, they retained their original natures, but they had acquired incorrect beliefs about their own functioning. They looked into the mirror of self-awareness, and saw something unlike their true selves: they reflected incorrectly.

Your brain doesn't treat words as logical definitions with no empirical consequences, and so neither should you. The mere act of creating a word can cause your mind to allocate a category, and thereby trigger unconscious inferences of similarity. Or block inferences of similarity; if I create two labels I can get your mind to allocate two categories. Notice how I said "you" and "your brain" as if they were different things?

Making errors about the inside of your head doesn't change what's there; otherwise Aristotle would have died when he concluded that the brain was an organ for cooling the blood. Philosophical mistakes usually don't interfere with blink-of-an-eye perceptual inferences.

But philosophical mistakes can severely mess up the deliberate thinking processes that we use to try to correct our first impressions. If you believe that you can "define a word any way you like," without realizing that your brain goes on categorizing without your conscious oversight, then you won't make the effort to choose your definitions wisely.

Extensions and Intensions



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“What is red?”

“Red is a color.”

“What’s a color?”

“A color is a property of a thing.”

But what is a thing? And what’s a property? Soon the two are lost in a maze of words defined in other words, the problem that Steven Harnad once described as trying to learn Chinese from a Chinese/Chinese dictionary.

Alternatively, if you asked me “What is red?” I could point to a stop sign, then to someone wearing a red shirt, and a traffic light that happens to be red, and blood from where I accidentally cut myself, and a red business card, and then I could call up a color wheel on my computer and move the cursor to the red area. This would probably be sufficient, though if you know what the word “No” means, the truly strict would insist that I point to the sky and say “No.”

I think I stole this example from S. I. Hayakawa—though I’m really not sure, because I heard this way back in the indistinct blur of my childhood. (When I was twelve, my father accidentally deleted all my computer files. I have no memory of anything before that.)

But that’s how I remember first learning about the difference between intensional and extensional definition. To give an “intensional definition” is to define a word or phrase in terms of other words, as a dictionary does. To give an “extensional definition” is to point to examples, as adults do when teaching children. The preceding sentence gives an intensional definition of “extensional definition,” which makes it an extensional example of “intensional definition.”

In Hollywood Rationality and popular culture generally, “rationalists” are depicted as word-obsessed, floating in endless verbal space disconnected from reality.

But the actual Traditional Rationalists have long insisted on maintaining a tight connection to experience:

If you look into a textbook of chemistry for a definition of lithium, you may be told that it is that element whose atomic weight is 7 very nearly. But if the author has a more logical mind he will tell you that if you search among minerals that are vitreous, translucent, grey or white, very hard, brittle, and insoluble, for one which imparts a crimson tinge to an unluminous flame, this mineral being triturated with lime or witherite rats-bane, and then fused, can be partly dissolved in muriatic acid; and if this solution be evaporated, and the residue be extracted with sulphuric acid, and duly purified, it can be converted by ordinary methods into a chloride, which being obtained in the solid state, fused, and electrolyzed with half a dozen powerful cells, will yield a globule of a pinkish silvery metal that will float on gasolene; and the material of that is a specimen of lithium.

—Charles Sanders Peirce¹

That's an example of "logical mind" as described by a genuine Traditional Rationalist, rather than a Hollywood scriptwriter.

But note: Peirce isn't *actually* showing you a piece of lithium. He didn't have pieces of lithium stapled to his book. Rather he's giving you a treasure map—an intensionally defined procedure which, when executed, will lead you to an extensional example of lithium. This is not the same as just tossing you a hunk of lithium, but it's not the same as saying "atomic weight 7" either. (Though if you had *sufficiently sharp* eyes, saying "3 protons" might let you pick out lithium at a glance ...)

So that is intensional and extensional *definition*, which is a way of telling someone else what you mean by a concept. When I talked about "definitions" above, I talked about a way of

1. Charles Sanders Peirce, *Collected Papers* (Harvard University Press, 1931).

communicating concepts—*telling someone else* what you mean by “red,” “tiger,” “human,” or “lithium.” Now let’s talk about the actual concepts themselves.

The actual intension of my “tiger” concept would be the neural pattern (in my temporal cortex) that inspects an incoming signal from the visual cortex to determine whether or not it is a tiger.

The actual extension of my “tiger” concept is everything I call a tiger.

Intensional definitions don’t capture entire intensions; extensional definitions don’t capture entire extensions. If I point to just one tiger and say the word “tiger,” the communication may fail if they think I mean “dangerous animal” or “male tiger” or “yellow thing.” Similarly, if I say “dangerous yellow-black striped animal,” without pointing to anything, the listener may visualize giant hornets.

You can’t capture in words all the details of the cognitive concept—as it exists in your mind—that lets you recognize things as tigers or nontigers. It’s too large. And you can’t point to all the tigers you’ve ever seen, let alone everything you *would* call a tiger.

The strongest definitions use a crossfire of intensional and extensional communication to nail down a concept. Even so, you only communicate *maps to* concepts, or instructions for building concepts—you don’t communicate the *actual* categories as they exist in your mind or in the world.

(Yes, with enough creativity you can construct exceptions to this rule, like “Sentences Eliezer Yudkowsky has published containing the term ‘huragalon’ as of Feb 4, 2008.” I’ve just shown you this concept’s entire extension. But except in mathematics, definitions are usually treasure maps, not treasure.)

So that’s another reason you can’t “define a word any way you like”: You can’t directly program concepts into someone else’s brain.

Even within the Aristotelian paradigm, where we pretend that the definitions are the actual concepts, you don’t have *simultaneous* freedom of intension and extension. Suppose I define Mars as “A huge red rocky sphere, around a tenth of Earth’s

mass and 50% further away from the Sun.” It’s then a separate matter to show that this intensional definition matches some particular extensional thing in my experience, or indeed, that it matches any real thing whatsoever. If instead I say “That’s Mars” and point to a red light in the night sky, it becomes a separate matter to show that this extensional light matches any particular intensional definition I may propose—or any intensional beliefs I may have—such as “Mars is the God of War.”

But most of the brain’s work of applying intensions happens sub-deliberately. We aren’t consciously aware that our identification of a red light as “Mars” is a separate matter from our verbal definition “Mars is the God of War.” No matter what kind of intensional definition I make up to describe Mars, my mind believes that “Mars” refers to this thingy, and that it is the fourth planet in the Solar System.

When you take into account the way the human mind actually, pragmatically works, the notion “I can define a word any way I like” soon becomes “I can believe anything I want about a fixed set of objects” or “I can move any object I want in or out of a fixed membership test.” Just as you can’t usually convey a concept’s whole intension in words because it’s a big complicated neural membership test, you can’t *control* the concept’s entire intension because it’s applied sub-deliberately. This is why arguing that XYZ is true “by definition” is so popular. If definition changes behaved like the empirical null-ops they’re supposed to be, no one would bother arguing them. But abuse definitions just a little, and they turn into magic wands—in arguments, of course; not in reality.

Similarity Clusters



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Once upon a time, the philosophers of Plato’s Academy claimed that the best definition of human was a “featherless biped.” Diogenes of Sinope, also called Diogenes the Cynic, is said to have promptly exhibited a plucked chicken and declared “Here is Plato’s man.” The Platonists promptly changed their definition to “a featherless biped with broad nails.”

No dictionary, no encyclopedia, has ever listed all the things that humans have in common. We have red blood, five fingers on each of two hands, bony skulls, 23 pairs of chromosomes—but the same might be said of other animal species. We make complex tools to make complex tools, we use syntactical combinatorial language, we harness critical fission reactions as a source of energy: these things may serve out to single out only humans, but not all humans—many of us have never built a fission reactor. With the right set of necessary-and-sufficient gene sequences you could single out all humans, and only humans—at least for now—but it would still be far from *all* that humans have in common.

But so long as you don’t happen to be near a plucked chicken, saying “Look for featherless bipeds” may serve to pick out a few dozen of the particular things that are humans, as opposed to houses, vases, sandwiches, cats, colors, or mathematical theorems.

Once the definition “featherless biped” has been bound to some *particular* featherless bipeds, you can look over the group, and begin harvesting some of the *other* characteristics—beyond mere featherfree twolegginess—that the “featherless bipeds” seem to share in common. The particular featherless bipeds that you see seem to also use language, build complex tools, speak combinatorial language with syntax, bleed red blood if poked, die when they drink hemlock.

Thus the category “human” grows richer, and adds more and more characteristics; and when Diogenes finally presents his

plucked chicken, we are not fooled: This plucked chicken is obviously not similar to the other “featherless bipeds.”

(If Aristotelian logic were a good model of human psychology, the Platonists would have looked at the plucked chicken and said, “Yes, that’s a human; what’s your point?”)

If the *first* featherless biped you see is a plucked chicken, then you may end up thinking that the verbal label “human” denotes a plucked chicken; so I can modify my treasure map to point to “featherless bipeds with broad nails,” and if I am wise, go on to say, “See Diogenes over there? That’s a human, and I’m a human, and you’re a human; and that chimpanzee is not a human, though fairly close.”

The initial clue only has to lead the user to the similarity cluster—the group of things that have many characteristics in common. After that, the initial clue has served its purpose, and I can go on to convey the new information “humans are currently mortal,” or whatever else I want to say about us featherless bipeds.

A dictionary is best thought of, not as a book of Aristotelian class definitions, but a book of hints for matching verbal labels to similarity clusters, or matching labels to properties that are useful in distinguishing similarity clusters.

Typicality and Asymmetrical Similarity



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Birds fly. Well, except ostriches don't. But which is a more typical bird—a robin, or an ostrich?

Which is a more typical chair: a desk chair, a rocking chair, or a beanbag chair?

Most people would say that a robin is a more typical bird, and a desk chair is a more typical chair. The cognitive psychologists who study this sort of thing experimentally, do so under the heading of “typicality effects” or “prototype effects.”¹ For example, if you ask subjects to press a button to indicate “true” or “false” in response to statements like “A robin is a bird” or “A penguin is a bird,” reaction times are faster for more central examples.² Typicality measures correlate well using different investigative methods—reaction times are one example; you can also ask people to directly rate, on a scale of 1 to 10, how well an example (like a specific robin) fits a category (like “bird”).

So we have a mental measure of typicality—which might, perhaps, function as a heuristic—but is there a corresponding bias we can use to pin it down?

Well, which of these statements strikes you as more natural: “98 is approximately 100,” or “100 is approximately 98”? If you're like most people, the first statement seems to make more sense.³ For similar reasons, people asked to rate how similar Mexico

1. Eleanor Rosch, “Principles of Categorization,” in *Cognition and Categorization*, ed. Eleanor Rosch and Barbara B. Lloyd (Hillsdale, NJ: Lawrence Erlbaum, 1978).
2. George Lakoff, *Women, Fire, and Dangerous Things: What Categories Reveal about the Mind* (Chicago: Chicago University Press, 1987).
3. Jerrold Sadock, “Truth and Approximations,” *Papers from the Third Annual Meeting of the Berkeley Linguistics Society* (1977): 430–439.

is to the United States, gave consistently higher ratings than people asked to rate how similar the United States is to Mexico.⁴

And if that still seems harmless, a study by Rips showed that people were more likely to expect a disease would spread from robins to ducks on an island, than from ducks to robins.⁵ Now this is not a *logical* impossibility, but in a pragmatic sense, whatever difference separates a duck from a robin and would make a disease less likely to spread from a duck to a robin, must also be a difference between a robin and a duck, and would make a disease less likely to spread from a robin to a duck.

Yes, you can come up with rationalizations, like “Well, there could be more neighboring species of the robins, which would make the disease more likely to spread initially, etc.,” but be careful not to try too hard to rationalize the probability ratings of subjects who didn’t even realize there was a comparison going on. And don’t forget that Mexico is more similar to the United States than the United States is to Mexico, and that 98 is closer to 100 than 100 is to 98. A simpler interpretation is that people are using the (demonstrated) similarity heuristic as a proxy for the probability that a disease spreads, and this heuristic is (demonstrably) asymmetrical.

Kansas is unusually close to the center of the United States, and Alaska is unusually far from the center of the United States; so Kansas is probably closer to most places in the US and Alaska is probably farther. It does not follow, however, that Kansas is closer to Alaska than is Alaska to Kansas. But people seem to reason (metaphorically speaking) as if closeness is an inherent property of Kansas and distance is an inherent property of Alaska; so that Kansas is still close, even to Alaska; and Alaska is still distant, even from Kansas.

So once again we see that Aristotle’s notion of categories—logical classes with membership determined by a collection of properties that are individually strictly necessary, and together

4. Amos Tversky and Itamar Gati, “Studies of Similarity,” in *Cognition and Categorization*, ed. Eleanor Rosch and Barbara Lloyd (Hillsdale, NJ: Lawrence Erlbaum Associates, Inc., 1978), 79–98.

5. Lance J. Rips, “Inductive Judgments about Natural Categories,” *Journal of Verbal Learning and Verbal Behavior* 14 (1975): 665–681.

strictly sufficient—is not a good model of human cognitive psychology. (Science’s view has changed somewhat over the last 2350 years? Who would’ve thought?) We don’t even reason as if set membership is a true-or-false property: statements of set membership can be more or less true. (Note: This is *not* the same thing as being more or less probable.)

One more reason not to pretend that you, or anyone else, is *really* going to treat words as Aristotelian logical classes.

The Cluster Structure of Thingspace

The notion of a “configuration space” is a way of translating object *descriptions* into object *positions*. It may seem like blue is “closer” to blue-green than to red, but how much closer? It’s hard to answer that question by just staring at the colors. But it helps to know that the (proportional) color coordinates in RGB are 0:0:5, 0:3:2, and 5:0:0. It would be even clearer if plotted on a 3D graph.

In the same way, you can see a robin as a robin—brown tail, red breast, standard robin shape, maximum flying speed when unladen, its species-typical DNA and individual alleles. Or you could see a robin as a single point in a configuration space whose dimensions described everything we knew, or could know, about the robin.

A robin is bigger than a virus, and smaller than an aircraft carrier—that might be the “volume” dimension. Likewise a robin weighs more than a hydrogen atom, and less than a galaxy; that might be the “mass” dimension. Different robins will have strong correlations between “volume” and “mass,” so the robin-points will be lined up in a fairly linear string, in those two dimensions—but the correlation won’t be exact, so we do need two separate dimensions.

This is the benefit of viewing robins as points in space: You couldn’t see the linear lineup as easily if you were just imagining the robins as cute little wing-flapping creatures.

A robin’s DNA is a highly multidimensional variable, but you can still think of it as part of a robin’s location in thingspace—millions of quaternary coordinates, one coordinate for each DNA base—or maybe a more sophisticated view than that. The shape of the robin, and its color (surface reflectance), you can likewise think of as part of the robin’s position in thingspace, even though they aren’t *single* dimensions.

Just like the coordinate point 0:0:5 contains the same information as the actual HTML color blue, we shouldn’t actually



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lose information when we see robins as points in space. We believe the same statement about the robin's mass whether we visualize a robin balancing the scales opposite a 0.07-kilogram weight, or a robin-point with a mass-coordinate of +70.

We can even imagine a configuration space with one or more dimensions for every distinct characteristic of an object, so that the *position* of an object's point in this space corresponds to *all* the information in the real object itself. Rather redundantly represented, too—dimensions would include the mass, the volume, and the density.

If you think that's extravagant, quantum physicists use an *infinite-dimensional* configuration space, and a single point in that space describes the location of every particle in the universe. So we're actually being comparatively conservative in our visualization of *thingspace*—a point in thingspace describes just one object, not the entire universe.

If we're not sure of the robin's exact mass and volume, then we can think of a little cloud in thingspace, a *volume of uncertainty*, within which the robin might be. The density of the cloud is the density of our belief that the robin has that particular mass and volume. If you're more sure of the robin's density than of its mass and volume, your probability-cloud will be highly concentrated in the density dimension, and concentrated around a slanting line in the subspace of mass/volume. (Indeed, the cloud here is actually a surface, because of the relation $VD = M$.)

"Radial categories" are how cognitive psychologists describe the non-Aristotelian boundaries of words. The central "mother" conceives her child, gives birth to it, and supports it. Is an egg donor who never sees her child a mother? She is the "genetic mother." What about a woman who is implanted with a foreign embryo and bears it to term? She is a "surrogate mother." And the woman who raises a child that isn't hers genetically? Why, she's an "adoptive mother." The Aristotelian syllogism would run, "Humans have ten fingers, Fred has nine fingers, therefore Fred is not a human," but the way we actually think is "Humans have ten fingers, Fred is a human, therefore Fred is a 'nine-fingered human.'"

We can think about the radial-ness of categories in intensional terms, as described above—properties that are usually present, but optionally absent. If we thought about the intension of the word “mother,” it might be like a distributed glow in thingspace, a glow whose intensity matches the degree to which that volume of thingspace matches the category “mother.” The glow is concentrated in the center of genetics and birth and child-raising; the volume of egg donors would also glow, but less brightly.

Or we can think about the radial-ness of categories extensionally. Suppose we mapped all the birds in the world into thingspace, using a distance metric that corresponds as well as possible to perceived similarity in humans: A robin is more similar to another robin, than either is similar to a pigeon, but robins and pigeons are all more similar to each other than either is to a penguin, et cetera.

Then the center of all birdness would be densely populated by many neighboring tight clusters, robins and sparrows and canaries and pigeons and many other species. Eagles and falcons and other large predatory birds would occupy a nearby cluster. Penguins would be in a more distant cluster, and likewise chickens and ostriches.

The result might look, indeed, something like an astronomical cluster: many galaxies orbiting the center, and a few outliers.

Or we could think simultaneously about both the intension of the cognitive category “bird,” and its extension in real-world birds: The central clusters of robins and sparrows glowing brightly with highly typical birdness; satellite clusters of ostriches and penguins glowing more dimly with atypical birdness, and Abraham Lincoln a few megaparsecs away and glowing not at all.

I prefer that last visualization—the glowing points—because as I see it, the structure of the cognitive intension followed from the extensional cluster structure. First came the structure-in-the-world, the empirical distribution of birds over thingspace; then, by observing it, we formed a category whose intensional glow roughly overlays this structure.

This gives us yet another view of why words are not Aristotelian classes: the empirical clustered structure of the real universe is not so crystalline. A natural cluster, a group of things highly similar to each other, may have *no* set of necessary and sufficient properties—*no* set of characteristics that all group members have, and no non-members have.

But even if a category is irrecoverably blurry and bumpy, there's no need to panic. I would not object if someone said that birds are "feathered flying things." *But penguins don't fly!*—well, fine. The usual rule has an exception; it's not the end of the world. Definitions can't be expected to exactly match the empirical structure of thingspace in any event, because the map is smaller and much less complicated than the territory. The point of the definition "feathered flying things" is to lead the listener to the bird cluster, not to give a total description of every existing bird down to the molecular level.

When you draw a boundary around a group of extensional points *empirically* clustered in thingspace, you may find at least one exception to every simple intensional rule you can invent.

But if a definition works well enough in practice to point out the intended empirical cluster, objecting to it may justly be called "nitpicking."

Disguised Queries

Imagine that you have a peculiar job in a peculiar factory: Your task is to take objects from a mysterious conveyor belt, and sort the objects into two bins. When you first arrive, Susan the Senior Sorter explains to you that blue egg-shaped objects are called “bleggs” and go in the “blegg bin,” while red cubes are called “rubes” and go in the “rube bin.”



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Once you start working, you notice that bleggs and rubes differ in ways besides color and shape. Bleggs have fur on their surface, while rubes are smooth. Bleggs flex slightly to the touch; rubes are hard. Bleggs are opaque, the rube’s surface slightly translucent.

Soon after you begin working, you encounter a blegg shaded an unusually dark blue—in fact, on closer examination, the color proves to be purple, halfway between red and blue.

Yet wait! Why are you calling this object a “blegg”? A “blegg” was originally defined as blue and egg-shaped—the qualification of blueness appears in the very name “blegg,” in fact. This object is not blue. One of the necessary qualifications is missing; you should call this a “purple egg-shaped object,” not a “blegg.”

But it so happens that, in addition to being purple and egg-shaped, the object is also furred, flexible, and opaque. So when you saw the object, you thought, “Oh, a strangely colored blegg.” It certainly isn’t a rube ...right?

Still, you aren’t quite sure what to do next. So you call over Susan the Senior Sorter.

“Oh, yes, it’s a blegg,” Susan says, “you can put it in the blegg bin.”

You start to toss the purple blegg into the blegg bin, but pause for a moment. “Susan,” you say, “how do you *know* this is a blegg?”

Susan looks at you oddly. "Isn't it obvious? This object may be purple, but it's still egg-shaped, furred, flexible, and opaque, like all the other bleggs. You've got to expect a few color defects. Or is this one of those philosophical conundrums, like 'How do you know the world wasn't created five minutes ago complete with false memories?' In a philosophical sense I'm not *absolutely certain* that this is a blegg, but it seems like a good guess."

"No, I mean ..." You pause, searching for words. "Why is there a blegg bin and a rube bin? What's the *difference* between bleggs and rubes?"

"Bleggs are blue and egg-shaped, rubes are red and cube-shaped," Susan says patiently. "You got the standard orientation lecture, right?"

"Why do bleggs and rubes *need* to be sorted?"

"Er ...because otherwise they'd be all mixed up?" says Susan. "Because nobody will pay us to sit around all day and *not* sort bleggs and rubes?"

"Who originally determined that the first blue egg-shaped object was a 'blegg,' and how did they determine that?"

Susan shrugs. "I suppose you could just as easily call the red cube-shaped objects 'bleggs' and the blue egg-shaped objects 'rubes,' but it seems easier to remember this way."

You think for a moment. "Suppose a completely mixed-up object came off the conveyor. Like, an orange sphere-shaped furred translucent object with writhing green tentacles. How could I tell whether it was a blegg or a rube?"

"Wow, no one's ever found an object *that* mixed up," says Susan, "but I guess we'd take it to the sorting scanner."

"How does the sorting scanner work?" you inquire. "X-rays? Magnetic resonance imaging? Fast neutron transmission spectroscopy?"

"I'm told it works by Bayes's Rule, but I don't quite understand how," says Susan. "I like to say it, though. Bayes Bayes Bayes Bayes Bayes."

"What does the sorting scanner *tell* you?"

"It tells you whether to put the object into the blegg bin or the rube bin. That's why it's called a sorting scanner."

At this point you fall silent.

"Incidentally," Susan says casually, "it may interest you to know that bleggs contain small nuggets of vanadium ore, and rubes contain shreds of palladium, both of which are useful industrially."

"Susan, you are pure evil."

"Thank you."

So now it seems we've discovered the heart and essence of bleggness: a blegg is an object that contains a nugget of vanadium ore. Surface characteristics, like blue color and furredness, do not *determine* whether an object is a blegg; surface characteristics only matter because they help you *infer* whether an object is a blegg, that is, whether the object contains vanadium.

Containing vanadium is a necessary and sufficient definition: all bleggs contain vanadium and everything that contains vanadium is a blegg: "blegg" is just a shorthand way of saying "vanadium-containing object." Right?

Not so fast, says Susan: Around 98% of bleggs contain vanadium, but 2% contain palladium instead. To be precise (Susan continues) around 98% of blue egg-shaped furred flexible opaque objects contain vanadium. For unusual bleggs, it may be a different percentage: 95% of purple bleggs contain vanadium, 92% of hard bleggs contain vanadium, etc.

Now suppose you find a blue egg-shaped furred flexible opaque object, an ordinary blegg in every visible way, and just for kicks you take it to the sorting scanner, and the scanner says "palladium"—this is one of the rare 2%. Is it a blegg?

At first you might answer that, since you intend to throw this object in the rube bin, you might as well call it a "rube." However, it turns out that almost all bleggs, if you switch off

the lights, glow faintly in the dark, while almost all rubes do not glow in the dark. And the percentage of bleggs that glow in the dark is not significantly different for blue egg-shaped furred flexible opaque objects that contain palladium, instead of vanadium. Thus, if you want to guess whether the object glows like a blegg, or remains dark like a rube, you should guess that it glows like a blegg.

So is the object *really* a blegg or a rube?

On one hand, you'll throw the object in the rube bin no matter what else you learn. On the other hand, if there are any unknown characteristics of the object you need to infer, you'll infer them as if the object were a blegg, not a rube—group it into the similarity cluster of blue egg-shaped furred flexible opaque things, and not the similarity cluster of red cube-shaped smooth hard translucent things.

The question “Is this object a blegg?” may stand in for different queries on different occasions.

If it weren't standing in for *some* query, you'd have no reason to care.

Is atheism a “religion”? Is transhumanism a “cult”? People who argue that atheism is a religion “because it states beliefs about God” are really trying to argue (I think) that the reasoning methods used in atheism are on a par with the reasoning methods used in religion, or that atheism is no safer than religion in terms of the probability of causally engendering violence, etc.... What's really at stake is an atheist's claim of substantial difference and superiority relative to religion, which the religious person is trying to reject by denying the difference rather than the superiority(!).

But that's not the a priori irrational part: The a priori irrational part is where, in the course of the argument, someone pulls out a dictionary and looks up the definition of “atheism” or “religion.” (And yes, it's just as silly whether an atheist or religiousist does it.) How could a dictionary *possibly* decide whether an empirical cluster of atheists is really substantially different from an empirical cluster of theologians? How can reality vary with the meaning of a word? The points in thingspace don't move around when we redraw a boundary.

But people often don't *realize* that their argument about where to draw a definitional boundary, is really a dispute over whether to infer a characteristic shared by most things inside an empirical cluster ...

Hence the phrase, "disguised query."

Neural Categories



discussion
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3-c33

In *Disguised Queries*, I talked about a classification task of “bleggs” and “rubes.” The typical blegg is blue, egg-shaped, furred, flexible, opaque, glows in the dark, and contains vanadium. The typical rube is red, cube-shaped, smooth, hard, translucent, unglowing, and contains palladium. For the sake of simplicity, let us forget the characteristics of flexibility/hardness and opaqueness/translucency. This leaves five dimensions in thingspace: color, shape, texture, luminance, and interior.

Suppose I want to create an Artificial Neural Network (ANN) to predict unobserved blegg characteristics from observed blegg characteristics. And suppose I’m fairly naive about ANNs: I’ve read excited popular science books about how neural networks are distributed, emergent, and parallel *just like the human brain!!* but I can’t derive the differential equations for gradient descent in a non-recurrent multilayer network with sigmoid units (which is actually a lot easier than it sounds).

Then I might design a neural network that looks something like Figure 161.1.

Network 1 is for classifying bleggs and rubes. But since “blegg” is an unfamiliar and synthetic concept, I’ve also included a similar Network 1b in Figure 161.2 for distinguishing humans from Space Monsters, with input from Aristotle (“All men are mortal”) and Plato’s Academy (“A featherless biped with broad nails”).

A neural network needs a learning rule. The obvious idea is that when two nodes are often active at the same time, we should strengthen the connection between them—this is one of the first rules ever proposed for training a neural network, known as Hebb’s Rule.

Thus, if you often saw things that were both blue and furred—thus simultaneously activating the “color” node in the + state and the “texture” node in the + state—the connection would

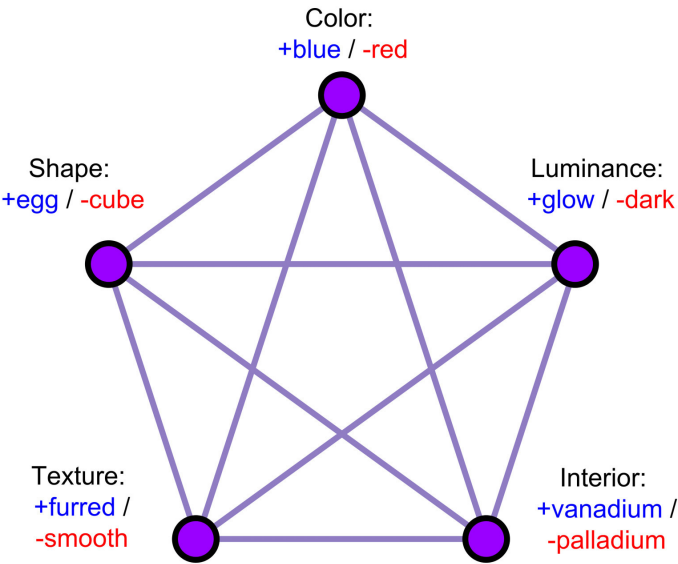


Figure 161.1 Network 1

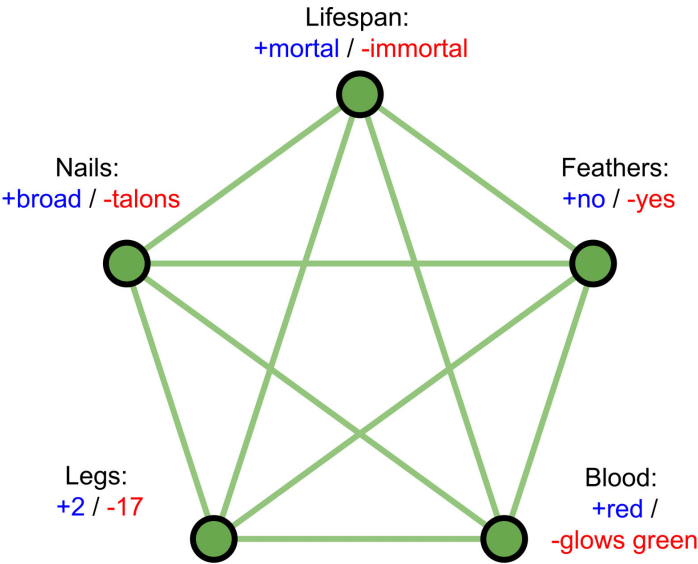


Figure 161.2 Network 1b

strengthen between color and texture, so that + colors activated + textures, and vice versa. If you saw things that were blue and egg-shaped and vanadium-containing, that would strengthen positive mutual connections between color and shape and interior.

Let's say you've already seen plenty of bleggs and rubes come off the conveyor belt. But now you see something that's furred, egg-shaped, and—gasp!—reddish purple (which we'll model as a “color” activation level of $-2/3$). You haven't yet tested the luminance, or the interior. What to predict, what to predict?

What happens then is that the activation levels in Network 1 bounce around a bit. Positive activation flows to luminance from shape, negative activation flows to interior from color, negative activation flows from interior to luminance ...Of course all these messages are passed in *parallel!!* and *asynchronously!!* just like the human brain ...

Finally Network 1 settles into a stable state, which has high positive activation for “luminance” and “interior.” The network may be said to “expect” (though it has not yet seen) that the object will glow in the dark, and that it contains vanadium.

And lo, Network 1 exhibits this behavior even though there's no explicit node that says whether the object is a blegg or not. The judgment is *implicit in the whole network!!* Bleggness is an *attractor!!* which arises as the result of *emergent behavior!!* from the *distributed!!* learning rule.

Now in real life, this kind of network design—however faddish it may sound—runs into *all sorts* of problems. Recurrent networks don't always settle right away: They can oscillate, or exhibit chaotic behavior, or just take a very long time to settle down. This is a Bad Thing when you see something big and yellow and striped, and you have to wait five minutes for your distributed neural network to settle into the “tiger” attractor. Asynchronous and parallel it may be, but it's not real-time.

And there are other problems, like double-counting the evidence when messages bounce back and forth: If you suspect that an object glows in the dark, your suspicion will activate

belief that the object contains vanadium, which in turn will activate belief that the object glows in the dark.

Plus if you try to scale up the Network 1 design, it requires $O(N^2)$ connections, where N is the total number of observables.

So what might be a more realistic neural network design?

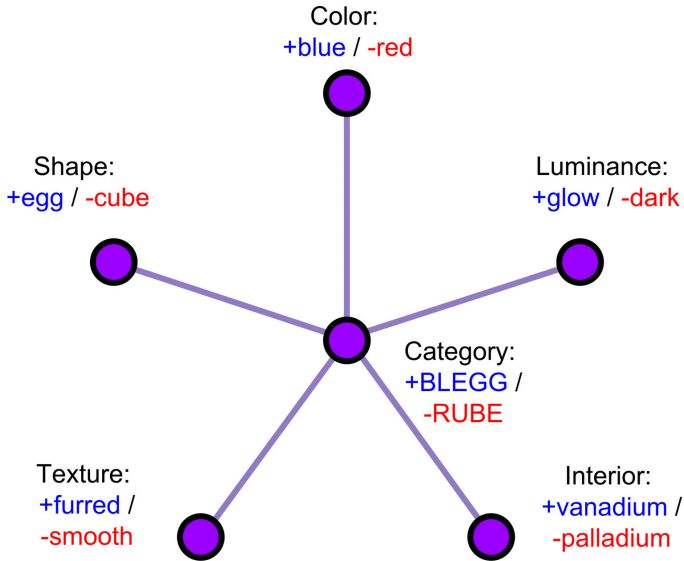


Figure 161.3 Network 2

In Network 2 of Figure 161.3, a wave of activation converges on the central node from any clamped (observed) nodes, and then surges back out again to any unclamped (unobserved) nodes. Which means we can compute the answer in one step, rather than waiting for the network to settle—an important requirement in biology when the neurons only run at 20Hz. And the network architecture scales as $O(N)$, rather than $O(N^2)$.

Admittedly, there are some things you can notice more easily with the first network architecture than the second. Network 1 has a direct connection between every two nodes. So if red objects *never* glow in the dark, but red furred objects usually have the other blegg characteristics like egg-shape and vanadium, Network 1 can easily represent this: it just takes a very strong direct negative connection from color to luminance, but

more powerful positive connections from texture to all other nodes except luminance.

Nor is this a “special exception” to the general rule that bleggs glow—remember, in Network 1, there is no unit that represents blegg-ness; blegg-ness emerges as an attractor in the distributed network.

So yes, those $O(N^2)$ connections were buying us something. But not very much. Network 1 is not *more* useful on most real-world problems, where you rarely find an animal stuck halfway between being a cat and a dog.

(There are also facts that you can’t easily represent in Network 1 *or* Network 2. Let’s say sea-blue color and spheroid shape, when found together, always indicate the presence of palladium; but when found individually, without the other, they are each very strong evidence for vanadium. This is hard to represent, in either architecture, without extra nodes. Both Network 1 and Network 2 embody implicit assumptions about what kind of environmental structure is likely to exist; the ability to read this off is what separates the adults from the babes, in machine learning.)

Make no mistake: Neither Network 1 nor Network 2 is biologically realistic. *But* it still seems like a fair guess that however the brain really works, it is in some sense closer to Network 2 than Network 1. Fast, cheap, scalable, works well to distinguish dogs and cats: natural selection goes for that sort of thing like water running down a fitness landscape.

It seems like an ordinary enough task to classify objects as either bleggs or rubes, tossing them into the appropriate bin. But would you notice if sea-blue objects never glowed in the dark?

Maybe, if someone presented you with twenty objects that were alike only in being sea-blue, and then switched off the light, and none of the objects glowed. If you got hit over the head with it, in other words. Perhaps by presenting you with all these sea-blue objects in a group, your brain forms a new subcategory, and can detect the “doesn’t glow” characteristic within that subcategory. But you probably wouldn’t notice if the sea-blue objects were scattered among a hundred other

bleggs and rubes. It wouldn't be *easy* or *intuitive* to notice, the way that distinguishing cats and dogs is easy and intuitive.

Or: "Socrates is human, all humans are mortal, therefore Socrates is mortal." How did Aristotle know that Socrates was human? Well, Socrates had no feathers, and broad nails, and walked upright, and spoke Greek, and, well, was generally shaped like a human and acted like one. So the brain decides, once and for all, that Socrates is human; and from there, infers that Socrates is mortal like all other humans thus yet observed. It doesn't seem easy or intuitive to ask how much wearing clothes, as opposed to using language, is associated with mortality. Just, "things that wear clothes and use language are human" and "humans are mortal."

Are there biases associated with trying to classify things into categories once and for all? Of course there are. See e.g. *Cultish Countercultishness*, vol. II, p. 273.

How An Algorithm Feels From Inside



discussion
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“If a tree falls in the forest, and no one hears it, does it make a sound?” I remember seeing an actual argument get started on this subject—a fully naive argument that went nowhere near Berkeleian subjectivism. Just:

“It makes a sound, just like any other falling tree!”

“But how can there be a sound that no one hears?”

The standard rationalist view would be that the first person is speaking as if “sound” means acoustic vibrations in the air; the second person is speaking as if “sound” means an auditory experience in a brain. If you ask “Are there acoustic vibrations?” or “Are there auditory experiences?” the answer is at once obvious. And so the argument is really about the definition of the word “sound.”

I think the standard analysis is essentially correct. So let’s accept that as a premise, and ask: Why do people get into such arguments? What’s the underlying psychology?

A key idea of the heuristics and biases program is that mistakes are often more revealing of cognition than correct answers. Getting into a heated dispute about whether, if a tree falls in a deserted forest, it makes a sound, is traditionally considered a mistake.

So what kind of mind design corresponds to that error?

In *Disguised Queries* I introduced the blegg/rube classification task, in which Susan the Senior Sorter explains that your job is to sort objects coming off a conveyor belt, putting the blue eggs or “bleggs” into one bin, and the red cubes or “rubes” into the rube bin. This, it turns out, is because bleggs contain small nuggets of vanadium ore, and rubes contain small shreds of palladium, both of which are useful industrially.

Except that around 2% of blue egg-shaped objects contain palladium instead. So if you find a blue egg-shaped thing that

contains palladium, should you call it a “rube” instead? You’re going to put it in the rube bin—why not call it a “rube”?

But when you switch off the light, nearly all bleggs glow faintly in the dark. And blue egg-shaped objects that contain palladium are just as likely to glow in the dark as any other blue egg-shaped object.

So if you find a blue egg-shaped object that contains palladium and you ask “Is it a blegg?,” the answer depends on what you have to do with the answer. If you ask “Which bin does the object go in?,” then you choose as if the object is a rube. But if you ask “If I turn off the light, will it glow?,” you predict as if the object is a blegg. In one case, the question “Is it a blegg?” stands in for the disguised query, “Which bin does it go in?” In the other case, the question “Is it a blegg?” stands in for the disguised query, “Will it glow in the dark?”

Now suppose that you have an object that is blue and egg-shaped and contains palladium; and you have already observed that it is furred, flexible, opaque, and glows in the dark.

This answers *every* query, observes every observable introduced. There’s nothing left for a disguised query to stand *for*.

So why might someone feel an impulse to go on arguing whether the object is *really* a blegg?

These diagrams from Neural Categories show two different neural networks that might be used to answer questions about bleggs and rubes. Network 1 (Figure 162.1) has a number of disadvantages—such as potentially oscillating/chaotic behavior, or requiring $O(N^2)$ connections—but Network 1’s structure does have one major advantage over Network 2: every unit in the network corresponds to a testable query. If you observe every observable, clamping every value, there are no units in the network left over.

Network 2 (Figure 162.2), however, is a far better candidate for being something vaguely like how the human brain works: It’s fast, cheap, scalable—and has an extra dangling unit in the center, whose activation can still vary, even after we’ve observed every single one of the surrounding nodes.

Which is to say that even after you know whether an object is blue or red, egg or cube, furred or smooth, bright or dark,

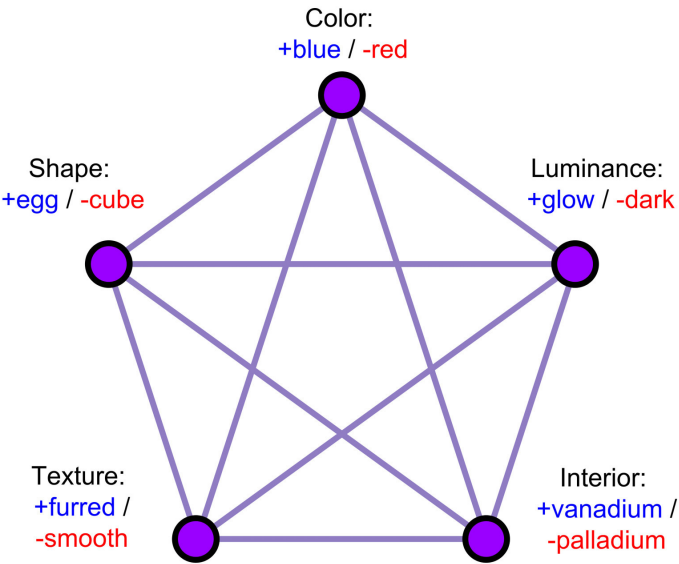


Figure 162.1 Network 1

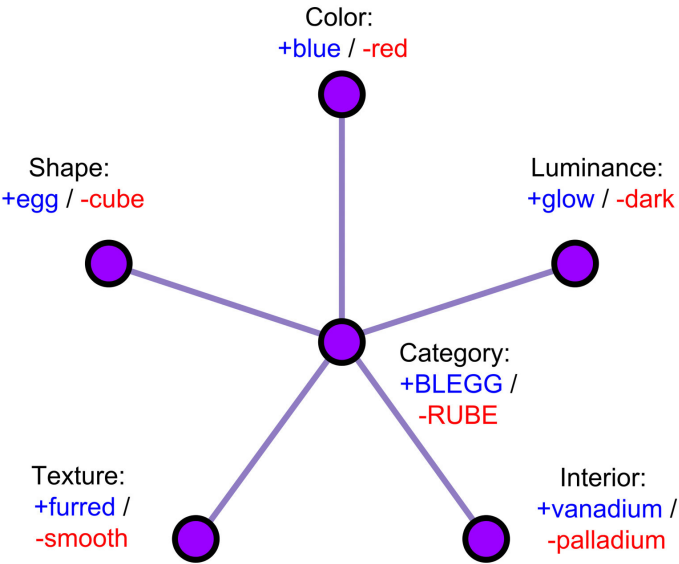


Figure 162.2 Network 2

and whether it contains vanadium or palladium, it *feels* like there's a leftover, unanswered question: *But is it really a blegg?*

Usually, in our daily experience, acoustic vibrations and auditory experience go together. But a tree falling in a deserted forest unbundles this common association. And even after you know that the falling tree creates acoustic vibrations but not auditory experience, it *feels* like there's a leftover question: *Did it make a sound?*

We know where Pluto is, and where it's going; we know Pluto's shape, and Pluto's mass—but is it a planet?

Now remember: When you look at Network 2, as I've laid it out here, you're seeing the algorithm from the outside. People don't think to themselves, "Should the central unit fire, or not?" any more than you think "Should neuron #12,234,320,242 in my visual cortex fire, or not?"

It takes a deliberate effort to visualize your brain from the outside—and then you still don't see your actual brain; you imagine what you *think* is there. Hopefully based on science, but regardless, you don't have any direct access to neural network structures from introspection. That's why the ancient Greeks didn't invent computational neuroscience.

When you look at Network 2, you are seeing from the *outside*; but the way that neural network structure feels from the *inside*, if you yourself *are* a brain running that algorithm, is that even after you know every characteristic of the object, you still find yourself wondering: "But is it a blegg, or not?"

This is a great gap to cross, and I've seen it stop people in their tracks. Because we don't instinctively see our intuitions as "intuitions," we just see them as the world. When you look at a green cup, you don't think of yourself as seeing a picture reconstructed in your visual cortex—although that *is* what you are seeing—you just see a green cup. You think, "Why, look, this cup is green," not, "The picture in my visual cortex of this cup is green."

And in the same way, when people argue over whether the falling tree makes a sound, or whether Pluto is a planet, they don't see themselves as arguing over whether a categorization

should be active in their neural networks. It seems like either the tree makes a sound, or not.

We know where Pluto is, and where it's going; we know Pluto's shape, and Pluto's mass—but is it a planet? And yes, there were people who said this was a fight over definitions—but even that is a Network 2 sort of perspective, because you're arguing about how the central unit ought to be wired up. If you were a mind constructed along the lines of Network 1, you wouldn't say "It depends on how you define 'planet,'" you would just say, "Given that we know Pluto's orbit and shape and mass, there is no question left to ask." Or, rather, that's how it would *feel*—it would *feel* like there was no question left—if you were a mind constructed along the lines of Network 1.

Before you can question your intuitions, you have to realize that what your mind's eye is looking at *is* an intuition—some cognitive algorithm, as seen from the inside—rather than a direct perception of the Way Things Really Are.

People cling to their intuitions, I think, not so much because they believe their cognitive algorithms are perfectly reliable, but because they can't see their intuitions *as the way their cognitive algorithms happen to look from the inside*.

And so everything you try to say about how the native cognitive algorithm goes astray, ends up being contrasted to their direct perception of the Way Things Really Are—and discarded as obviously wrong.

Disputing Definitions

I have watched more than one conversation—even conversations supposedly about cognitive science—go the route of disputing over definitions. Taking the classic example to be “If a tree falls in a forest, and no one hears it, does it make a sound?” the dispute often follows a course like this:



discussion
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If a tree falls in the forest, and no one hears it, does it make a sound?

ALBERT: “Of course it does. What kind of silly question is that? Every time I’ve listened to a tree fall, it made a sound, so I’ll guess that other trees falling also make sounds. I don’t believe the world changes around when I’m not looking.”

BARRY: “Wait a minute. If no one hears it, how can it be a sound?”

In this example, Barry is arguing with Albert because of a genuinely different intuition about what constitutes a sound. But there’s more than one way the Standard Dispute can start. Barry could have a motive for rejecting Albert’s conclusion. Or Barry could be a skeptic who, upon hearing Albert’s argument, reflexively scrutinized it for possible logical flaws; and then, on finding a counterargument, automatically accepted it without applying a second layer of search for a counter-counterargument; thereby arguing himself into the opposite position. This doesn’t require that Barry’s *prior* intuition—the intuition Barry would have had, if we’d asked him before Albert spoke—differs from Albert’s.

Well, if Barry didn’t have a differing intuition before, he sure has one now.

ALBERT: “What do you mean, there’s no sound?
The tree’s roots snap, the trunk comes crashing

down and hits the ground. This generates vibrations that travel through the ground and the air. That's where the energy of the fall goes, into heat and sound. Are you saying that if people leave the forest, the tree violates Conservation of Energy?"

BARRY: "But no one hears anything. If there are no humans in the forest, or, for the sake of argument, anything else with a complex nervous system capable of 'hearing,' then no one hears a sound."

Albert and Barry recruit arguments that *feel* like support for their respective positions, describing in more detail the thoughts that caused their "sound"-detectors to fire or stay silent. But so far the conversation has still focused on the forest, rather than definitions. And note that they don't actually disagree on anything that happens in the forest.

ALBERT: "This is the dumbest argument I've ever been in. You're a niddlewicking fallumphing pickleplumber."

BARRY: "Yeah? Well, you look like your face caught on fire and someone put it out with a shovel."

Insult has been proffered and accepted; now neither party can back down without losing face. Technically, this isn't part of the *argument*, as rationalists account such things; but it's such an important part of the Standard Dispute that I'm including it anyway.

ALBERT: "The tree produces acoustic vibrations. By definition, that is a sound."

BARRY: "No one hears anything. By definition, that is not a sound."

The argument starts shifting to focus on definitions. Whenever you feel tempted to say the words "by definition" in an argument that is not literally about pure mathematics, remember that anything which is true "by definition" is true in all

possible worlds, and so observing its truth can never constrain *which* world you live in.

ALBERT: "My computer's microphone can record a sound without anyone being around to hear it, store it as a file, and it's called a 'sound file.' And what's stored in the file is the pattern of vibrations in air, not the pattern of neural firings in anyone's brain. 'Sound' means a pattern of vibrations."

Albert deploys an argument that *feels* like support for the word "sound" *having a particular meaning*. This is a different kind of question from whether acoustic vibrations take place in a forest—but the shift usually passes unnoticed.

BARRY: "Oh, yeah? Let's just see if the dictionary agrees with you."

There's a lot of things I could be curious about in the falling-tree scenario. I could go into the forest and look at trees, or learn how to derive the wave equation for changes of air pressure, or examine the anatomy of an ear, or study the neuroanatomy of the auditory cortex. Instead of doing any of these things, I am to consult a dictionary, apparently. Why? Are the editors of the dictionary expert botanists, expert physicists, expert neuroscientists? Looking in an encyclopedia might make sense, but why a *dictionary*?

ALBERT: "Hah! Definition 2c in Merriam-Webster: 'Sound: Mechanical radiant energy that is transmitted by longitudinal pressure waves in a material medium (as air).'"

BARRY: "Hah! Definition 2b in Merriam-Webster: 'Sound: The sensation perceived by the sense of hearing.'"

ALBERT AND BARRY, CHORUS: "Consarned dictionary! This doesn't help at all!"

Dictionary editors are historians of usage, not legislators of language. Dictionary editors find words in current usage, then write down the words next to (a small part of) what people seem to mean by them. If there's more than one usage, the editors write down more than one definition.

ALBERT: "Look, suppose that I left a microphone in the forest and recorded the pattern of the acoustic vibrations of the tree falling. If I played that back to someone, they'd call it a 'sound'! That's the common usage! Don't go around making up your own wacky definitions!"

BARRY: "One, I can define a word any way I like so long as I use it consistently. Two, the meaning I gave was in the dictionary. Three, who gave you the right to decide what is or isn't common usage?"

There's quite a lot of rationality errors in the Standard Dispute. Some of them I've already covered, and some of them I've yet to cover; likewise the remedies.

But for now, I would just like to point out—in a mournful sort of way—that Albert and Barry seem to agree on virtually every question of what is *actually* going on inside the forest, and yet it doesn't seem to generate any feeling of agreement.

Arguing about definitions is a garden path; people wouldn't go down the path if they saw at the outset where it led. If you asked Albert (Barry) why he's still arguing, he'd probably say something like: "Barry (Albert) is trying to sneak in his own definition of 'sound,' the scurvey scoundrel, to support his ridiculous point; and I'm here to defend the standard definition."

But suppose I went back in time to before the start of the argument:

(Eliezer appears from nowhere in a peculiar conveyance that looks just like the time machine from the original The Time Machine movie.)

BARRY: "Gosh! A time traveler!"

ELIEZER: "I am a traveler from the future! Hear my words! I have traveled far into the past—around fifteen minutes—"

ALBERT: "Fifteen *minutes?*"

ELIEZER: "—to bring you this message!"

(There is a pause of mixed confusion and expectancy.)

ELIEZER: "Do you think that 'sound' should be defined to require both acoustic vibrations (pressure waves in air) and also auditory experiences (someone to listen to the sound), or should 'sound' be defined as meaning only acoustic vibrations, or only auditory experience?"

BARRY: "You went back in time to ask us *that?*"

ELIEZER: "My purposes are my own! Answer!"

ALBERT: "Well...I don't see why it would matter. You can pick any definition so long as you use it consistently."

BARRY: "Flip a coin. Er, flip a coin twice."

ELIEZER: "Personally I'd say that if the issue arises, both sides should switch to describing the event in unambiguous lower-level constituents, like acoustic vibrations or auditory experiences. Or each side could designate a new word, like 'alberzle' and 'bargulum,' to use for what they respectively used to call 'sound'; and then both sides could use the new words consistently. That way neither side has to back down or lose face, but they can still communicate. And of course you should try to keep track, at all times, of some testable proposition that the argument is actually about. Does that sound right to you?"

ALBERT: "I guess..."

BARRY: "Why are we talking about this?"

ELIEZER: "To preserve your friendship against a contingency you will, now, never know. For the future has already changed!"

(Eliezer and the machine vanish in a puff of smoke.)

BARRY: "Where were we again?"

ALBERT: "Oh, yeah: If a tree falls in the forest,
and no one hears it, does it make a sound?"

BARRY: "It makes an alberzle but not a bargu-
lum. What's the next question?"

This remedy doesn't destroy *every* dispute over categoriza-
tions. But it destroys a substantial fraction.

Feel the Meaning

When I hear someone say, “Oh, look, a butterfly,” the spoken phonemes “butterfly” enter my ear and vibrate on my ear drum, being transmitted to the cochlea, tickling auditory nerves that transmit activation spikes to the auditory cortex, where phoneme processing begins, along with recognition of words, and reconstruction of syntax (a by no means serial process), and all manner of other complications.

But at the end of the day, or rather, at the end of the second, I am primed to look where my friend is pointing and see a visual pattern that I will recognize as a butterfly; and I would be quite surprised to see a wolf instead.

My friend looks at a butterfly, his throat vibrates and lips move, the pressure waves travel invisibly through the air, my ear hears and my nerves transduce and my brain reconstructs, and lo and behold, I know what my friend is looking at. Isn’t that marvelous? If we didn’t know about the pressure waves in the air, it would be a tremendous discovery in all the newspapers: Humans are telepathic! Human brains can transfer thoughts to each other!

Well, we *are* telepathic, in fact; but magic isn’t exciting when it’s merely *real*, and all your friends can do it too.

Think telepathy is simple? Try building a computer that will be telepathic with you. Telepathy, or “language,” or whatever you want to call our partial thought transfer ability, is more complicated than it looks.

But it would be quite inconvenient to go around thinking, “Now I shall partially transduce some features of my thoughts into a linear sequence of phonemes which will invoke similar thoughts in my conversational partner ...”

So the brain hides the complexity—or rather, never represents it in the first place—which leads people to think some peculiar thoughts about words.



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As I remarked earlier, when a large yellow striped object leaps at me, I think “Yikes! A tiger!” not “Hm...objects with the properties of largeness, yellowness, and stripedness have previously often possessed the properties ‘hungry’ and ‘dangerous,’ and therefore, although it is not logically necessary, *auughhhh CRUNCH CRUNCH GULP.*”

Similarly, when someone shouts “Yikes! A tiger!,” natural selection would not favor an organism that thought, “Hm...I have just heard the syllables ‘Tie’ and ‘Grr’ which my fellow tribe members associate with their internal analogues of my own *tiger* concept, and which they are more likely to utter if they see an object they categorize as *aiiiieee CRUNCH CRUNCH help it’s got my arm CRUNCH GULP.*”

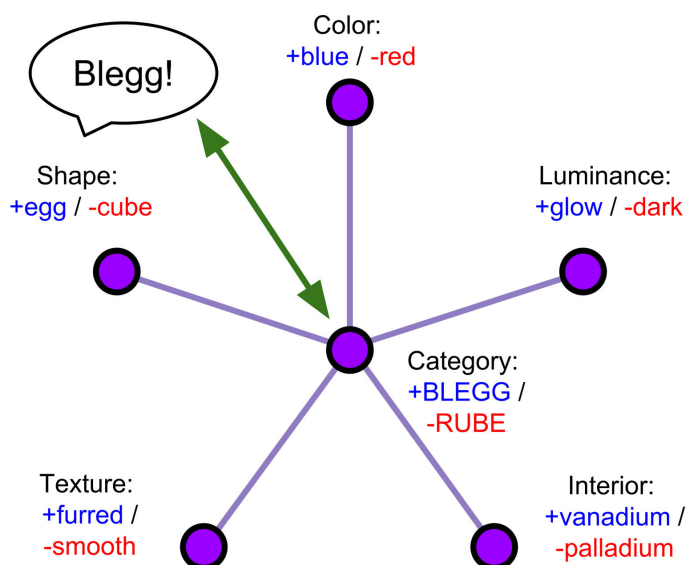


Figure 164.1 Network 3

Considering this as a design constraint on the human cognitive architecture, you wouldn't want *any* extra steps between when your auditory cortex recognizes the syllables “tiger,” and when the tiger concept gets activated.

Going back to the parable of bleggs and rubes, and the centralized network that categorizes quickly and cheaply, you

might visualize a direct connection running from the unit that recognizes the syllable “blegg” to the unit at the center of the blegg network. The central unit, the blegg concept, gets activated almost as soon as you hear Susan the Senior Sorter say, “Blegg!”

Or, for purposes of talking—which also shouldn’t take eons—as soon as you see a blue egg-shaped thing and the central blegg unit fires, you holler “Blegg!” to Susan.

And what that algorithm feels like from inside is that the label, and the concept, are very nearly *identified*; the meaning *feels like* an intrinsic property of the word itself.

The cognoscenti will recognize this as a case of E. T. Jaynes’s “Mind Projection Fallacy.” It feels like a word *has a* meaning, as a property of the word itself; just like how redness is a property of a red apple, or mysteriousness is a property of a mysterious phenomenon.

Indeed, on most occasions, the brain will not distinguish at all between the word and the meaning—only bothering to separate the two while learning a new language, perhaps. And even then, you’ll see Susan pointing to a blue egg-shaped thing and saying “Blegg!,” and you’ll think, *I wonder what “blegg” means*, and not, *I wonder what mental category Susan associates to the auditory label “blegg.”*

Consider, in this light, the part of the Standard Dispute of Definitions where the two parties argue about what the word “sound” *really* means—the same way they might argue whether a particular apple is *really* red or green:

ALBERT: “My computer’s microphone can record a sound without anyone being around to hear it, store it as a file, and it’s called a ‘sound file.’ And what’s stored in the file is the pattern of vibrations in air, not the pattern of neural firings in anyone’s brain. ‘Sound’ means a pattern of vibrations.”

BARRY: “Oh, yeah? Let’s just see if the dictionary agrees with you.”

Albert feels intuitively that the word “sound” *has a meaning* and that the meaning *is* acoustic vibrations. Just as Albert feels that a tree falling in the forest *makes a sound* (rather than causing an event that *matches the sound category*).

Barry likewise *feels* that:

```
sound.meaning == auditory experiences

forest.sound == false.
```

Rather than:

```
myBrain.FindConcept("sound") ==
    concept_AuditoryExperience

concept_AuditoryExperience.match(forest)
    == false.
```

Which is closer to what’s *really* going on; but humans have not evolved to know this, anymore than humans instinctively know the brain is made of neurons.

Albert and Barry’s conflicting intuitions provide the fuel for continuing the argument in the phase of arguing over what the word “sound” means—which *feels* like arguing over a fact like any other fact, like arguing over whether the sky is blue or green.

You may not even notice that anything has gone astray, until you try to perform the rationalist ritual of stating a testable experiment whose result depends on the facts you’re so heatedly disputing ...

The Argument from Common Usage

Part of the Standard Definitional Dispute runs as follows:

ALBERT: “Look, suppose that I left a microphone in the forest and recorded the pattern of the acoustic vibrations of the tree falling. If I played that back to someone, they’d call it a ‘sound’! That’s the common usage! Don’t go around making up your own wacky definitions!”

BARRY: “One, I can define a word any way I like so long as I use it consistently. Two, the meaning I gave was in the dictionary. Three, who gave you the right to decide what is or isn’t common usage?”



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Not all definitional disputes progress as far as recognizing the notion of common usage. More often, I think, someone picks up a dictionary because they believe that words have meanings, and the dictionary faithfully records what this meaning is. Some people even seem to believe that the dictionary *determines* the meaning—that the dictionary editors are the Legislators of Language. Maybe because back in elementary school, their authority-teacher said that they had to obey the dictionary, that it was a mandatory rule rather than an optional one?

Dictionary editors read what other people write, and record what the words seem to mean; they are historians. The Oxford English Dictionary may be *comprehensive*, but never *authoritative*.

But surely there is a social imperative to use words in a commonly understood way? Does not our human telepathy, our valuable power of language, rely on mutual coordination to work? Perhaps we should voluntarily treat dictionary editors as supreme arbiters—even if *they* prefer to think of themselves as historians—in order to maintain the quiet cooperation on which all speech depends.

The phrase “authoritative dictionary” is almost never used correctly, an example of proper usage being *The Authoritative Dictionary of IEEE Standards Terms*. The IEEE is a body of voting members who have a professional need for exact agreement on terms and definitions, and so *The Authoritative Dictionary of IEEE Standards Terms* is actual, negotiated legislation, which exerts whatever authority one regards as residing in the IEEE.

In everyday life, shared language usually does not arise from a deliberate agreement, as of the IEEE. It’s more a matter of infection, as words are invented and diffuse through the culture. (A “meme,” one might say, following Richard Dawkins forty years ago—but you already know what I mean, and if not, you can look it up on Google, and then you too will have been infected.)

Yet as the example of the IEEE shows, agreement on language can also be a cooperatively established public good. If you and I wish to undergo an exchange of thoughts via language, the human telepathy, then it is in our mutual interest that we use the *same* word for similar concepts—preferably, concepts similar to the limit of resolution in our brain’s representation thereof—even though we have no obvious mutual interest in using any *particular* word for a concept.

We have no obvious mutual interest in using the word “oto” to mean sound, or “sound” to mean oto; but we have a mutual interest in using the *same* word, whichever word it happens to be. (Preferably, words we use frequently should be short, but let’s not get into information theory just yet.)

But, while we have a mutual interest, it is not strictly *necessary* that you and I use the similar labels *internally*; it is only convenient. If I know that, to you, “oto” means sound—that is, you associate “oto” to a concept very similar to the one I associate to “sound”—then I can say “Paper crumpling makes a crackling oto.” It requires extra thought, but I can do it if I want.

Similarly, if you say “What is the walking-stick of a bowling ball dropping on the floor?” and I know which concept *you* associate with the syllables “walking-stick,” then I can figure out what you mean. It may require some thought, and give

me pause, because I ordinarily associate “walking-stick” with a different concept. But I can do it just fine.

When humans really *want* to communicate with each other, we’re hard to stop! If we’re stuck on a deserted island with no common language, we’ll take up sticks and draw pictures in sand.

Albert’s appeal to the Argument from Common Usage assumes that agreement on language is a cooperatively established public good. Yet Albert assumes this for the sole purpose of rhetorically accusing Barry of breaking the agreement, and endangering the public good. Now the falling-tree argument has gone all the way from botany to semantics to politics; and so Barry responds by challenging Albert for the authority to define the word.

A rationalist, with the discipline of hugging the query active, would notice that the conversation had gone rather far astray.

Oh, dear reader, is it all really necessary? Albert knows what Barry means by “sound.” Barry knows what Albert means by “sound.” Both Albert and Barry have access to words, such as “acoustic vibrations” or “auditory experience,” which they already associate to the same concepts, and which can describe events in the forest without ambiguity. If they were stuck on a deserted island, trying to communicate with each other, their work would be *done*.

When both sides *know* what the other side *wants* to say, and both sides accuse the other side of defecting from “common usage,” then whatever it is they are about, it is clearly not *working out a way to communicate with each other*. But this is the whole benefit that common usage provides in the first place.

Why would you argue about the meaning of a word, two sides trying to wrest it back and forth? If it’s just a namespace conflict that has gotten blown out of proportion, and nothing more is at stake, then the two sides need merely generate two new words and use them consistently.

Yet often categorizations function as hidden inferences and disguised queries. Is atheism a “religion”? If someone is arguing that the reasoning methods used in atheism are on a par with

the reasoning methods used in Judaism, or that atheism is on a par with Islam in terms of causally engendering violence, then they have a clear argumentative stake in lumping it all together into an indistinct gray blur of “faith.”

Or consider the fight to blend together blacks and whites as “people.” This would not be a time to generate two words—what’s at stake is exactly the idea that you shouldn’t draw a moral distinction.

But once any empirical proposition is at stake, *or* any moral proposition, you can no longer appeal to common usage.

If the question is how to cluster together similar things for purposes of inference, empirical predictions will depend on the answer; which means that definitions can be *wrong*. A conflict of predictions cannot be settled by an opinion poll.

If you want to know whether atheism should be clustered with supernaturalist religions for purposes of some particular empirical inference, the dictionary can’t answer you.

If you want to know whether blacks are people, the dictionary can’t answer you.

If everyone believes that the red light in the sky is Mars the God of War, the dictionary will define “Mars” as the God of War. If everyone believes that fire is the release of phlogiston, the dictionary will define “fire” as the release of phlogiston.

There is an art to using words; even when definitions are not literally true or false, they are often wiser or more foolish. Dictionaries are mere histories of past usage; if you treat them as supreme arbiters of meaning, it binds you to the wisdom of the past, forbidding you to do better.

Though do take care to ensure (if you must depart from the wisdom of the past) that people can figure out what you’re trying to swim.

Empty Labels

Consider (yet again) the Aristotelian idea of categories. Let's say that there's some object with properties *A*, *B*, *C*, *D*, and *E*, or at least it looks *E*-ish.



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FRED: "You mean that thing over there is blue, round, fuzzy, and—"

ME: "In Aristotelian logic, it's not supposed to make a difference what the properties are, or what I call them. That's why I'm just using the letters."

Next, I invent the Aristotelian category "zawa," which describes those objects, all those objects, and only those objects, that have properties *A*, *C*, and *D*.

ME: "Object 1 is zawa, *B*, and *E*."

FRED: "And it's blue—I mean, *A*—too, right?"

ME: "That's implied when I say it's zawa."

FRED: "Still, I'd like you to say it explicitly."

ME: "Okay. Object 1 is *A*, *B*, zawa, and *E*."

Then I add another word, "yokie," which describes all and only objects that are *B* and *E*; and the word "xippo," which describes all and only objects which are *E* but not *D*.

ME: "Object 1 is zawa and yokie, but not xippo."

FRED: "Wait, is it luminescent? I mean, is it *E*?"

ME: "Yes. That is the only possibility on the information given."

FRED: "I'd rather you spelled it out."

ME: "Fine. Object 1 is *A*, zawa, *B*, yokie, *C*, *D*, *E*, and not xippo."

FRED: "Amazing! You can tell all that just by looking?"

Impressive, isn't it? Let's invent even more new words: "Bolo" is A, C , and yokie; "mun" is A, C , and xippo; and "merlacdonian" is bolo and mun.

Pointlessly confusing? I think so too. Let's replace the labels with the definitions:

"Zawa, B , and E " becomes $[A, C, D], B, E$
 "Bolo and A " becomes $[A, C, [B, E]], A$
 "Merlacdonian" becomes $[A, C, [B, E]],$
 $[A, C, [E, \neg D]]$.

And the thing to remember about the Aristotelian idea of categories is that $[A, C, D]$ is the *entire* information of "zawa." It's not just that I can vary the label, but that I can get along just fine without any label at all—the rules for Aristotelian classes work purely on structures like $[A, C, D]$. To call one of these structures "zawa," or attach any other label to it, is a human convenience (or inconvenience) which makes not the slightest difference to the Aristotelian rules.

Let's say that "human" is to be defined as a mortal featherless biped. Then the classic syllogism would have the form:

All $[\text{mortal}, \neg\text{feathers}, \text{bipedal}]$ are mortal.
 Socrates is a $[\text{mortal}, \neg\text{feathers}, \text{bipedal}]$.
 Therefore, Socrates is mortal.

The feat of reasoning looks a lot less impressive now, doesn't it?

Here the *illusion of inference* comes from the labels, which conceal the premises, and pretend to novelty in the conclusion. Replacing labels with definitions reveals the illusion, making visible the tautology's empirical unhelpfulness. You can never say that Socrates is a $[\text{mortal}, \neg\text{feathers}, \text{biped}]$ until you have observed him to be mortal.

There's an idea, which you may have noticed I hate, that "you can define a word any way you like." This idea came from the Aristotelian notion of categories; since, if you follow the Aristotelian rules *exactly* and *without flaw*—which humans never do; Aristotle knew perfectly well that Socrates was human,

even though that wasn't justified under his rules—but, *if* some imaginary nonhuman entity were to follow the rules exactly, they would never arrive at a contradiction. They wouldn't arrive at much of anything: they couldn't say that Socrates is a [mortal, ¬feathers, biped] until they observed him to be mortal.

But it's not so much that labels are *arbitrary* in the Aristotelian system, as that the Aristotelian system works fine without *any labels at all*—it cranks out exactly the same stream of tautologies, they just look a lot less impressive. The labels are only there to create the *illusion* of inference.

So if you're going to have an Aristotelian proverb at all, the proverb should be, not "I can define a word any way I like," nor even, "Defining a word never has any consequences," but rather, "Definitions don't need words."

Taboo Your Words



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In the game Taboo (by Hasbro), the objective is for a player to have their partner guess a word written on a card, without using that word or five additional words listed on the card. For example, you might have to get your partner to say “baseball” without using the words “sport,” “bat,” “hit,” “pitch,” “base” or of course “baseball.”

As soon as I see a problem like that, I at once think, “An artificial group conflict in which you use a long wooden cylinder to whack a thrown spheroid, and then run between four safe positions.” It might not be the most efficient strategy to convey the word “baseball” under the stated rules—that might be, “It’s what the Yankees play”—but the general skill of *blanking a word out of my mind* was one I’d practiced for years, albeit with a different purpose.

In the previous essay we saw how replacing terms with definitions could reveal the empirical unproductivity of the classical Aristotelian syllogism. All humans are mortal (and also, apparently, featherless bipeds); Socrates is human; therefore Socrates is mortal. When we replace the word “human” by its apparent definition, the following underlying reasoning is revealed:

All [mortal, ¬feathers, biped] are mortal;
Socrates is a [mortal, ¬feathers, biped];
Therefore Socrates is mortal.

But the principle of replacing words by definitions applies much more broadly:

ALBERT: “A tree falling in a deserted forest makes a sound.”

BARRY: “A tree falling in a deserted forest does not make a sound.”

Clearly, since one says “sound” and one says “not sound,” we must have a contradiction, right? But suppose that they both dereference their pointers before speaking:

ALBERT: “A tree falling in a deserted forest matches [membership test: this event generates acoustic vibrations].”

BARRY: “A tree falling in a deserted forest does not match [membership test: this event generates auditory experiences].”

Now there is no longer an apparent collision—all they had to do was prohibit themselves from using the word *sound*. If “acoustic vibrations” came into dispute, we would just play Taboo again and say “pressure waves in a material medium”; if necessary we would play Taboo again on the word “wave” and replace it with the wave equation. (Play Taboo on “auditory experience” and you get “That form of sensory processing, within the human brain, that takes as input a linear time series of frequency mixes ...”)

But suppose, on the other hand, that Albert and Barry were to have the argument:

ALBERT: “Socrates matches the concept [membership test: this person will die after drinking hemlock].”

BARRY: “Socrates matches the concept [membership test: this person will not die after drinking hemlock].”

Now Albert and Barry have a substantive clash of expectations; a difference in what they anticipate seeing after Socrates drinks hemlock. But they might not notice this, if they happened to use the same word “human” for their different concepts.

You get a very different picture of what people agree or disagree about, depending on whether you take a label’s-eye-view (Albert says “sound” and Barry says “not sound,” so they must

disagree) or taking the test's-eye-view (Albert's membership test is acoustic vibrations, Barry's is auditory experience).

Get together a pack of *soi-disant* futurists and ask them if they believe we'll have Artificial Intelligence in thirty years, and I would guess that at least half of them will say yes. If you leave it at that, they'll shake hands and congratulate themselves on their consensus. But make the term "Artificial Intelligence" taboo, and ask them to describe *what* they expect to see, without ever using words like "computers" or "think," and you might find quite a conflict of expectations hiding under that featureless standard word. See also Shane Legg's compilation of 71 definitions of "intelligence."¹

The illusion of unity across religions can be dispelled by making the term "God" taboo, and asking them to say what it is they believe in; or making the word "faith" taboo, and asking them why they believe it. Though mostly they won't be able to answer at all, because it is mostly profession in the first place, and you cannot cognitively zoom in on an audio recording.

When you find yourself in philosophical difficulties, *the first line of defense is not to define your problematic terms, but to see whether you can think without using those terms at all*. Or any of their short synonyms. And be careful not to let yourself invent a new word to use instead. Describe outward observables and interior mechanisms; don't use a single handle, whatever that handle may be.

Albert says that people have "free will." Barry says that people don't have "free will." Well, that will certainly generate an apparent conflict. Most philosophers would advise Albert and Barry to try to define exactly what they mean by "free will," on which topic they will certainly be able to discourse at great length. I would advise Albert and Barry to describe what it is that they think people do, or do not have, without using the phrase "free will" at all. (If you want to try this at

1. sharov.me/r/3-15
[0706.3639] *A Collection of Definitions of Intelligence* •
arxiv.org



home, you should also avoid the words “choose,” “act,” “decide,” “determined,” “responsible,” or any of their synonyms.)

This is one of the nonstandard tools in my toolbox, and in my humble opinion, it works *way way* better than the standard one. It also requires more effort to use; you get what you pay for.

Replace the Symbol with the Substance



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What does it take to—as in the previous essay’s example—see a “baseball game” as “An artificial group conflict in which you use a long wooden cylinder to whack a thrown spheroid, and then run between four safe positions”? What does it take to play the rationalist version of Taboo, in which the goal is not to find a synonym that isn’t on the card, but to find a way of describing without the standard concept-handle?

You have to visualize. You have to make your mind’s eye see the details, as though looking for the first time. You have to perform an Original Seeing.

Is that a “bat”? No, it’s a long, round, tapering, wooden rod, narrowing at one end so that a human can grasp and swing it.

Is that a “ball”? No, it’s a leather-covered spheroid with a symmetrical stitching pattern, hard but not metal-hard, which someone can grasp and throw, or strike with the wooden rod, or catch.

Are those “bases”? No, they’re fixed positions on a game field, that players try to run to as quickly as possible because of their safety within the game’s artificial rules.

The chief obstacle to performing an original seeing is that your mind already has a nice neat summary, a nice little easy-to-use concept handle. Like the word “baseball,” or “bat,” or “base.” It takes an effort to stop your mind from sliding down the familiar path, the easy path, the path of least resistance, where the small featureless word rushes in and obliterates the details you’re trying to see. A word itself can have the destructive force of cliché; a word itself can carry the poison of a cached thought.

Playing the game of Taboo—being able to describe without using the standard pointer/label/handle—is one of the *fundamental* rationalist capacities. It occupies the same primordial

level as the habit of constantly asking “Why?” or “What does this belief make me anticipate?”

The art is closely related to:

- Pragmatism, because seeing in this way often gives you a much closer connection to anticipated experience, rather than propositional belief;
- Reductionism, because seeing in this way often forces you to drop down to a lower level of organization, look at the parts instead of your eye skipping over the whole;
- Hugging the query, because words often distract you from the question you really want to ask;
- Avoiding cached thoughts, which will rush in using standard words, so you can block them by tabooing standard words;
- The writer’s rule of “Show, don’t tell!,” which has power among rationalists;
- And not losing sight of your original purpose.

How could tabooing a word help you keep your purpose?

From Lost Purposes:

As you read this, some young man or woman is sitting at a desk in a university, earnestly studying material they have no intention of ever using, and no interest in knowing for its own sake. They want a high-paying job, and the high-paying job requires a piece of paper, and the piece of paper requires a previous master’s degree, and the master’s degree requires a bachelor’s degree, and the university that grants the bachelor’s degree requires you to take a class in twelfth-century knitting patterns to graduate. So they diligently study, intending to forget it all the moment the final exam is administered, but still seriously working away, because they *want* that piece of paper.

Why are you going to “school”? To get an “education” ending in a “degree.” Blank out the forbidden words and all their obvious synonyms, visualize the actual details, and you’re much more likely to notice that “school” currently seems to consist of sitting next to bored teenagers listening to material you already know, that a “degree” is a piece of paper with some writing on it, and that “education” is forgetting the material as soon as you’re tested on it.

Leaky generalizations often manifest through categorizations: People who actually learn in classrooms are categorized as “getting an education,” so “getting an education” must be good; but then anyone who actually shows up at a college will also match against the concept “getting an education,” whether or not they learn.

Students who understand math will do well on tests, but if you require schools to produce good test scores, they’ll spend all their time teaching to the test. A *mental category*, that imperfectly matches your goal, can produce the same kind of incentive failure *internally*. You want to learn, so you need an “education”; and then as long as you’re getting anything that matches against the category “education,” you may not notice whether you’re learning or not. Or you’ll notice, but you won’t realize you’ve lost sight of your original purpose, because you’re “getting an education” and that’s how you mentally described your goal.

To categorize is to throw away information. If you’re told that a falling tree makes a “sound,” you don’t know what the actual sound is; you haven’t actually heard the tree falling. If a coin lands “heads,” you don’t know its radial orientation. A blue egg-shaped thing may be a “blegg,” but what if the exact egg shape varies, or the exact shade of blue? You want to use categories to throw away irrelevant information, to sift gold from dust, but often the standard categorization ends up throwing out relevant information too. And when you end up in that sort of mental trouble, the first and most obvious solution is to play Taboo.

For example: “Play Taboo” is itself a leaky generalization. Hasbro’s version is not the rationalist version; they only list

five additional banned words on the card, and that's not nearly enough coverage to exclude thinking in familiar old words. What rationalists do would count as playing Taboo—it would match against the “play Taboo” concept—but not everything that counts as playing Taboo works to force original seeing. If you just think “play Taboo to force original seeing,” you'll start thinking that anything that counts as playing Taboo must count as original seeing.

The rationalist version isn't a game, which means that you can't win by trying to be clever and stretching the rules. You have to play Taboo with a voluntary handicap: Stop yourself from using synonyms that aren't on the card. You also have to stop yourself from inventing a new simple word or phrase that functions as an equivalent mental handle to the old one. You are trying to zoom in on your map, not rename the cities; dereference the pointer, not allocate a new pointer; see the events as they happen, not rewrite the cliché in a different wording.

By visualizing the problem in more detail, you can see the lost purpose: Exactly what do you do when you “play Taboo”? What purpose does each and every part serve?

If you see your activities and situation originally, you will be able to originally see your goals as well. If you can look with fresh eyes, as though for the first time, you will see yourself doing things that you would never dream of doing if they were not habits.

Purpose is lost whenever the substance (learning, knowledge, health) is displaced by the symbol (a degree, a test score, medical care). To heal a lost purpose, or a lossy categorization, you must do the reverse:

Replace the symbol with the substance; replace the signifier with the signified; replace the property with the membership test; replace the word with the meaning; replace the label with the concept; replace the summary with the details; replace the proxy question with the real question; dereference the pointer; drop into a lower level of organization; mentally simulate the process instead of naming it; zoom in on your map.

The Simple Truth was generated by an exercise of this discipline to describe “truth” on a lower level of organization, without invoking terms like “accurate,” “correct,” “represent,” “reflect,” “semantic,” “believe,” “knowledge,” “map,” or “real.” (And remember that the goal is not *really* to play Taboo—the word “true” appears in the text, but *not* to define truth. It would get a buzzer in Hasbro’s game, but we’re not *actually* playing that game. Ask yourself whether the document fulfilled its purpose, not whether it followed the rules.)

Bayes’s Rule itself describes “evidence” in pure math, without using words like “implies,” “means,” “supports,” “proves,” or “justifies.” Set out to *define* such philosophical terms, and you’ll just go in circles.

And then there’s the most important word of all to Taboo. I’ve often warned that you should be careful not to overuse it, or even avoid the concept in certain cases. Now you know the real reason why. It’s not a bad subject to think about. But your true understanding is measured by your ability to describe what you’re doing and why, *without* using that word or any of its synonyms.

Fallacies of Compression

“The map is not the territory,” as the saying goes. The only life-size, atomically detailed, 100% accurate map of California is California. But California has important regularities, such as the shape of its highways, that can be described using vastly less information—not to mention vastly less *physical material*—than it would take to describe every atom within the state borders. Hence the *other* saying: “The map is not the territory, but you can’t fold up the territory and put it in your glove compartment.”

A paper map of California, at a scale of 10 kilometers to 1 centimeter (a million to one), doesn’t have room to show the distinct position of two fallen leaves lying a centimeter apart on the sidewalk. Even if the map tried to show the leaves, the leaves would appear as the same point on the map; or rather the map would need a feature size of 10 nanometers, which is a finer resolution than most book printers handle, not to mention human eyes.

Reality is very large—just the part we can see is billions of lightyears across. But your map of reality is written on a few pounds of neurons, folded up to fit inside your skull. I don’t mean to be insulting, but your skull is tiny. Comparatively speaking.

Inevitably, then, certain things that are distinct in reality, will be compressed into the same point on your map.

But what this feels like from inside is not that you say, “Oh, look, I’m compressing two things into one point on my map.” What it *feels* like from inside is that there is just *one* thing, and you are seeing it.

A sufficiently young child, or a sufficiently ancient Greek philosopher, would not know that there were such things as “acoustic vibrations” or “auditory experiences.” There would



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just be a single thing that happened when a tree fell; a single event called “sound.”

To realize that there are *two* distinct events, underlying *one* point on your map, is an essentially *scientific* challenge—a big, difficult scientific challenge.

Sometimes fallacies of compression result from confusing two known things under the same label—you know about acoustic vibrations, and you know about auditory processing in brains, but you call them both “sound” and so confuse yourself. But the more dangerous fallacy of compression arises from having *no idea whatsoever* that two distinct entities even *exist*. There is just one mental folder in the filing system, labeled “sound,” and everything thought about “sound” drops into that one folder. It’s not that there are two folders with the same label; there’s just a single folder. By default, the map is compressed; why would the brain create two mental buckets where one would serve?

Or think of a mystery novel in which the detective’s critical insight is that one of the suspects has an identical twin. In the course of the detective’s ordinary work, their job is just to observe that Carol is wearing red, that she has black hair, that her sandals are leather—but all these are *facts about* Carol. It’s easy enough to question an individual fact, like WearsRed(Carol) or BlackHair(Carol). Maybe BlackHair(Carol) is false. Maybe Carol dyes her hair. Maybe BrownHair(Carol). But it takes a subtler detective to wonder if the Carol in WearsRed(Carol) and BlackHair(Carol)—the Carol file into which their observations drop—should be split into *two* files. Maybe there are two Carols, so that the Carol who wore red is not the same woman as the Carol who had black hair.

Here it is the very act of *creating* two different buckets that is the stroke of genius insight. ’Tis easier to question one’s facts than one’s ontology.

The map of reality contained in a human brain, unlike a paper map of California, can expand dynamically when we write down more detailed descriptions. But what this feels like from inside is not so much zooming in on a map, as fissioning

an indivisible atom—taking *one thing* (it felt like one thing) and splitting it into two or more things.

Often this manifests in the creation of new words, like “acoustic vibrations” and “auditory experiences” instead of just “sound.” Something about creating the new name seems to allocate the new bucket. The detective is liable to start calling one of their suspects “Carol-2” or “the Other Carol” almost as soon as they realize that there are two Carols.

But expanding the map isn’t always as simple as generating new city names. It is a stroke of scientific insight to realize that such things as acoustic vibrations, or auditory experiences, even *exist*.

The obvious modern-day illustration would be words like “intelligence” or “consciousness.” Every now and then one sees a press release claiming that a research study has “explained consciousness” because a team of neurologists investigated a 40Hz electrical rhythm that might have something to do with cross-modality binding of sensory information, or because they investigated the reticular activating system that keeps humans awake. That’s an extreme example, and the usual failures are more subtle, but they are of the same kind. The part of “consciousness” that people find most interesting is reflectivity, self-awareness, realizing that the person I see in the mirror is “me”; that and the hard problem of subjective experience as distinguished by David Chalmers. We also label “conscious” the state of being awake, rather than asleep, in our daily cycle. But they are all different concepts going under the same name, and the underlying phenomena are different scientific puzzles. You can explain being awake without explaining reflectivity or subjectivity.

Fallacies of compression also underlie the bait-and-switch technique in philosophy—you argue about “consciousness” under one definition (like the ability to think about thinking) and then apply the conclusions to “consciousness” under a different definition (like subjectivity). Of course it may be that the two are the same thing, but if so, genuinely *understanding* this fact would require *first* a conceptual split and *then* a genius stroke of reunification.

Expanding your map is (I say again) a *scientific* challenge: part of the art of science, the skill of inquiring into the world. (And of course you cannot solve a scientific challenge by appealing to dictionaries, nor master a complex skill of inquiry by saying “I can define a word any way I like.”) Where you see a single confusing thing, with protean and self-contradictory attributes, it is a good guess that your map is cramming too much into one point—you need to pry it apart and allocate some new buckets. This is not like *defining* the single thing you see, but it *does* often follow from figuring out how to talk about the thing without using a single mental handle.

So the skill of prying apart the map is linked to the rationalist version of Taboo, and to the wise use of words; because words often represent the points on our map, the labels under which we file our propositions and the buckets into which we drop our information. Avoiding a single word, or allocating new ones, is often part of the skill of expanding the map.

Categorizing Has Consequences

Among the many genetic variations and mutations you carry in your genome, there are a very few alleles you probably know—including those determining your blood type: the presence or absence of the A, B, and + antigens. If you receive a blood transfusion containing an antigen you don't have, it will trigger an allergic reaction. It was Karl Landsteiner's discovery of this fact, and how to test for compatible blood types, that made it possible to transfuse blood without killing the patient. (1930 Nobel Prize in Medicine.) Also, if a mother with blood type A (for example) bears a child with blood type A+, the mother may acquire an allergic reaction to the + antigen; if she has another child with blood type A+, the child will be in danger, unless the mother takes an allergic suppressant during pregnancy. Thus people learn their blood types before they marry.

Oh, and *also*: people with blood type A are earnest and creative, while people with blood type B are wild and cheerful. People with type O are agreeable and sociable, while people with type AB are cool and controlled. (You would think that O would be the absence of A and B, while AB would just be A plus B, but no ...) All this, according to the Japanese blood type theory of personality. It would seem that blood type plays the role in Japan that astrological signs play in the West, right down to blood type horoscopes in the daily newspaper.

This fad is especially odd because blood types have *never been* mysterious, not in Japan and not anywhere. We only know blood types even *exist* thanks to Karl Landsteiner. No mystic witch doctor, no venerable sorcerer, ever said a word about blood types; there are no ancient, dusty scrolls to shroud the error in the aura of antiquity. If the medical profession claimed tomorrow that it had all been a colossal hoax, we layfolk would not have one scrap of evidence from our unaided senses to contradict them.



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There's never been a war between blood types. There's never even been a political conflict between blood types. The stereotypes must have arisen *strictly* from the *mere existence* of the labels.

Now, someone is bound to point out that this is a story of categorizing humans. Does the same thing happen if you categorize plants, or rocks, or office furniture? I can't recall reading about such an experiment, but of course, that doesn't mean one hasn't been done. (I'd expect the chief difficulty of doing such an experiment would be finding a protocol that didn't mislead the subjects into thinking that, since the label was given you, it must be significant somehow.) So while I don't mean to update on imaginary evidence, I would predict a positive result for the experiment: I would expect them to find that mere labeling had power over all things, at least in the human imagination.

You can see this in terms of similarity clusters: once you draw a boundary around a group, the mind starts trying to harvest similarities from the group. And unfortunately the human pattern-detectors seem to operate in such overdrive that we see patterns whether they're there or not; a weakly negative correlation can be mistaken for a strong positive one with a bit of selective memory.

You can see this in terms of neural algorithms: creating a name for a set of things is like allocating a subnetwork to find patterns in them.

You can see this in terms of a compression fallacy: things given the same name end up dumped into the same mental bucket, blurring them together into the same point on the map.

Or you can see this in terms of the boundless human ability to make stuff up out of thin air and believe it because no one can prove it's wrong. As soon as you name the category, you can start making up stuff about it. The named thing doesn't have to be perceptible; it doesn't have to exist; it doesn't even have to be coherent.

And no, it's not just Japan: Here in the West, a blood-type-based diet book called *Eat Right 4 Your Type* was a bestseller.

Any way you look at it, drawing a boundary in thingspace is not a neutral act. Maybe a more cleanly designed, more purely Bayesian AI could ponder an arbitrary class and not be influenced by it. But you, a human, do not have that option. Categories are not static things in the context of a human brain; as soon as you actually think of them, they exert force on your mind. One more reason not to believe you can define a word any way you like.

Sneaking in Connotations



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In the previous essay, we saw that in Japan, blood types have taken the place of astrology—if your blood type is AB, for example, you’re supposed to be “cool and controlled.”

So suppose we decided to invent a new word, “wiggin,” and *defined* this word to mean people with green eyes and black hair—

A green-eyed man with black hair walked into a restaurant.

“Ha,” said Danny, watching from a nearby table, “did you see that? A wiggin just walked into the room. Bloody wiggins. Commit all sorts of crimes, they do.”

His sister Erda sighed. “You haven’t *seen* him commit any crimes, have you, Danny?”

“Don’t need to,” Danny said, producing a dictionary. “See, it says right here in the Oxford English Dictionary. ‘Wiggin. (1) A person with green eyes and black hair.’ He’s got green eyes and black hair, he’s a wiggin. You’re not going to argue with the Oxford English Dictionary, are you? *By definition*, a green-eyed black-haired person is a wiggin.”

“But you called him a wiggin,” said Erda. “That’s a nasty thing to say about someone you don’t even know. You’ve got no evidence that he puts too much ketchup on his burgers, or that as a kid he used his slingshot to launch baby squirrels.”

“But he *is* a wiggin,” Danny said patiently. “He’s got green eyes and black hair, right? Just you watch, as soon as his burger arrives, he’s reaching for the ketchup.”

The human mind passes from observed characteristics to inferred characteristics via the medium of words. In “All humans

are mortal, Socrates is a human, therefore Socrates is mortal,” the observed characteristics are Socrates’s clothes, speech, tool use, and generally human shape; the categorization is “human”; the inferred characteristic is poisonability by hemlock.

Of course there’s no hard distinction between “observed characteristics” and “inferred characteristics.” If you hear someone speak, they’re probably shaped like a human, all else being equal. If you see a human figure in the shadows, then *ceteris paribus* it can probably speak.

And yet some properties do tend to be more inferred than observed. You’re more likely to decide that someone is human, and will therefore burn if exposed to open flame, than carry through the inference the other way around.

If you look in a dictionary for the definition of “human,” you’re more likely to find characteristics like “intelligence” and “featherless biped”—characteristics that are useful for quickly eyeballing what is and isn’t a human—rather than the ten thousand connotations, from vulnerability to hemlock, to overconfidence, that we can infer from someone’s being human. Why? Perhaps dictionaries are intended to let you match up labels to similarity groups, and so are designed to quickly isolate clusters in thingspace. Or perhaps the big, distinguishing characteristics are the most salient, and therefore first to pop into a dictionary editor’s mind. (I’m not sure how aware dictionary editors are of what they *really* do.)

But the upshot is that when Danny pulls out his OED to look up “wiggin,” he sees listed only the first-glance characteristics that distinguish a wiggin: Green eyes and black hair. The OED doesn’t list the many minor *connotations* that have come to attach to this term, such as criminal proclivities, culinary peculiarities, and some unfortunate childhood activities.

How did those connotations get there in the first place? Maybe there was once a famous wiggin with those properties. Or maybe someone made stuff up at random, and wrote a series of bestselling books about it (*The Wiggin*, *Talking to Wiggins*, *Raising Your Little Wiggin*, *Wiggins in the Bedroom*). Maybe even the wiggins believe it now, and act accordingly. As soon

as you call some people “wiggins,” the word will begin acquiring connotations.

But remember the Parable of Hemlock: If we go by the logical class definitions, we can never class Socrates as a “human” until after we observe him to be mortal. Whenever someone pulls a dictionary, they’re generally trying to sneak in a *connotation*, not the actual definition written down in the dictionary.

After all, if the *only* meaning of the word “wiggin” is “green-eyed black-haired person,” then why not just call those people “green-eyed black-haired people”? And if you’re wondering whether someone is a ketchup-reacher, why not ask directly, “Is he a ketchup-reacher?” rather than “Is he a wiggin?” (Note substitution of substance for symbol.)

Oh, but arguing the *real* question would require *work*. You’d have to actually watch the wiggin to see if he reached for the ketchup. Or maybe see if you can find statistics on how many green-eyed black-haired people actually like ketchup. At any rate, you wouldn’t be able to do it sitting in your living room with your eyes closed. And people are lazy. They’d rather argue “by definition,” especially since they think “you can define a word any way you like.”

But of course the *real* reason they care whether someone is a “wiggin” is a connotation—a feeling that comes along with the word—that isn’t in the definition they *claim* to use.

Imagine Danny saying, “Look, he’s got green eyes and black hair. He’s a wiggin! It says so right there in the dictionary!—*therefore*, he’s got black hair. Argue with that, if you can!”

Doesn’t have much of a triumphant ring to it, does it? If the real point of the argument actually *was* contained in the dictionary definition—if the argument genuinely *was* logically valid—then the argument would *feel* empty; it would either say nothing new, or beg the question.

It’s only the attempt to smuggle in connotations *not* explicitly listed in the definition, that makes anyone feel they can *score a point* that way.

Arguing “By Definition”

“This plucked chicken has two legs and no feathers—therefore, *by definition*, it is a human!”

When people argue definitions, they usually start with some visible, known, or at least widely believed set of characteristics; then pull out a dictionary, and point out that these characteristics fit the dictionary definition; and so conclude, “Therefore, *by definition*, atheism is a religion!”

But visible, known, widely believed characteristics are rarely the real point of a dispute. Just the fact that someone thinks Socrates’s two legs are evident enough to make a good premise for the argument, “Therefore, *by definition*, Socrates is human!” indicates that bipedalism probably isn’t *really* what’s at stake—or the listener would reply, “Whaddaya mean Socrates is bipedal? That’s what we’re arguing about in the first place!”

Now there is an important sense in which we can legitimately move from evident characteristics to not-so-evident ones. You can, legitimately, see that Socrates is human-shaped, and predict his vulnerability to hemlock. But this *probabilistic* inference does not rely on dictionary definitions or common usage; it relies on the universe containing empirical clusters of similar things.

This cluster structure is not going to change depending on how you define your words. Even if you look up the dictionary definition of “human” and it says “all featherless bipeds except Socrates,” that isn’t going to change the *actual* degree to which Socrates is similar to the rest of us featherless bipeds.

When you are arguing *correctly* from cluster structure, you’ll say something like, “Socrates has two arms, two feet, a nose and tongue, speaks fluent Greek, uses tools, and in every aspect I’ve been able to observe him, seems to have every major and minor property that characterizes *Homo sapiens*; so I’m going to guess that he has human DNA, human biochemistry, and is



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vulnerable to hemlock just like all other *Homo sapiens* in whom hemlock has been clinically tested for lethality.”

And suppose I reply, “But I saw Socrates out in the fields with some herbologists; I think they were trying to prepare an antidote. Therefore I *don’t* expect Socrates to keel over after he drinks the hemlock—he will be an exception to the general behavior of objects in his cluster: they did not take an antidote, and he did.”

Now there’s not much point in arguing over whether Socrates is “human” or not. The conversation has to move to a more detailed level, poke around *inside* the details that make up the “human” category—talk about human biochemistry, and specifically, the neurotoxic effects of coniine.

If you go on insisting, “But Socrates is a human and humans, *by definition*, are mortal!” then what you’re really trying to do is blur out everything you know about Socrates *except* the fact of his humanity—insist that the only correct prediction is the one you would make if you knew nothing about Socrates *except* that he was human.

Which is like insisting that a coin is 50% likely to be showing heads or tails, because it is a “fair coin,” after you’ve *actually looked at the coin* and it’s showing heads. It’s like insisting that Frodo has ten fingers, because most hobbits have ten fingers, after you’ve *already looked at his hands* and seen nine fingers. Naturally this is illegal under Bayesian probability theory: You can’t just refuse to condition on new evidence.

And you can’t just keep one categorization and make estimates based on that, while deliberately throwing out everything else you know.

Not every piece of new evidence makes a significant difference, of course. If I see that Socrates has nine fingers, this isn’t going to noticeably change my estimate of his vulnerability to hemlock, because I’ll expect that the way Socrates lost his finger didn’t change the rest of his biochemistry. And this is true, *whether or not* the dictionary’s definition says that human beings have ten fingers. The legal inference is based on the cluster structure of the environment, and the causal structure

of biology; *not* what the dictionary editor writes down, nor even “common usage.”

Now ordinarily, when you’re doing this *right*—in a *legitimate* way—you just say, “The coniine alkaloid found in hemlock produces muscular paralysis in humans, resulting in death by asphyxiation.” Or more simply, “Humans are vulnerable to hemlock.” That’s how it’s usually said in a *legitimate* argument.

When would someone feel the need to *strengthen* the argument with the emphatic phrase “by definition”? (E.g. “Humans are vulnerable to hemlock *by definition!*”) Why, when the inferred characteristic has been called into doubt—Socrates has been seen consulting herbologists—and so the speaker feels the need to tighten the vise of logic.

So when you see “by definition” used like this, it usually means: “Forget what you’ve heard about Socrates consulting herbologists—humans, *by definition*, are mortal!”

People feel the need to squeeze the argument onto a single course by saying “Any *P*, by definition, has property *Q!*,” on exactly those occasions when they see, and prefer to dismiss out of hand, *additional arguments* that call into doubt the default inference based on clustering.

So too with the argument “*X*, *by definition*, is a *Y!*” E.g., “Atheists believe that God doesn’t exist; therefore atheists have beliefs about God, because a negative belief is still a belief; therefore atheism asserts answers to theological questions; therefore atheism is, *by definition*, a religion.”

You wouldn’t feel the need to say, “Hinduism, *by definition*, is a religion!” because, well, of course Hinduism is a religion. It’s not just a religion “by definition,” it’s, like, an *actual* religion.

Atheism does not resemble the central members of the “religion” cluster, so if it wasn’t for the fact that atheism is a religion *by definition*, you might go around thinking that atheism *wasn’t* a religion. That’s why you’ve got to crush all opposition by pointing out that “Atheism is a religion” is true *by definition*, because it isn’t true any other way.

Which is to say: People insist that “*X*, *by definition*, is a *Y!*” on those occasions when they’re trying to sneak in a connotation

of Y that isn't directly in the definition, and X doesn't look all that much like other members of the Y cluster.

Over the last thirteen years I've been keeping track of how often this phrase is used correctly versus incorrectly—though not with literal statistics, I fear. But eyeballing suggests that using the phrase *by definition*, anywhere outside of math, is among the most alarming signals of flawed argument I've ever found. It's right up there with "Hitler," "God," "absolutely certain," and "can't prove that."

This heuristic of failure is not perfect—the first time I ever spotted a correct usage outside of math, it was by Richard Feynman; and since then I've spotted more. But you're probably better off just deleting the phrase "by definition" from your vocabulary—and *always* on any occasion where you might be tempted to say it in italics or followed with an exclamation mark. That's a bad idea *by definition!*

Where to Draw the Boundary?

The one comes to you and says:¹

Long have I pondered the meaning of the word “Art,” and at last I’ve found what seems to me a satisfactory definition: “Art is that which is designed for the purpose of creating a reaction in an audience.”

*Just because there’s a word “art” doesn’t mean that it **has a meaning**, floating out there in the void, which you can **discover** by finding the right definition.*

It feels that way, but it is not so.

Wondering how to *define a word* means you’re looking at the problem the wrong way—searching for the mysterious essence of what is, in fact, a communication signal.

Now, there *is* a real challenge which a rationalist may legitimately attack, but the challenge is not to find a satisfactory definition of a word. The real challenge can be played as a single-player game, without speaking aloud. The challenge is figuring out which things are similar to each other—which things are clustered together—and sometimes, which things have a common cause.

If you define “electromagnetism” to include lightning, include compasses, exclude light, and include Mesmer’s “animal magnetism” (what we now call hypnosis), then you will have some trouble asking “How does electromagnetism work?” You have lumped together things which do not belong together, and excluded others that would be needed to complete a set.



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(This example is historically plausible; Mesmer came before Faraday.)

We could say that electromagnetism is a *wrong word*, a boundary in thingspace that loops around and swerves through the clusters, a cut that fails to carve reality along its natural joints.

Figuring where to cut reality in order to carve along the joints—*this* is the problem worthy of a rationalist. It is what people *should* be trying to do, when they set out in search of the floating essence of a word.

And make no mistake: it is a *scientific* challenge to realize that you need a single word to describe breathing and fire. So do not think to consult the dictionary editors, for that is not their job.

What is “art”? But there is no essence of the word, floating in the void.

Perhaps you come to me with a long list of the things that you call “art” and “not art”:

The *Little Fugue in G Minor*: Art.

A punch in the nose: Not art.

Escher’s *Relativity*: Art.

A flower: Not art.

The Python programming language: Art.

A cross floating in urine: Not art.

Jack Vance’s *Tschai* novels: Art.

Modern Art: Not art.

And you say to me: “It feels intuitive to me to draw this boundary, but I don’t know why—can you find me an intension that matches this extension? Can you give me a *simple* description of this boundary?”

So I reply: “I think it has to do with admiration of craftsmanship: work going in and wonder coming out. What the included items have in common is the similar aesthetic emotions that they inspire, and the deliberate human effort that went into them with the intent of producing such an emotion.”

Is this helpful, or is it just cheating at Taboo? I would argue that the list of which human emotions are or are not *aesthetic* is far more compact than the list of everything that is or isn't art. You might be able to see those emotions lighting up an fMRI scan—I say this by way of emphasizing that emotions are not ethereal.

But of course my definition of art is not the real point. The real point is that you could well dispute either the intension or the extension of my definition.

You could say, "Aesthetic emotion is *not* what these things have in common; what they have in common is an intent to inspire *any* complex emotion for the sake of inspiring it." That would be disputing my intension, my attempt to draw a curve through the data points. You would say, "Your equation may roughly fit those points, but it is not the true generating distribution."

Or you could dispute my extension by saying, "Some of these things do belong together—I can see what you're getting at—but the Python language shouldn't be on the list, and Modern Art should be." (This would mark you as a philistine, but you could argue it.) Here, the presumption is that there is indeed an underlying curve that generates this apparent list of similar and dissimilar things—that there is a rhyme and reason, *even though you haven't said yet where it comes from*—but I have unwittingly lost the rhythm and included some data points from a different generator.

Long before you *know* what it is that electricity and magnetism have in common, you might still suspect—based on surface appearances—that "animal magnetism" does not belong on the list.

Once upon a time it was thought that the word "fish" included dolphins. Now you could play the oh-so-clever arguer, and say, "The list: {Salmon, guppies, sharks, dolphins, trout} is just a list—you can't say that a list is *wrong*. I can prove in set theory that this list exists. So my definition of *fish*, which is simply this extensional list, cannot possibly be 'wrong' as you claim."

Or you could stop playing games and admit that dolphins don't belong on the fish list.

You come up with a list of things that *feel* similar, and take a guess at why this is so. But when you finally discover what they *really* have in common, it may turn out that your guess was wrong. It may even turn out that your list was wrong.

You cannot hide behind a comforting shield of correct-by-definition. Both extensional definitions and intensional definitions can be wrong, can fail to carve reality at the joints.

Categorizing is a guessing endeavor, in which you can make mistakes; so it's wise to be able to admit, from a theoretical standpoint, that your definition-guesses can be "mistaken."

Entropy, and Short Codes

(If you aren't familiar with Bayesian inference, this may be a good time to read *An Intuitive Explanation of Bayes's Theorem*, p. 259.)

Suppose you have a system X that's equally likely to be in any of 8 possible states:

$$\{X_1, X_2, X_3, X_4, X_5, X_6, X_7, X_8\}.$$

There's an extraordinarily ubiquitous quantity—in physics, mathematics, and even biology—called *entropy*; and the entropy of X is 3 bits. This means that, on average, we'll have to ask 3 yes-or-no questions to find out X 's value. For example, someone could tell us X 's value using this code:

$$\begin{array}{llll} X_1 : 001 & X_2 : 010 & X_3 : 011 & X_4 : 100 \\ X_5 : 101 & X_6 : 110 & X_7 : 111 & X_8 : 000. \end{array}$$

So if I asked “Is the first symbol 1?” and heard “yes,” then asked “Is the second symbol 1?” and heard “no,” then asked “Is the third symbol 1?” and heard “no,” I would know that X was in state 4.

Now suppose that the system Y has four possible states with the following probabilities:

$$\begin{array}{ll} Y_1 : 1/2(50\%) & Y_2 : 1/4(25\%) \\ Y_3 : 1/8(12.5\%) & Y_4 : 1/8(12.5\%). \end{array}$$

Then the entropy of Y would be 1.75 bits, meaning that we can find out its value by asking 1.75 yes-or-no questions.

What does it mean to talk about asking one and three-fourths of a question? Imagine that we designate the states of Y using the following code:

$$Y_1 : 1 \quad Y_2 : 01 \quad Y_3 : 001 \quad Y_4 : 000.$$



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First you ask, “Is the first symbol 1?” If the answer is “yes,” you’re done: Y is in state 1. This happens half the time, so 50% of the time, it takes 1 yes-or-no question to find out Y ’s state.

Suppose that instead the answer is “No.” Then you ask, “Is the second symbol 1?” If the answer is “yes,” you’re done: Y is in state 2. The system Y is in state 2 with probability $1/4$, and each time Y is in state 2 we discover this fact using two yes-or-no questions, so 25% of the time it takes 2 questions to discover Y ’s state.

If the answer is “No” twice in a row, you ask “Is the third symbol 1?” If “yes,” you’re done and Y is in state 3; if “no,” you’re done and Y is in state 4. The $1/8$ of the time that Y is in state 3, it takes three questions; and the $1/8$ of the time that Y is in state 4, it takes three questions.

$$\begin{aligned} & (1/2 \times 1) + (1/4 \times 2) + (1/8 \times 3) + (1/8 \times 3) \\ &= 0.5 + 0.5 + 0.375 + 0.375 \\ &= 1.75. \end{aligned}$$

The general formula for the entropy $H(S)$ of a system S is the sum, over all S_i , of $-P(S_i) \log_2(P(S_i))$.

For example,¹ the log (base 2) of $1/8$ is -3 . So $-(1/8 \times -3) = 0.375$ is the contribution of state S_4 to the total entropy: $1/8$ of the time, we have to ask 3 questions.

You can’t always devise a perfect code for a system, but if you have to tell someone the state of arbitrarily many copies of S in a single message, you can get arbitrarily close to a perfect code. (Google “arithmetic coding” for a simple method.)

Now, you might ask: “Why not use the code 10 for Y_4 , instead of 000? Wouldn’t that let us transmit messages more quickly?”

But if you use the code 10 for Y_4 , then when someone answers “Yes” to the question “Is the first symbol 1?,” you won’t know yet whether the system state is Y_1 (1) or Y_4 (10). In fact, if you change the code this way, the whole system falls apart—because if you hear “1001,” you don’t know if it means “ Y_4 , followed by Y_2 ” or “ Y_1 , followed by Y_3 .”

1. Note for calculation: $\log_2(x) = \log_{10}(x) / \log_{10}(2)$

The moral is that *short words are a conserved resource*.

The key to creating a good code—a code that transmits messages as compactly as possible—is to reserve short words for things that you’ll need to say frequently, and use longer words for things that you won’t need to say as often.

When you take this art to its limit, the length of the message you need to describe something corresponds exactly or almost exactly to its probability. This is the Minimum Description Length or Minimum Message Length formalization of Occam’s Razor.

And so even the *labels* that we use for words are not quite arbitrary. The sounds that we attach to our concepts can be better or worse, wiser or more foolish. Even apart from considerations of common usage!

I say all this, because the idea that “You can *X* any way you like” is a huge obstacle to learning how to *X* wisely. “It’s a free country; I have a right to my own opinion” obstructs the art of finding truth. “I can define a word any way I like” obstructs the art of carving reality at its joints. And even the sensible-sounding “The labels we attach to words are arbitrary” obstructs awareness of compactness. Prosody too, for that matter—Tolkien once observed what a beautiful sound the phrase “cellar door” makes; that is the kind of awareness it takes to use language like Tolkien.

The length of words also plays a nontrivial role in the cognitive science of language:

Consider the phrases “recliner,” “chair,” and “furniture.” Recliner is a more specific category than chair; furniture is a more general category than chair. But the vast majority of chairs have a common use—you use the same sort of motor actions to sit down in them, and you sit down in them for the same sort of purpose (to take your weight off your feet while you eat, or read, or type, or rest). Recliners do not depart from this theme. “Furniture,” on the other hand, includes things like beds and tables which have different uses, and call up different motor functions, from chairs.

In the terminology of cognitive psychology, “chair” is a *basic-level category*.

People have a tendency to talk, and presumably think, at the basic level of categorization—to draw the boundary around “chairs,” rather than around the more specific category “recliner,” or the more general category “furniture.” People are more likely to say “You can sit in that chair” than “You can sit in that recliner” or “You can sit in that furniture.”

And it is no coincidence that the word for “chair” contains fewer syllables than either “recliner” or “furniture.” Basic-level categories, in general, tend to have short names; and nouns with short names tend to refer to basic-level categories. Not a perfect rule, of course, but a definite tendency. Frequent use goes along with short words; short words go along with frequent use.

Or as Douglas Hofstadter put it, there’s a reason why the English language uses “the” to mean “the” and “antidisestablishmentarianism” to mean “antidisestablishmentarianism” instead of antidisestablishmentarianism other way around.

Mutual Information, and Density in Thingspace

Suppose you have a system X that can be in any of 8 states, which are all equally probable (relative to your current state of knowledge), and a system Y that can be in any of 4 states, all equally probable.

The entropy of X , as defined in the previous essay, is 3 bits; we'll need to ask 3 yes-or-no questions to find out X 's exact state. The entropy of Y is 2 bits; we have to ask 2 yes-or-no questions to find out Y 's exact state. This may seem obvious since $2^3 = 8$ and $2^2 = 4$, so 3 questions can distinguish 8 possibilities and 2 questions can distinguish 4 possibilities; but remember that if the possibilities were not all equally likely, we could use a more clever code to discover Y 's state using e.g. 1.75 questions on average. In this case, though, X 's *probability mass* is *evenly distributed* over all its possible states, and likewise Y , so we can't use any clever codes.

What is the entropy of the combined system (X, Y) ?

You might be tempted to answer, "It takes 3 questions to find out X , and then 2 questions to find out Y , so it takes 5 questions total to find out the state of X and Y ."

But what if the two variables are entangled, so that learning the state of Y tells us something about the state of X ?

In particular, let's suppose that X and Y are either both odd or both even.

Now if we receive a 3-bit message (ask 3 questions) and learn that X is in state X_5 , we know that Y is in state Y_1 or state Y_3 , but not state Y_2 or state Y_4 . So the single additional question "Is Y in state Y_3 ?" answered "No," tells us the entire state of (X, Y) : $X = X_5, Y = Y_1$. And we learned this with a total of 4 questions.

Conversely, if we learn that Y is in state Y_4 using two questions, it will take us only an additional two questions to learn



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whether X is in state X_2 , X_4 , X_6 , or X_8 . Again, four questions to learn the state of the joint system.

The *mutual information* of two variables is defined as the difference between the entropy of the joint system and the entropy of the independent systems: $I(X; Y) = H(X) + H(Y) - H(X, Y)$.

Here there is one bit of mutual information between the two systems: Learning X tells us one bit of information about Y (cuts down the space of possibilities from 4 possibilities to 2, a factor-of-2 decrease in the volume) and learning Y tells us one bit of information about X (cuts down the possibility space from 8 possibilities to 4).

What about when probability mass is not evenly distributed? Last essay, for example, we discussed the case in which Y had the probabilities $1/2$, $1/4$, $1/8$, $1/8$ for its four states. Let us take this to be our probability distribution over Y , considered independently—if we saw Y , without seeing anything else, this is what we'd expect to see. And suppose the variable Z has two states, Z_1 and Z_2 , with probabilities $3/8$ and $5/8$ respectively.

Then if and only if the joint distribution of Y and Z is as follows, there is zero mutual information between Y and Z :

$$\begin{array}{ll} Z_1 Y_1 : 3/16 & Z_1 Y_2 : 3/32 \\ Z_1 Y_3 : 3/64 & Z_1 Y_4 : 3/64 \\ Z_2 Y_1 : 5/16 & Z_2 Y_2 : 5/32 \\ Z_2 Y_3 : 5/64 & Z_2 Y_4 : 5/64. \end{array}$$

This distribution obeys the law

$$P(Y, Z) = P(Y)P(Z).$$

For example, $P(Z_1 Y_2) = P(Z_1)P(Y_2) = 3/8 \times 1/4 = 3/32$.

And observe that we can recover the marginal (independent) probabilities of Y and Z just by looking at the joint distribution:

$$\begin{aligned}
P(Y_1) &= \text{total probability of all the different} \\
&\quad \text{ways } Y_1 \text{ can happen} \\
&= P(Z_1 Y_1) + P(Z_2 Y_1) \\
&= 3/16 + 5/16 \\
&= 1/2.
\end{aligned}$$

So, just by inspecting the joint distribution, we can determine whether the marginal variables Y and Z are independent; that is, whether the joint distribution factors into the product of the marginal distributions; whether, for all Y and Z , we have $P(Y, Z) = P(Y)P(Z)$.

This last is significant because, by Bayes's Rule,

$$\begin{aligned}
P(Z_j Y_i) &= P(Y_i)P(Z_j) \\
P(Z_j Y_i)/P(Z_j) &= P(Y_i) \\
P(Y_i|Z_j) &= P(Y_i).
\end{aligned}$$

In English: "After you learn Z_j , your belief about Y_i is just what it was before."

So when the distribution factorizes—when $P(Y, Z) = P(Y)P(Z)$ —this is *equivalent* to "Learning about Y never tells us anything about Z or vice versa."

From which you might suspect, correctly, that there is no mutual information between Y and Z . Where there is no mutual information, there is no Bayesian evidence, and vice versa.

Suppose that in the distribution (Y, Z) above, we treated each possible combination of Y and Z as a separate event—so that the distribution (Y, Z) would have a total of 8 possibilities, with the probabilities shown—and then we calculated the entropy of the distribution (Y, Z) the same way we would calculate the entropy of any distribution:

$$\begin{aligned}
& P(Z_1Y_1) \log_2(P(Z_1Y_1)) + P(Z_1Y_2) \log_2(P(Z_1Y_2)) + \\
& P(Z_1Y_3) \log_2(P(Z_1Y_3)) + \dots + \\
& P(Z_2Y_4) \log_2(P(Z_2Y_4)) \\
& = (3/16) \log_2(3/16) + (3/32) \log_2(3/32) + \\
& (3/64) \log_2(3/64) + \dots + (5/64) \log_2(5/64).
\end{aligned}$$

You would end up with the same total you would get if you separately calculated the entropy of Y plus the entropy of Z . There is no mutual information between the two variables, so our uncertainty about the joint system is not any less than our uncertainty about the two systems considered separately. (I am not showing the calculations, but you are welcome to do them; and I am not showing the proof that this is true in general, but you are welcome to Google on “Shannon entropy” and “mutual information.”)

What if the joint distribution doesn’t factorize? For example:

$$\begin{array}{ll}
Z_1Y_1 : 12/64 & Z_1Y_2 : 8/64 \\
Z_1Y_3 : 1/64 & Z_1Y_4 : 3/64 \\
Z_2Y_1 : 20/64 & Z_2Y_2 : 8/64 \\
Z_2Y_3 : 7/64 & Z_2Y_4 : 5/64.
\end{array}$$

If you add up the joint probabilities to get marginal probabilities, you should find that $P(Y_1) = 1/2$, $P(Z_1) = 3/8$, and so on—the marginal probabilities are the same as before.

But the joint probabilities do not always equal the product of the marginal probabilities. For example, the probability $P(Z_1Y_2)$ equals $8/64$, where $P(Z_1)P(Y_2)$ would equal $3/8 \times 1/4 = 6/64$. That is, the probability of running into Z_1Y_2 together is greater than you’d expect based on the probabilities of running into Z_1 or Y_2 separately.

Which in turn implies:

$$\begin{aligned}
P(Z_1Y_2) &> P(Z_1)P(Y_2) \\
P(Z_1Y_2)/P(Y_2) &> P(Z_1) \\
P(Z_1|Y_2) &> P(Z_1).
\end{aligned}$$

Since there's an "unusually high" probability for $P(Z_1 Y_2)$ —defined as a probability higher than the marginal probabilities would indicate by default—it follows that observing Y_2 is evidence that increases the probability of Z_1 . And by a symmetrical argument, observing Z_1 must favor Y_2 .

As there are at least some values of Y that tell us about Z (and vice versa) there must be mutual information between the two variables; and so you will find—I am confident, though I haven't actually checked—that calculating the entropy of (Y, Z) yields less total uncertainty than the sum of the independent entropies of Y and Z . That is, $H(Y, Z) = H(Y) + H(Z) - I(Y; Z)$, with all quantities necessarily positive.

(I digress here to remark that the symmetry of the expression for the mutual information shows that Y *must* tell us as much about Z , on average, as Z tells us about Y . I leave it as an exercise to the reader to reconcile this with anything they were taught in logic class about how, if all ravens are black, being allowed to reason $\text{Raven}(x) \Rightarrow \text{Black}(x)$ doesn't mean you're allowed to reason $\text{Black}(x) \Rightarrow \text{Raven}(x)$. How different seem the symmetrical probability flows of the Bayesian, from the sharp lurches of logic—even though the latter is just a degenerate case of the former.)

"But," you ask, "what has all this to do with the proper use of words?"

In Empty Labels and then Replace the Symbol with the Substance, we saw the technique of replacing a word with its definition—the example being given:

All [mortal, $\neg$ feathers, bipedal] are mortal.

Socrates is a [mortal, $\neg$ feathers, bipedal].

Therefore, Socrates is mortal.

Why, then, would you even want to have a word for "human"? Why not just say "Socrates is a mortal featherless biped"?

Because it's helpful to have shorter words for things that you encounter often. If your code for describing single properties is already efficient, then there will not be an advantage to having a special word for a conjunction—like "human" for "mortal featherless biped"—unless things that are mortal *and*

featherless *and* bipedal, are found *more often* than the marginal probabilities would lead you to expect.

In efficient codes, word length corresponds to probability—so the code for Z_1Y_2 will be just as long as the code for Z_1 plus the code for Y_2 , unless $P(Z_1Y_2) > P(Z_1)P(Y_2)$, in which case the code for the word can be shorter than the codes for its parts.

And this in turn corresponds exactly to the case where we can infer some of the properties of the thing from seeing its other properties. It must be more likely than the default that featherless bipedal things will also be mortal.

Of course the word “human” really describes many, many more properties—when you see a human-shaped entity that talks and wears clothes, you can infer whole hosts of biochemical and anatomical and cognitive facts about it. To replace the word “human” with a description of everything we know about humans would require us to spend an inordinate amount of time talking. But this is true *only* because a featherless talking biped is far more likely than default to be poisonable by hemlock, or have broad nails, or be overconfident.

Having a word for a thing, rather than just listing its properties, is a more compact code precisely in those cases where we can infer some of those properties from the other properties. (With the exception perhaps of very primitive words, like “red,” that we would use to send an entirely uncompressed description of our sensory experiences. But by the time you encounter a bug, or even a rock, you’re dealing with nonsimple property collections, far above the primitive level.)

So having a word “wigin” for green-eyed black-haired people is more useful than just saying “green-eyed black-haired person” precisely when:

1. Green-eyed people are more likely than average to be black-haired (and vice versa), meaning that we can probabilistically infer green eyes from black hair or vice versa;
or
2. Wiggins share other properties that can be inferred at greater-than-default probability. In this case we have to separately observe the green eyes and black hair; but

then, after observing both these properties independently, we can probabilistically infer other properties (like a taste for ketchup).

One may even consider the act of defining a word as a promise to this effect. Telling someone, “I define the word ‘wiggin’ to mean a person with green eyes and black hair,” by Gricean implication, asserts that the word “wiggin” will somehow help you make inferences / shorten your messages.

If green-eyes and black hair have no greater than default probability to be found together, nor does any other property occur at greater than default probability along with them, then the word “wiggin” is a lie: The word claims that certain people are worth distinguishing as a group, but they’re not.

In this case the word “wiggin” does not help describe reality more compactly—it is not defined by someone sending the shortest message—it has no role in the simplest explanation. Equivalently, the word “wiggin” will be of no help to you in doing any Bayesian inference. Even if you do not call the word a lie, it is surely an error.

And the way to carve reality at its joints is to draw your boundaries around concentrations of unusually high probability density in Thingspace.

Superexponential Conceptspace, and Simple Words



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Thingspace, you might think, is a rather huge space. Much larger than reality, for where reality only contains things that actually exist, Thingspace contains everything that *could* exist.

Actually, the way I “defined” Thingspace to have dimensions for every possible attribute—including correlated attributes like density and volume and mass—Thingspace may be too poorly defined to have anything you could call a *size*. But it’s important to be able to visualize Thingspace *anyway*. Surely, no one can *really* understand a flock of sparrows if all they see is a cloud of flapping cawing things, rather than a cluster of points in Thingspace.

But as vast as Thingspace may be, it doesn’t hold a candle to the size of Conceptspace.

“Concept,” in machine learning, means a rule that includes or excludes examples. If you see the data {2:+, 3:-, 14:+, 23:-, 8:+, 9:-} then you might guess that the concept was “even numbers.” There is a rather large literature (as one might expect) on how to learn concepts from data...given random examples, given chosen examples...given possible errors in classification...and most importantly, given different spaces of possible rules.

Suppose, for example, that we want to learn the concept “good days on which to play tennis.” The possible attributes of Days are

```
Sky: {Sunny, Cloudy, Rainy}
AirTemp: {Warm, Cold}
Humidity: {Normal, High}
Wind: {Strong, Weak}.
```

We're then presented with the following data, where + indicates a positive example of the concept, and - indicates a negative classification:

```
+ Sky: Sunny; AirTemp: Warm;
  Humidity: High; Wind: Strong.
- Sky: Rainy; AirTemp: Cold;
  Humidity: High; Wind: Strong.
+ Sky: Sunny; AirTemp: Warm;
  Humidity: High; Wind: Weak.
```

What should an algorithm infer from this?

A machine learner might represent *one* concept that fits this data as follows:

```
{ Sky: ?; AirTemp: Warm; Humidity:
  High; Wind: ?}.
```

In this format, to determine whether this concept accepts or rejects an example, we compare element-by-element: ? accepts anything, but a specific value accepts only that specific value.

So the concept above will accept only Days with AirTemp = Warm and Humidity = High, but the Sky and the Wind can take on any value. This fits both the negative and the positive classifications in the data so far—though it isn't the *only* concept that does so.

We can also simplify the above concept representation to

```
{?, Warm, High, ?}.
```

Without going into details, the classic algorithm would be:

- Maintain the set of the most general hypotheses that fit the data—those that positively classify as many examples as possible, while still fitting the facts.
- Maintain another set of the most specific hypotheses that fit the data—those that negatively classify as many examples as possible, while still fitting the facts.

- Each time we see a new negative example, we strengthen all the most general hypotheses as little as possible, so that the new set is again as general as possible while fitting the facts.
- Each time we see a new positive example, we relax all the most specific hypotheses as little as possible, so that the new set is again as specific as possible while fitting the facts.
- We continue until we have only a single hypothesis left. This will be the answer *if* the target concept was in our hypothesis space at all.

In the case above, the set of most general hypotheses would be

$$\{\{?, \text{Warm}, ?, ?\}, \{\text{Sunny}, ?, ?, ?\}\},$$

while the set of most specific hypotheses contains the single member $\{\text{Sunny}, \text{Warm}, \text{High}, ?\}$.

Any other concept you can find that fits the data will be strictly more specific than one of the most general hypotheses, and strictly more general than the most specific hypothesis.

(For more on this, I recommend Tom Mitchell's *Machine Learning*, from which this example was adapted.¹)

Now you may notice that the format above *cannot* represent all possible concepts. E.g., "Play tennis when the sky is sunny *or* the air is warm." That fits the data, but in the concept representation defined above, there's no quadruplet of values that describes the rule.

Clearly our machine learner is not very general. Why not allow it to represent *all possible* concepts, so that it can learn with the greatest possible flexibility?

Days are composed of these four variables, one variable with 3 values and three variables with 2 values. So there are $3 \times 2 \times 2 \times 2 = 24$ possible Days that we could encounter.

1. Tom M. Mitchell, *Machine Learning* (McGraw-Hill Science/Engineering/Math, 1997).

The format given for representing Concepts allows us to require any of these values for a variable, or leave the variable open. So there are $4 \times 3 \times 3 \times 3 = 108$ concepts in that representation. For the most-general/most-specific algorithm to work, we need to start with the most specific hypothesis “no example is ever positively classified.” If we add that, it makes a total of 109 concepts.

Is it suspicious that there are more possible concepts than possible Days? Surely not: After all, a concept can be viewed as a *collection* of Days. A concept can be viewed as the set of days that it classifies positively, or isomorphically, the set of days that it classifies negatively.

So the space of *all possible* concepts that classify Days is the set of all possible sets of Days, whose size is $2^{24} = 16,777,216$.

This complete space includes all the concepts we have discussed so far. But it also includes concepts like “Positively classify only the examples {Sunny, Warm, High, Strong} and {Sunny, Warm, High, Weak} and reject everything else” or “Negatively classify only the example {Rainy, Cold, High, Strong} and accept everything else.” It includes concepts with no compact representation, just a flat list of what is and isn’t allowed.

That’s the problem with trying to build a “fully general” inductive learner: They can’t learn concepts until they’ve seen every possible example in the instance space.

If we add on more attributes to Days—like the Water temperature, or the Forecast for tomorrow—then the number of possible days will grow exponentially in the number of attributes. But this isn’t a problem with our restricted concept space, because you can narrow down a large space using a logarithmic number of examples.

Let’s say we add the Water: {Warm, Cold} attribute to days, which will make for 48 possible Days and 325 possible concepts. Let’s say that each Day we see is, usually, classified positive by around half of the currently-plausible concepts, and classified negative by the other half. Then when we learn the actual classification of the example, it will cut the space of

compatible concepts in half. So it might only take 9 examples ($2^9 = 512$) to narrow 325 possible concepts down to one.

Even if Days had forty binary attributes, it should still only take a manageable amount of data to narrow down the possible concepts to one. Sixty-four examples, if each example is classified positive by half the remaining concepts. *Assuming*, of course, that the *actual* rule is one we can represent at all!

If you want to think of all the possibilities, well, good luck with that. The space of *all possible* concepts grows *superexponentially* in the number of attributes.

By the time you're talking about data with forty binary attributes, the number of possible examples is past a trillion—but the number of possible *concepts* is past two-to-the-trillionth-power. To narrow down that *superexponential* concept space, you'd have to see over a trillion examples before you could say what was In, and what was Out. You'd have to see every possible example, in fact.

That's with forty binary attributes, mind you. Forty bits, or 5 bytes, to be classified simply "Yes" or "No." Forty bits implies 2^{40} possible examples, and $2^{2^{40}}$ possible concepts that classify those examples as positive or negative.

So, here in the real world, where objects take more than 5 bytes to describe *and* a trillion examples are not available *and* there is noise in the training data, we only even *think* about *highly regular* concepts. A human mind—or the whole observable universe—is not nearly large enough to consider all the other hypotheses.

From this perspective, learning doesn't just *rely on* inductive bias, it is *nearly all* inductive bias—when you compare the number of concepts ruled out a priori, to those ruled out by mere evidence.

But what has this (you inquire) to do with the proper use of words?

It's the whole reason that words have intensions as well as extensions.

In the last essay, I concluded:

The way to carve reality at its joints is to draw boundaries around concentrations of unusually high probability density.

I deliberately left out a key qualification in that (slightly edited) statement, because I couldn't explain it until now. A better statement would be:

The way to carve reality at its joints, is to draw *simple* boundaries around concentrations of unusually high probability density in Thingspace.

Otherwise you would just gerrymander Thingspace. You would create really odd noncontiguous boundaries that collected the observed examples, examples that couldn't be described in any shorter message than your observations themselves, and say: "This is what I've seen before, and what I expect to see more of in the future."

In the real world, nothing above the level of molecules repeats itself *exactly*. Socrates is shaped a lot like all those other humans who were vulnerable to hemlock, but he isn't shaped *exactly* like them. So your guess that Socrates is a "human" relies on drawing *simple* boundaries around the human cluster in Thingspace. Rather than, "Things shaped exactly like [5-megabyte shape specification 1] and with [lots of other characteristics], or exactly like [5-megabyte shape specification 2] and [lots of other characteristics], ..., are human."

If you don't draw *simple* boundaries around your experiences, you can't do inference with them. So you try to describe "art" with intensional definitions like "that which is intended to inspire any complex emotion for the sake of inspiring it," rather than just pointing at a long list of things that are, or aren't art.

In fact, the above statement about "how to carve reality at its joints" is a bit chicken-and-eggish: You can't assess the *density* of actual observations until you've already done at least a little carving. And the probability distribution comes from drawing the boundaries, not the other way around—if you already *had* the probability distribution, you'd have everything necessary for inference, so why would you bother drawing boundaries?

And this suggests another—yes, yet another—reason to be suspicious of the claim that “you can define a word any way you like.” When you consider the superexponential size of Conceptspace, it becomes clear that singling out one particular concept for consideration is an act of no small audacity—not just for us, but for any mind of bounded computing power.

Presenting us with the word “wiggin,” defined as “a black-haired green-eyed person,” without some reason for raising *this particular concept* to the level of our deliberate attention, is rather like a detective saying: “Well, I haven’t the slightest shred of support one way or the other for who could’ve murdered those orphans ...not even an intuition, mind you ...but have we considered John Q. Wiffleheim of 1234 Norkle Rd as a suspect?”

Conditional Independence, and Naive Bayes

Previously I spoke of mutual information between X and Y , written $I(X; Y)$, which is the difference between the entropy of the joint probability distribution, $H(X, Y)$, and the entropies of the marginal distributions, $H(X) + H(Y)$.

I gave the example of a variable X , having eight states, X_1 through X_8 , which are all equally probable if we have not yet encountered any evidence; and a variable Y , with states Y_1 through Y_4 , which are all equally probable if we have not yet encountered any evidence. Then if we calculate the marginal entropies $H(X)$ and $H(Y)$, we will find that X has 3 bits of entropy, and Y has 2 bits.

However, we also know that X and Y are both even or both odd; and this is all we know about the relation between them. So for the joint distribution (X, Y) there are only 16 possible states, all equally probable, for a joint entropy of 4 bits. This is a 1-bit entropy defect, compared to 5 bits of entropy if X and Y were independent. This entropy defect is the mutual information—the information that X tells us about Y , or vice versa, so that we are not as uncertain about one after having learned the other.

Suppose, however, that there exists a third variable Z . The variable Z has two states, “even” and “odd,” perfectly correlated to the evenness or oddness of (X, Y) . In fact, we’ll suppose that Z is just the question “Are X and Y even or odd?”

If we have no evidence about X and Y , then Z itself necessarily has 1 bit of entropy on the information given. There is 1 bit of mutual information between Z and X , and 1 bit of mutual information between Z and Y . And, as previously noted, 1 bit of mutual information between X and Y . So how much entropy for the whole system (X, Y, Z) ? You might naively expect that



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$$H(X, Y, Z) = H(X) + H(Y) + H(Z) \\ - I(X; Z) - I(Z; Y) - I(X; Y),$$

but this turns out not to be the case.

The joint system (X, Y, Z) only has 16 possible states—since Z is just the question “Are X and Y even or odd?”—so $H(X, Y, Z) = 4$ bits.

But if you calculate the formula just given, you get

$$(3 + 2 + 1 - 1 - 1 - 1) \text{ bits} = 3 \text{ bits} = \text{WRONG!}$$

Why? Because if you have the mutual information between X and Z , and the mutual information between Z and Y , that may include some of the *same* mutual information that we’ll calculate exists between X and Y . In this case, for example, knowing that X is even tells us that Z is even, and knowing that Z is even tells us that Y is even, but this is the same information that X would tell us about Y . We double-counted some of our knowledge, and so came up with too little entropy.

The correct formula is (I believe):

$$H(X, Y, Z) = H(X) + H(Y) + H(Z) \\ - I(X; Z) - I(Z; Y) - I(X; Y|Z).$$

Here the last term, $I(X; Y|Z)$, means, “the information that X tells us about Y , given that we already know Z .” In this case, X doesn’t tell us anything about Y , given that we already know Z , so the term comes out as zero—and the equation gives the correct answer. There, isn’t that nice?

“No,” you correctly reply, “for you have not told me how to *calculate* $I(X; Y|Z)$, only given me a verbal argument that it ought to be zero.”

We calculate $I(X; Y|Z)$ just the way you would expect. We know $I(X; Y) = H(X) + H(Y) - H(X, Y)$, so

$$I(X; Y|Z) = H(X|Z) + H(Y|Z) - H(X, Y|Z).$$

And now, I suppose, you want to know how to calculate the conditional entropy? Well, the *original* formula for the entropy is

$$H(S) = \sum_i -P(S_i) \times \log_2(P(S_i)).$$

If we then learned a new fact Z_0 , our remaining uncertainty about S would be

$$H(S|Z_0) = \sum_i -P(S_i|Z_0) \log_2(P(S_i|Z_0)).$$

So if we're going to learn a new fact Z , but we don't know which Z yet, then, on average, we expect to be around this uncertain of S afterward:

$$H(S|Z) = \sum_j \left(P(Z_j) \sum_i -P(S_i|Z_j) \log_2(P(S_i|Z_j)) \right)$$

And that's how one calculates conditional entropies; from which, in turn, we can get the conditional mutual information.

There are *all sorts* of ancillary theorems here, like

$$H(X|Y) = H(X, Y) - H(Y)$$

and

$$\begin{aligned} &\text{if } I(X; Z) = 0 \text{ and } I(Y; X|Z) = 0 \\ &\text{then } I(X; Y) = 0, \end{aligned}$$

but I'm not going to go into those.

"But," you ask, "what does *this* have to do with the nature of words and their hidden Bayesian structure?"

I am just so *unspeakably* glad that you asked that question, because I was planning to tell you whether you liked it or not. But first there are a couple more preliminaries.

You will remember—yes, you *will* remember—that there is a duality between mutual information and Bayesian evidence.

Mutual information is positive if and only if the probability of at least some joint events $P(x, y)$ does not equal the product of the probabilities of the separate events $P(x)P(y)$. This, in turn, is exactly equivalent to the condition that Bayesian evidence exists between x and y :

$$\begin{aligned} I(X; Y) > 0 &\Rightarrow \\ P(x, y) &\neq P(x)P(y) \\ P(x, y)/P(y) &\neq P(x) \\ P(x|y) &\neq P(x). \end{aligned}$$

If you're conditioning on Z , you just adjust the whole derivation accordingly:

$$\begin{aligned} I(X; Y|Z) > 0 &\Rightarrow \\ P(x, y|z) &\neq P(x|z)P(y|z) \\ P(x, y|z)/P(y|z) &\neq P(x|z) \\ (P(x, y, z)/P(z))/(P(y, z)/P(z)) &\neq P(x|z) \\ P(x, y, z)/P(y, z) &\neq P(x|z) \\ P(x|y, z) &\neq P(x|z). \end{aligned}$$

Which last line reads “Even knowing Z , learning Y still changes our beliefs about X .”

Conversely, as in our original case of Z being “even” or “odd,” Z screens off X from Y —that is, if we know that Z is “even,” learning that Y is in state Y_4 tells us *nothing more* about whether X is X_2 , X_4 , X_6 , or X_8 . Or if we know that Z is “odd,” then learning that X is X_5 tells us nothing more about whether Y is Y_1 or Y_3 . Learning Z has rendered X and Y *conditionally independent*.

Conditional independence is a hugely important concept in probability theory—to cite just one example, without conditional independence, the universe would have no structure.

Here, though, I only intend to talk about one particular kind of conditional independence—the case of a central variable that screens off other variables surrounding it, like a central body with tentacles.

Let there be five variables U , V , W , X , and Y ; and moreover, suppose that for every pair of these variables, one variable is evidence about the other. If you select U and W , for example, then learning $U = U_1$ will tell you something you didn't know before about the probability that $W = W_1$.

An unmanageable inferential mess? Evidence gone wild? Not necessarily.

Maybe U is "Speaks a language," V is "Two arms and ten digits," W is "Wears clothes," X is "Poisonable by hemlock," and Y is "Red blood." Now if you encounter a thing-in-the-world, that might be an apple and might be a rock, and you learn that this thing speaks Chinese, you are liable to assess a much higher probability that it wears clothes; and if you learn that the thing is not poisonable by hemlock, you will assess a somewhat lower probability that it has red blood.

Now some of these rules are stronger than others. There is the case of Fred, who is missing a finger due to a volcano accident, and the case of Barney the Baby who doesn't speak yet, and the case of Irving the IRCBot who emits sentences but has no blood. So if we learn that a certain thing is not wearing clothes, that doesn't screen off everything that its speech capability can tell us about its blood color. If the thing doesn't wear clothes but *does* talk, maybe it's Nude Nellie.

This makes the case more interesting than, say, five integer variables that are all odd or all even, but otherwise uncorrelated. In that case, knowing *any* one of the variables would screen off everything that knowing a second variable could tell us about a third variable.

But here, we have dependencies that don't go away as soon as we learn just one variable, as the case of Nude Nellie shows. So is it an unmanageable inferential inconvenience?

Fear not! For there may be some *sixth* variable Z , which, if we knew it, really *would* screen off every pair of variables from each other. There may be some variable Z —even if we have to *construct* Z rather than observing it directly—such that:

$$P(U|V, W, X, Y, Z) = P(U|Z)$$

$$P(V|U, W, X, Y, Z) = P(V|Z)$$

$$P(W|U, V, X, Y, Z) = P(W|Z)$$

$$\vdots$$

Perhaps, *given that* a thing is “human,” then the probabilities of it speaking, wearing clothes, and having the standard number of fingers, are all independent. Fred may be missing a finger—but he is no more likely to be a nudist than the next person; Nude Nellie never wears clothes, but knowing this doesn’t make it any less likely that she speaks; and Baby Barney doesn’t talk yet, but is not missing any limbs.

This is called the “Naive Bayes” method, because it usually isn’t quite true, but *pretending* that it’s true can simplify the living daylights out of your calculations. We don’t keep separate track of the influence of clothed-ness on speech capability given finger number. We just use all the information we’ve observed to keep track of the probability that this thingy is a human (or alternatively, something else, like a chimpanzee or robot) and then use our beliefs about the central class to predict anything we haven’t seen yet, like vulnerability to hemlock.

Any observations of U , V , W , X , and Y just act as evidence for the central class variable Z , and then we use the posterior distribution on Z to make any predictions that need making about unobserved variables in U , V , W , X , and Y .

Sound familiar? It should; see Figure 177.1.

As a matter of fact, if you use the right kind of neural network units, this “neural network” ends up *exactly, mathematically* equivalent to Naive Bayes. The central unit just needs a logistic threshold—an S-curve response—and the weights of the inputs just need to match the logarithms of the likelihood ratios, et cetera. In fact, it’s a good guess that this is one of the reasons why logistic response often works so well in neural networks—it lets the algorithm sneak in a little Bayesian reasoning while the designers aren’t looking.

Just because someone is presenting you with an algorithm that they call a “neural network” with buzzwords like “scruffy”

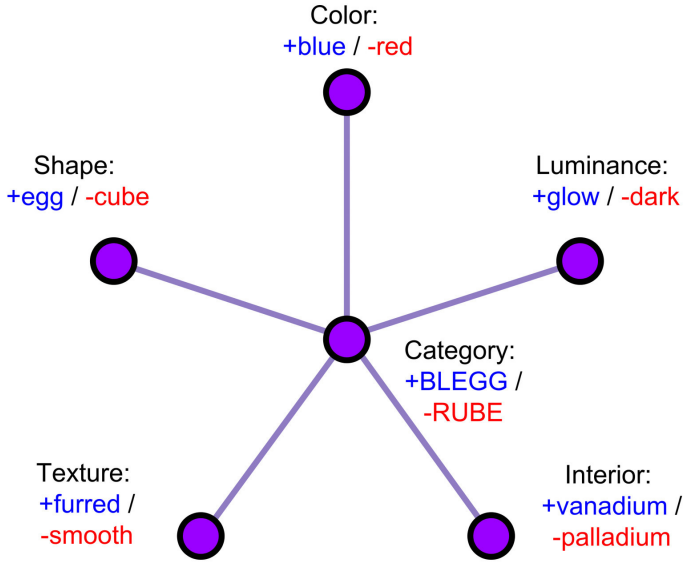


Figure 177.1 Network 2

and “emergent” plastered all over it, disclaiming proudly that they have no idea how the learned network works—well, don’t assume that their little AI algorithm *really is* Beyond the Realms of Logic. For this paradigm of adhockery, if it works, will turn out to have Bayesian structure; it may even be exactly equivalent to an algorithm of the sort called “Bayesian.”

Even if it doesn’t *look* Bayesian, on the surface.

And then you just *know* that the Bayesians are going to start explaining exactly how the algorithm works, what underlying assumptions it reflects, which environmental regularities it exploits, where it works and where it fails, and even attaching understandable meanings to the learned network weights.

Disappointing, isn’t it?

Words as Mental Paintbrush Handles



discussion
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3-c50

Suppose I tell you: “It’s the strangest thing: The lamps in this hotel have triangular lightbulbs.”

You may or may not have visualized it—if you haven’t done it yet, do so now—what, in your mind’s eye, does a “triangular lightbulb” look like?

In your mind’s eye, did the glass have sharp edges, or smooth?

When the phrase “triangular lightbulb” first crossed my mind—no, the hotel doesn’t have them—then as best as my introspection could determine, I first saw a pyramidal lightbulb with sharp edges, then (almost immediately) the edges were smoothed, and then my mind generated a loop of fluorescent bulb in the shape of a smooth triangle as an alternative.

As far as I can tell, no deliberative/verbal thoughts were involved—just wordless reflex flinch away from the imaginary mental vision of sharp glass, which design problem was solved before I could even think in words.

Believe it or not, for some decades, there was a serious debate about whether people *really* had mental images in their mind—an actual *picture* of a chair somewhere—or if people just naively *thought* they had mental images (having been misled by “introspection,” a very bad forbidden activity), while actually just having a little “chair” label, like a LISP token, active in their brain.

I am trying hard not to say anything like “How spectacularly silly,” because there is always the hindsight effect to consider, but: how spectacularly silly.

This academic paradigm, I think, was mostly a deranged legacy of behaviorism, which denied the existence of thoughts in humans, and sought to explain all human phenomena as “reflex,” including speech. Behaviorism probably deserves its own write at some point, as it was a perversion of rationalism; but this is not that write.

“You call it ‘silly,’” you inquire, “but how do you *know* that your brain represents visual images? Is it merely that you can close your eyes and see them?”

This question *used* to be harder to answer, back in the day of the controversy. If you wanted to prove the existence of mental imagery “scientifically,” rather than just by introspection, you had to infer the existence of mental imagery from experiments like this: Show subjects two objects and ask them if one can be rotated into correspondence with the other. The response time is linearly proportional to the angle of rotation required. This is easy to explain if you are actually visualizing the image and continuously rotating it at a constant speed, but hard to explain if you are just checking propositional features of the image.

Today we can actually neuroimage the little pictures in the visual cortex. So, yes, your brain really does represent a detailed image of what it sees or imagines. See Stephen Kosslyn’s *Image and Brain: The Resolution of the Imagery Debate*.¹

Part of the reason people get in trouble with words, is that they do not realize how much complexity lurks behind words.

Can you visualize a “green dog”? Can you visualize a “cheese apple”?

“Apple” isn’t just a sequence of two syllables or five letters. That’s a shadow. That’s the tip of the tiger’s tail.

Words, or rather the concepts behind them, are paintbrushes—you can use them to draw images in your own mind. Literally draw, if you employ concepts to make a picture in your visual cortex. And by the use of shared labels, you can reach into someone else’s mind, and grasp their paintbrushes to draw pictures in *their* minds—sketch a little green dog in their visual cortex.

But don’t think that, because you send syllables through the air, or letters through the Internet, it is the syllables or the letters that draw pictures in the visual cortex. That takes some complex instructions that wouldn’t fit in the sequence of

1. Stephen M. Kosslyn, *Image and Brain: The Resolution of the Imagery Debate* (Cambridge, MA: MIT Press, 1994).

letters. “Apple” is 5 bytes, and drawing a picture of an apple from scratch would take more data than that.

“Apple” is merely the tag attached to the true and wordless *apple* concept, which can paint a picture in your visual cortex, or collide with “cheese,” or recognize an apple when you see one, or taste its archetype in apple pie, maybe even send out the motor behavior for eating an apple ...

And it’s not as simple as just calling up a picture from memory. Or how would you be able to visualize combinations like a “triangular lightbulb”—imposing triangelness on lightbulbs, keeping the essence of both, even if you’ve never seen such a thing in your life?

Don’t make the mistake the behaviorists made. There’s far more to speech than sound in air. The labels are just pointers—“look in memory area 1387540.” Sooner or later, when you’re handed a pointer, it comes time to dereference it, and actually look in memory area 1387540.

What does a word point to?

Variable Question Fallacies

ALBERT: “Every time I’ve listened to a tree fall, it made a sound, so I’ll guess that other trees falling also make sounds. I don’t believe the world changes around when I’m not looking.”

BARRY: “Wait a minute. If no one hears it, how can it be a sound?”



discussion
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3-c51

While writing the dialogue of Albert and Barry in their dispute over whether a falling tree in a deserted forest makes a sound, I sometimes found myself losing empathy with my characters. I would start to lose the gut feel of why *anyone* would ever argue like that, even though I’d seen it happen many times.

On these occasions, I would repeat to myself, “Either the falling tree makes a sound, or it does not!” to restore my borrowed sense of indignation.

(P or $\neg P$) is not always a reliable heuristic, if you substitute arbitrary English sentences for P . “This sentence is false” cannot be consistently viewed as true or false. And then there’s the old classic, “Have you stopped beating your wife?”

Now if you are a mathematician, and one who believes in classical (rather than intuitionistic) logic, there are ways to continue insisting that (P or $\neg P$) is a theorem: for example, saying that “This sentence is false” is not a sentence.

But such resolutions are subtle, which suffices to demonstrate a need for subtlety. You cannot just bull ahead on every occasion with “Either it does or it doesn’t!”

So does the falling tree make a sound, or not, or ...?

Surely, $2 + 2 = X$ or it does not? Well, maybe, *if* it’s *really* the same X , the same 2, and the same $+$ and $=$. If X evaluates to 5 on some occasions and 4 on another, your indignation may be misplaced.

To even begin claiming that $(P \text{ or } \neg P)$ ought to be a necessary truth, the symbol P must stand for *exactly* the same thing in both halves of the dilemma. “Either the fall makes a sound, or not!”—but if Albert::sound is not the same as Barry::sound, there is nothing paradoxical about the tree making an Albert::sound but not a Barry::sound.

(The :: idiom is something I picked up in my C++ days for avoiding namespace collisions. If you’ve got two different packages that define a class Sound, you can write `Package1::Sound` to specify which Sound you mean. The idiom is not widely known, I think; which is a pity, because I often wish I could use it in writing.)

The variability may be subtle: Albert and Barry may carefully verify that it is the same tree, in the same forest, and the same occasion of falling, just to ensure that they really do have a substantive disagreement about exactly the same event. And then forget to check that they are matching this event against exactly the same concept.

Think about the grocery store that you visit most often: Is it on the left side of the street, or the right? But of course there is no “*the* left side” of the street, only *your* left side, as you travel along it from some particular direction. Many of the words we use are really functions of implicit variables supplied by context.

It’s actually one heck of a pain, requiring one heck of a lot of work, to handle this kind of problem in an Artificial Intelligence program intended to parse language—the phenomenon going by the name of “speaker deixis.”

“Martin told Bob the building was on his left.” But “left” is a function-word that evaluates with a speaker-dependent variable invisibly grabbed from the surrounding context. Whose “left” is meant, Bob’s or Martin’s?

The variables in a variable question fallacy often aren’t neatly labeled—it’s not as simple as “Say, do you think $Z + 2$ equals 6?”

If a namespace collision introduces two different concepts that look like “the same concept” because they have the same name—or a map compression introduces two different events

that look like the same event because they don't have separate mental files—or the same function evaluates in different contexts—then reality itself becomes protean, changeable. At least that's what the algorithm feels like from inside. Your mind's eye sees the map, not the territory directly.

If you have a question with a hidden variable, that evaluates to different expressions in different contexts, it *feels like* reality itself is unstable—what your mind's eye sees, shifts around depending on where it looks.

This often confuses undergraduates (and postmodernist professors) who discover a sentence with more than one interpretation; they think they have discovered an unstable portion of reality.

“Oh my gosh! ‘The Sun goes around the Earth’ is true for Hunga Huntergatherer, but for Amara Astronomer, ‘The Sun goes around the Earth’ is false! There is no fixed truth!” The deconstruction of this sophomoric nitwittery is left as an exercise to the reader.

And yet, even I initially found myself writing “If X is 5 on some occasions and 4 on another, the sentence ‘ $2 + 2 = X$ ’ may have no fixed truth-value.” There is not *one* sentence with a *variable* truth-value. “ $2 + 2 = X$ ” *has no* truth-value. It is not a *proposition*, not yet, not as mathematicians define proposition-ness, any more than “ $2 + 2 =$ ” is a proposition, or “Fred jumped over the” is a grammatical sentence.

But this fallacy tends to sneak in, even when you allegedly know better, because, well, that's how the algorithm feels from inside.

37 Ways That Words Can Be Wrong



discussion
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3-c52

Some reader is bound to declare that a better title for this essay would be “37 Ways That You Can Use Words Unwisely,” or “37 Ways That Suboptimal Use Of Categories Can Have Negative Side Effects On Your Cognition.”

But one of the primary lessons of this gigantic list is that saying “There’s no way my choice of *X* can be ‘wrong’” is nearly always an error in practice, whatever the theory. You can always be wrong. Even when it’s theoretically impossible to be wrong, you can still be wrong. There is never a Get Out of Jail Free card for anything you do. That’s life.

Besides, I can define the word “wrong” to mean anything I like—it’s not like a word can be *wrong*.

Personally, I think it quite justified to use the word “wrong” when:

1. *A word fails to connect to reality in the first place.* Is Socrates a framster? Yes or no? (The Parable of the Dagger)
2. *Your argument, if it worked, could coerce reality to go a different way by choosing a different word definition.* Socrates is a human, and humans, by definition, are mortal. So if you defined humans to not be mortal, would Socrates live forever? (The Parable of Hemlock)
3. *You try to establish any sort of empirical proposition as being true “by definition.”* Socrates is a human, and humans, by definition, are mortal. So is it a *logical truth* if we empirically predict that Socrates should keel over if he drinks hemlock? It seems like there are logically possible, non-self-contradictory worlds where Socrates doesn’t keel over—where he’s immune to hemlock by a quirk of biochemistry, say. Logical truths are true in all possible worlds, and so never tell you *which* possible

world you live in—and anything you can establish “by definition” is a logical truth. (The Parable of Hemlock)

4. *You unconsciously slap the conventional label on something, without actually using the verbal definition you just gave.* You know perfectly well that Bob is “human,” even though, by your definition, you can never call Bob “human” without first observing him to be mortal. (The Parable of Hemlock)
5. *The act of labeling something with a word disguises a challengeable inductive inference you are making.* If the last 11 egg-shaped objects drawn have been blue, and the last 8 cubes drawn have been red, it is a matter of induction to say this rule will hold in the future. But if you call the blue eggs “bleggs” and the red cubes “rubes,” you may reach into the barrel, feel an egg shape, and think “Oh, a blegg.” (Words as Hidden Inferences)
6. *You try to define a word using words, in turn defined with ever-more-abstract words, without being able to point to an example.* “What is red?” “Red is a color.” “What’s a color?” “It’s a property of a thing.” “What’s a thing? What’s a property?” It never occurs to you to point to a stop sign and an apple. (Extensions and Intensions)
7. *The extension doesn’t match the intension.* We aren’t consciously aware of our identification of a red light in the sky as “Mars,” which will probably happen regardless of your attempt to define “Mars” as “The God of War.” (Extensions and Intensions)
8. *Your verbal definition doesn’t capture more than a tiny fraction of the category’s shared characteristics, but you try to reason as if it does.* When the philosophers of Plato’s Academy claimed that the best definition of a human was a “featherless biped,” Diogenes the Cynic is said to have exhibited a plucked chicken and declared “Here is Plato’s Man.” The Platonists promptly changed their definition

to “a featherless biped with broad nails.” (Similarity Clusters)

9. *You try to treat category membership as all-or-nothing, ignoring the existence of more and less typical subclusters.* Ducks and penguins are less typical birds than robins and pigeons. Interestingly, a between-groups experiment showed that subjects thought a disease was more likely to spread from robins to ducks on an island, than from ducks to robins. (Typicality and Asymmetrical Similarity)
10. *A verbal definition works well enough in practice to point out the intended cluster of similar things, but you nitpick exceptions.* Not every human has ten fingers, or wears clothes, or uses language; but if you look for an empirical cluster of things which share these characteristics, you’ll get enough information that the occasional nine-fingered human won’t fool you. (The Cluster Structure of Thingspace)
11. *You ask whether something “is” or “is not” a category member but can’t name the question you really want answered.* What is a “man”? Is Barney the Baby Boy a “man”? The “correct” answer may depend considerably on whether the query you *really* want answered is “Would hemlock be a good thing to feed Barney?” or “Will Barney make a good husband?” (Disguised Queries)
12. *You treat intuitively perceived hierarchical categories like the only correct way to parse the world, without realizing that other forms of statistical inference are possible even though your brain doesn’t use them.* It’s much easier for a human to notice whether an object is a “blegg” or “rube”; than for a human to notice that red objects never glow in the dark, but red furred objects have all the other characteristics of bleggs. Other statistical algorithms work differently. (Neural Categories)
13. *You talk about categories as if they are manna fallen from the Platonic Realm, rather than inferences implemented in*

a real brain. The ancient philosophers said “Socrates is a man,” not, “My brain perceptually classifies Socrates as a match against the ‘human’ concept.” (How An Algorithm Feels From Inside)

14. *You argue about a category membership even after screening off all questions that could possibly depend on a category-based inference.* After you observe that an object is blue, egg-shaped, furred, flexible, opaque, luminescent, and palladium-containing, what’s *left* to ask by arguing, “Is it a blegg?” But if your brain’s categorizing neural network contains a (metaphorical) central unit corresponding to the inference of blegg-ness, it may still *feel* like there’s a leftover question. (How An Algorithm Feels From Inside)
15. *You allow an argument to slide into being about definitions, even though it isn’t what you originally wanted to argue about.* If, before a dispute started about whether a tree falling in a deserted forest makes a “sound,” you asked the two soon-to-be arguers whether they thought a “sound” should be defined as “acoustic vibrations” or “auditory experiences,” they’d probably tell you to flip a coin. Only after the argument starts does the definition of a word become politically charged. (Disputing Definitions)
16. *You think a word has a meaning, as a property of the word itself; rather than there being a label that your brain associates to a particular concept.* When someone shouts “Yikes! A tiger!,” evolution would not favor an organism that thinks, “Hm...I have just heard the syllables ‘Tie’ and ‘Grr’ which my fellow tribemembers associate with their internal analogues of my own *tiger* concept and which *aiiiieee CRUNCH CRUNCH GULP*.” So the brain takes a shortcut, and it seems that the meaning of tigerness is a property of the label itself. People argue about the *correct meaning* of a label like “sound.” (Feel the Meaning)
17. *You argue over the meanings of a word, even after all sides understand perfectly well what the other sides are trying to*

say. The human ability to associate labels to concepts is a tool for communication. When people *want* to communicate, we're hard to stop; if we have no common language, we'll draw pictures in sand. When you each understand what is in the other's mind, you are *done*. (The Argument From Common Usage)

18. *You pull out a dictionary in the middle of an empirical or moral argument.* Dictionary editors are historians of usage, not legislators of language. If the common definition contains a problem—if “Mars” is defined as the God of War, or a “dolphin” is defined as a kind of fish, or “Negroes” are defined as a separate category from humans, the dictionary will reflect the standard mistake. (The Argument From Common Usage)
19. *You pull out a dictionary in the middle of any argument ever.* Seriously, what the heck makes you think that dictionary editors are an authority on whether “atheism” is a “religion” or whatever? If you have any substantive issue whatsoever at stake, do you really think dictionary editors have access to ultimate wisdom that settles the argument? (The Argument From Common Usage)
20. *You defy common usage without a reason, making it gratuitously hard for others to understand you.* Fast stand up plutonium, with bagels without handle. (The Argument From Common Usage)
21. *You use complex renamings to create the illusion of inference.* Is a “human” defined as a “mortal featherless biped”? Then write: “All [mortal featherless bipeds] are mortal; Socrates is a [mortal featherless biped]; therefore, Socrates is mortal.” Looks less impressive that way, doesn't it? (Empty Labels)
22. *You get into arguments that you could avoid if you just didn't use the word.* If Albert and Barry aren't allowed to use the word “sound,” then Albert will have to say “A tree falling in a deserted forest generates acoustic vibrations,”

and Barry will say “A tree falling in a deserted forest generates no auditory experiences.” When a word poses a problem, the simplest solution is to eliminate the word and its synonyms. (Taboo Your Words)

23. *The existence of a neat little word prevents you from seeing the details of the thing you’re trying to think about.* What actually goes on in schools once you stop calling it “education”? What’s a degree, once you stop calling it a “degree”? If a coin lands “heads,” what’s its radial orientation? What is “truth,” if you can’t say “accurate” or “correct” or “represent” or “reflect” or “semantic” or “believe” or “knowledge” or “map” or “real” or any other simple term? (Replace the Symbol with the Substance)
24. *You have only one word, but there are two or more different things-in-reality, so that all the facts about them get dumped into a single undifferentiated mental bucket.* It’s part of a detective’s ordinary work to observe that Carol wore red last night, or that she has black hair; and it’s part of a detective’s ordinary work to wonder if maybe Carol dyes her hair. But it takes a subtler detective to wonder if there are two Carols, so that the Carol who wore red is not the same as the Carol who had black hair. (Fallacies of Compression)
25. *You see patterns where none exist, harvesting other characteristics from your definitions even when there is no similarity along that dimension.* In Japan, it is thought that people of blood type A are earnest and creative, blood type Bs are wild and cheerful, blood type Os are agreeable and sociable, and blood type ABs are cool and controlled. (Categorizing Has Consequences)
26. *You try to sneak in the connotations of a word, by arguing from a definition that doesn’t include the connotations.* A “wiggyn” is defined in the dictionary as a person with green eyes and black hair. The word “wiggyn” also carries the connotation of someone who commits crimes and launches cute baby squirrels, but that part isn’t in the

dictionary. So you point to someone and say: “Green eyes? Black hair? See, told you he’s a wiggin! Watch, next he’s going to steal the silverware.” (Sneaking in Connotations)

27. *You claim “X, by definition, is a Y!” On such occasions you’re almost certainly trying to sneak in a connotation of Y that wasn’t in your given definition.* You define “human” as a “featherless biped,” and point to Socrates and say, “No feathers—two legs—he must be human!” But what you *really* care about is something else, like mortality. If what was in dispute was Socrates’s number of legs, the other fellow would just reply, “Whaddaya mean, Socrates’s got two legs? That’s what we’re arguing about in the first place!” (Arguing “By Definition”)
28. *You claim “Ps, by definition, are Qs!”* If you see Socrates out in the field with some biologists, gathering herbs that might confer resistance to hemlock, there’s no point in arguing “Men, by definition, are mortal!” The main time you feel the need to tighten the vise by insisting that something is true “by definition” is when there’s other information that calls the default inference into doubt. (Arguing “By Definition”)
29. *You try to establish membership in an empirical cluster “by definition.”* You wouldn’t feel the need to say, “Hinduism, *by definition*, is a religion!” because, well, of course Hinduism is a religion. It’s not just a religion “by definition,” it’s, like, an *actual* religion. Atheism does not resemble the central members of the “religion” cluster, so if it wasn’t for the fact that atheism is a religion *by definition*, you might go around thinking that atheism *wasn’t* a religion. That’s why you’ve got to crush all opposition by pointing out that “Atheism is a religion” is true *by definition*, because it isn’t true any other way. (Arguing “By Definition”)
30. *Your definition draws a boundary around things that don’t really belong together.* You can claim, if you like, that you

are *defining* the word “fish” to refer to salmon, guppies, sharks, dolphins, and trout, but not jellyfish or algae. You can claim, if you like, that this is merely a list, and there is no way a list can be “wrong.” Or you can stop playing games and admit that you made a mistake and that dolphins don’t belong on the fish list. (Where to Draw the Boundary?)

31. *You use a short word for something that you won’t need to describe often, or a long word for something you’ll need to describe often. This can result in inefficient thinking, or even misapplications of Occam’s Razor, if your mind thinks that short sentences sound “simpler.”* Which sounds more plausible, “God did a miracle” or “A supernatural universe-creating entity temporarily suspended the laws of physics”? (Entropy, and Short Codes)
32. *You draw your boundary around a volume of space where there is no greater-than-usual density, meaning that the associated word does not correspond to any performable Bayesian inferences.* Since green-eyed people are not more likely to have black hair, or vice versa, and they don’t share any other characteristics in common, why have a word for “wiggin”? (Mutual Information, and Density in Thingspace)
33. *You draw an unsimple boundary without any reason to do so.* The act of defining a word to refer to all humans, except black people, seems kind of suspicious. If you don’t present reasons to draw that particular boundary, trying to create an “arbitrary” word in that location is like a detective saying: “Well, I haven’t the slightest shred of support one way or the other for who could’ve murdered those orphans...but have we considered John Q. Wif-fleheim as a suspect?” (Superexponential Conceptsace, and Simple Words)
34. *You use categorization to make inferences about properties that don’t have the appropriate empirical structure, namely, conditional independence given knowledge of the class, to*

be well-approximated by Naive Bayes. No way am I trying to summarize this one. Just read the essay. (Conditional Independence, and Naive Bayes)

35. *You think that words are like tiny little LISP symbols in your mind, rather than words being labels that act as handles to direct complex mental paintbrushes that can paint detailed pictures in your sensory workspace.* Visualize a “triangular lightbulb.” What did you see? (Words as Mental Paintbrush Handles)
36. *You use a word that has different meanings in different places as though it meant the same thing on each occasion, possibly creating the illusion of something protean and shifting.* “Martin told Bob the building was on his left.” But “left” is a function-word that evaluates with a speaker-dependent variable grabbed from the surrounding context. Whose “left” is meant, Bob’s or Martin’s? (Variable Question Fallacies)
37. *You think that definitions can’t be “wrong,” or that “I can define a word any way I like!”* This kind of attitude teaches you to indignantly defend your past actions, instead of paying attention to their consequences, or fessing up to your mistakes. (37 Ways That Suboptimal Use Of Categories Can Have Negative Side Effects On Your Cognition)

Everything you do in the mind has an effect, and your brain races ahead unconsciously without your supervision.

Saying “Words are arbitrary; I can define a word any way I like” makes around as much sense as driving a car over thin ice with the accelerator floored and saying, “Looking at this steering wheel, I can’t see why one radial angle is special—so I can turn the steering wheel any way I like.”

If you’re trying to go anywhere, or even just trying to *survive*, you had better start paying attention to the three or six dozen optimality criteria that control how you use words, definitions, categories, classes, boundaries, labels, and concepts.

Interlude: An Intuitive Explanation of Bayes's Theorem

Editor's Note: This is an abridgement of the original¹ version of this essay, which contained many interactive elements.



discussion
sharov.me/r/
3-c53

Your friends and colleagues are talking about something called “Bayes’s Theorem” or “Bayes’s Rule,” or something called Bayesian reasoning. They sound really enthusiastic about it, too, so you google and find a web page about Bayes’s Theorem and ...

It’s this equation. That’s all. Just one equation. The page you found gives a definition of it, but it doesn’t say what it is, or why it’s useful, or why your friends would be interested in it. It looks like this random statistics thing.

Why does a mathematical concept generate this strange enthusiasm in its students? What is the so-called Bayesian Revolution now sweeping through the sciences, which claims to subsume even the experimental method itself as a special case? What is the secret that the adherents of Bayes know? What is the light that they have seen?

Soon you will know. Soon you will be one of us.

While there are a few existing online explanations of Bayes’s Theorem, my experience with trying to introduce people to Bayesian reasoning is that the existing online explanations are too abstract. Bayesian reasoning is very *counterintuitive*. People do not employ Bayesian reasoning intuitively, find it very difficult to learn Bayesian reasoning when tutored, and rapidly forget Bayesian methods once the tutoring is over. This holds equally true for novice students and highly trained professionals in a field. Bayesian reasoning is apparently one of those things which, like quantum mechanics or the Wason

1. sharov.me/r/3-17

An Intuitive Explanation of Bayes's Theorem – Eliezer S. Yudkowsky • yudkowsky.net



Selection Test, is inherently difficult for humans to grasp with our built-in mental faculties.

Or so they claim. Here you will find an attempt to offer an *intuitive* explanation of Bayesian reasoning—an excruciatingly gentle introduction that invokes all the human ways of grasping numbers, from natural frequencies to spatial visualization. The intent is to convey, not abstract rules for manipulating numbers, but what the numbers mean, and why the rules are what they are (and cannot possibly be anything else). When you are finished reading this, you will see Bayesian problems in your dreams.

And let's begin.

Here's a story problem about a situation that doctors often encounter:

1% of women at age forty who participate in routine screening have breast cancer. 80% of women with breast cancer will get positive mammographies. 9.6% of women without breast cancer will also get positive mammographies. A woman in this age group had a positive mammography in a routine screening. What is the probability that she actually has breast cancer?

What do you think the answer is? If you haven't encountered this kind of problem before, please take a moment to come up with your own answer before continuing.

Next, suppose I told you that most doctors get the same wrong answer on this problem—usually, only around 15% of doctors get it right. (“Really? 15%? Is that a real number, or an urban legend based on an Internet poll?” It’s a real number.

See Casscells, Schoenberger, and Graboys 1978;² Eddy 1982;³ Gigerenzer and Hoffrage 1995;⁴ and many other studies. It's a surprising result which is easy to replicate, so it's been extensively replicated.)

On the story problem above, most doctors estimate the probability to be between 70% and 80%, which is wildly incorrect.

Here's an alternate version of the problem on which doctors fare somewhat better:

10 out of 1,000 women at age forty who participate in routine screening have breast cancer. 800 out of 1,000 women with breast cancer will get positive mammographies. 96 out of 1,000 women without breast cancer will also get positive mammographies. If 1,000 women in this age group undergo a routine screening, about what fraction of women with positive mammographies will actually have breast cancer?

And finally, here's the problem on which doctors fare best of all, with 46%—nearly half—arriving at the correct answer:

100 out of 10,000 women at age forty who participate in routine screening have breast cancer. 80 of every 100 women with breast cancer will get a positive mammography. 950 out of 9,900 women without breast cancer will also get a positive mammography. If 10,000 women in this age group undergo a routine screening, about what fraction of

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2. Ward Casscells, Arno Schoenberger, and Thomas Graboys, "Interpretation by Physicians of Clinical Laboratory Results," *New England Journal of Medicine* 299 (1978): 999–1001.
 3. David M. Eddy, "Probabilistic Reasoning in Clinical Medicine: Problems and Opportunities," in *Judgement Under Uncertainty: Heuristics and Biases*, ed. Daniel Kahneman, Paul Slovic, and Amos Tversky (Cambridge University Press, 1982).
 4. Gerd Gigerenzer and Ulrich Hoffrage, "How to Improve Bayesian Reasoning without Instruction: Frequency Formats," *Psychological Review* 102 (1995): 684–704.

women with positive mammographies will actually have breast cancer?

The correct answer is 7.8%, obtained as follows: Out of 10,000 women, 100 have breast cancer; 80 of those 100 have positive mammographies. From the same 10,000 women, 9,900 will not have breast cancer and of those 9,900 women, 950 will also get positive mammographies. This makes the total number of women with positive mammographies $950 + 80$ or 1,030. Of those 1,030 women with positive mammographies, 80 will have cancer. Expressed as a proportion, this is $80/1,030$ or 0.07767 or 7.8%.

To put it another way, before the mammography screening, the 10,000 women can be divided into two groups:

- Group 1: 100 women *with* breast cancer.
- Group 2: 9,900 women *without* breast cancer.

Summing these two groups gives a total of 10,000 patients, confirming that none have been lost in the math. After the mammography, the women can be divided into four groups:

- Group A: 80 women *with* breast cancer and a *positive* mammography.
- Group B: 20 women *with* breast cancer and a *negative* mammography.
- Group C: 950 women *without* breast cancer and a *positive* mammography.
- Group D: 8,950 women *without* breast cancer and a *negative* mammography.

The sum of groups A and B, the groups with breast cancer, corresponds to group 1; and the sum of groups C and D, the groups without breast cancer, corresponds to group 2. If you administer a mammography to 10,000 patients, then out

of the 1,030 with positive mammographies, eighty of those positive-mammography patients will have cancer. This is the correct answer, the answer a doctor should give a positive-mammography patient if she asks about the chance she has breast cancer; if thirteen patients ask this question, roughly one out of those thirteen will have cancer.

The most common mistake is to ignore the original fraction of women with breast cancer, and the fraction of women without breast cancer who receive false positives, and focus only on the fraction of women with breast cancer who get positive results. For example, the vast majority of doctors in these studies seem to have thought that if around 80% of women with breast cancer have positive mammographies, then the probability of a women with a positive mammography having breast cancer must be around 80%.

Figuring out the final answer always requires *all three* pieces of information—the percentage of women with breast cancer, the percentage of women without breast cancer who receive false positives, and the percentage of women with breast cancer who receive (correct) positives.

The original proportion of patients with breast cancer is known as the *prior probability*. The chance that a patient with breast cancer gets a positive mammography, and the chance that a patient without breast cancer gets a positive mammography, are known as the two *conditional probabilities*. Collectively, this initial information is known as *the priors*. The final answer—the estimated probability that a patient has breast cancer, given that we know she has a positive result on her mammography—is known as the *revised probability* or the *posterior probability*. What we've just seen is that the posterior probability depends in part on the prior probability.

To see that the final answer always depends on the original fraction of women with breast cancer, consider an alternate universe in which only one woman out of a million has breast cancer. Even if mammography in this world detects breast cancer in 8 out of 10 cases, while returning a false positive on

a woman without breast cancer in only 1 out of 10 cases, there will still be a hundred thousand false positives for every real case of cancer detected. The original probability that a woman has cancer is so extremely low that, although a positive result on the mammography does *increase* the estimated probability, the probability isn't increased to certainty or even "a noticeable chance"; the probability goes from 1:1,000,000 to 1:100,000.

What this demonstrates is that the mammography result doesn't *replace* your old information about the patient's chance of having cancer; the mammography *slides* the estimated probability in the direction of the result. A positive result slides the original probability upward; a negative result slides the probability downward. For example, in the original problem where 1% of the women have cancer, 80% of women with cancer get positive mammographies, and 9.6% of women without cancer get positive mammographies, a positive result on the mammography *slides* the 1% chance upward to 7.8%.

Most people encountering problems of this type for the first time carry out the mental operation of *replacing* the original 1% probability with the 80% probability that a woman with cancer gets a positive mammography. It may seem like a good idea, but it just doesn't work. "The probability that a woman with a positive mammography has breast cancer" is not at all the same thing as "the probability that a woman with breast cancer has a positive mammography"; they are as unlike as apples and cheese.

Q. Why did the Bayesian reasoner cross the road?

A. You need more information to answer this question.

Suppose that a barrel contains many small plastic eggs. Some eggs are painted red and some are painted blue. 40% of the eggs in the bin contain pearls, and 60% contain nothing. 30% of eggs

containing pearls are painted blue, and 10% of eggs containing nothing are painted blue. What is the probability that a blue egg contains a pearl? For this example the arithmetic is simple enough that you may be able to do it in your head, and I would suggest trying to do so.

A more compact way of specifying the problem:

$$P(\text{pearl}) = 40\%$$

$$P(\text{blue}|\text{pearl}) = 30\%$$

$$P(\text{blue}|\neg\text{pearl}) = 10\%$$

$$P(\text{pearl}|\text{blue}) = ?$$

The symbol “ $\neg$ ” is shorthand for “not,” so $\neg\text{pearl}$ reads “not pearl.”

The notation $P(\text{blue}|\text{pearl})$ is shorthand for “the probability of blue given pearl” or “the probability that an egg is painted blue, given that the egg contains a pearl.” The item on the right side is what you *already know* or the *premise*, and the item on the left side is the *implication* or *conclusion*. If we have $P(\text{blue}|\text{pearl}) = 30\%$, and we *already know* that some egg contains a pearl, then we can *conclude* there is a 30% chance that the egg is painted blue. Thus, the final fact we’re looking for—“the chance that a blue egg contains a pearl” or “the probability that an egg contains a pearl, if we know the egg is painted blue”—reads $P(\text{pearl}|\text{blue})$.

40% of the eggs contain pearls, and 60% of the eggs contain nothing. 30% of the eggs containing pearls are painted blue, so 12% of the eggs altogether contain pearls and are painted blue. 10% of the eggs containing nothing are painted blue, so altogether 6% of the eggs contain nothing and are painted blue. A total of 18% of the eggs are painted blue, and a total of 12% of the eggs are painted blue and contain pearls, so the chance a blue egg contains a pearl is 12/18 or 2/3 or around 67%.

As before, we can see the necessity of all three pieces of information by considering extreme cases. In a (large) barrel in which only one egg out of a thousand contains a pearl, knowing that an egg is painted blue slides the probability from

0.1% to 0.3% (instead of sliding the probability from 40% to 67%). Similarly, if 999 out of 1,000 eggs contain pearls, knowing that an egg is blue slides the probability from 99.9% to 99.966%; the probability that the egg does *not* contain a pearl goes from 1/1,000 to around 1/3,000.

On the pearl-egg problem, most respondents unfamiliar with Bayesian reasoning would probably respond that the probability a blue egg contains a pearl is 30%, or perhaps 20% (the 30% chance of a true positive minus the 10% chance of a false positive). Even if this mental operation seems like a good idea at the time, it makes no sense in terms of the question asked. It's like the experiment in which you ask a second-grader: "If eighteen people get on a bus, and then seven more people get on the bus, how old is the bus driver?" Many second-graders will respond: "Twenty-five." They understand when they're being prompted to carry out a particular mental procedure, but they haven't quite connected the procedure to reality. Similarly, to find the probability that a woman with a positive mammography has breast cancer, it makes no sense whatsoever to *replace* the original probability that the woman has cancer with the probability that a woman with breast cancer gets a positive mammography. Neither can you subtract the probability of a false positive from the probability of the true positive. These operations are as wildly irrelevant as adding the number of people on the bus to find the age of the bus driver.

A study by Gigerenzer and Hoffrage in 1995 showed that some ways of phrasing story problems are much more evocative of correct Bayesian reasoning.⁵ The *least* evocative phrasing used probabilities. A slightly more evocative phrasing used frequencies instead of probabilities; the problem remained the same, but instead of saying that 1% of women had breast cancer, one would say that 1 out of 100 women had breast cancer, that 80 out of 100 women with breast cancer would get a positive mammography, and so on. Why did a higher proportion of

5. Ibid.

subjects display Bayesian reasoning on this problem? Probably because saying “1 out of 100 women” encourages you to concretely visualize X women with cancer, leading you to visualize X women with cancer and a positive mammography, etc.

The most effective presentation found so far is what's known as *natural frequencies*—saying that 40 out of 100 eggs contain pearls, 12 out of 40 eggs containing pearls are painted blue, and 6 out of 60 eggs containing nothing are painted blue. A *natural frequencies* presentation is one in which the information about the prior probability is included in presenting the conditional probabilities. If you were just learning about the eggs' conditional probabilities through natural experimentation, you would—in the course of cracking open a hundred eggs—crack open around 40 eggs containing pearls, of which 12 eggs would be painted blue, while cracking open 60 eggs containing nothing, of which about 6 would be painted blue. In the course of learning the conditional probabilities, you'd see examples of blue eggs containing pearls about twice as often as you saw examples of blue eggs containing nothing.

Unfortunately, while natural frequencies are a step in the right direction, it probably won't be enough. When problems are presented in natural frequencies, the proportion of people using Bayesian reasoning rises to around half. A big improvement, but not big enough when you're talking about real doctors and real patients.

Q. How can I find the priors for a problem?

A. Many commonly used priors are listed in the *Handbook of Chemistry and Physics*.

Q. Where do priors *originally* come from?

A. Never ask that question.

Q. Uh huh. Then where do scientists get their priors?

A. Priors for scientific problems are established by annual vote of the AAAS. In recent years the vote has become fractious and controversial, with

widespread acrimony, factional polarization, and several outright assassinations. This may be a front for infighting within the Bayes Council, or it may be that the disputants have too much spare time. No one is really sure.

Q. I see. And where does everyone else get their priors?

A. They download their priors from Kazaa.

Q. What if the priors I want aren't available on Kazaa?

A. There's a small, cluttered antique shop in a back alley of San Francisco's Chinatown. *Don't ask about the bronze rat.*

Actually, priors are true or false just like the final answer—they reflect reality and can be judged by comparing them against reality. For example, if you think that 920 out of 10,000 women in a sample have breast cancer, and the actual number is 100 out of 10,000, then your priors are wrong. For our particular problem, the priors might have been established by three studies—a study on the case histories of women with breast cancer to see how many of them tested positive on a mammography, a study on women without breast cancer to see how many of them test positive on a mammography, and an epidemiological study on the prevalence of breast cancer in some specific demographic.

The probability $P(A, B)$ is the same as $P(B, A)$, but $P(A|B)$ is not the same thing as $P(B|A)$, and $P(A, B)$ is completely different from $P(A|B)$. It's a common confusion to mix up some or all of these quantities.

To get acquainted with all the relationships between them, we'll play "follow the degrees of freedom." For example, the two quantities $P(\text{cancer})$ and $P(\neg\text{cancer})$ have one degree of freedom between them, because of the general law $P(A) + P(\neg A) = 1$. If you know that $P(\neg\text{cancer}) = 0.99$, you can obtain $P(\text{cancer}) = 1 - P(\neg\text{cancer}) = 0.01$.

The quantities $P(\text{positive}|\text{cancer})$ and $P(\neg\text{positive}|\text{cancer})$ also have only one degree of freedom between them; either a woman with breast cancer gets a positive mammography or she doesn't. On the other hand, $P(\text{positive}|\text{cancer})$ and $P(\text{positive}|\neg\text{cancer})$ have *two* degrees of freedom. You can have a mammography test that returns positive for 80% of cancer patients and 9.6% of healthy patients, or that returns positive for 70% of cancer patients and 2% of healthy patients, or even a health test that returns "positive" for 30% of cancer patients and 92% of healthy patients. The two quantities, the output of the mammography test for cancer patients and the output of the mammography test for healthy patients, are in mathematical terms independent; one cannot be obtained from the other in any way, and so they have two degrees of freedom between them.

What about $P(\text{positive}, \text{cancer})$, $P(\text{positive}|\text{cancer})$, and $P(\text{cancer})$? Here we have three quantities; how many degrees of freedom are there? In this case the equation that must hold is

$$P(\text{positive}, \text{cancer}) = P(\text{positive}|\text{cancer}) \times P(\text{cancer}).$$

This equality reduces the degrees of freedom by one. If we know the fraction of patients with cancer, and the chance that a cancer patient has a positive mammography, we can deduce the fraction of patients who have breast cancer *and* a positive mammography by multiplying.

Similarly, if we know the number of patients with breast cancer and positive mammographies, and also the number of patients with breast cancer, we can estimate the chance that a woman with breast cancer gets a positive mammography by dividing: $P(\text{positive}|\text{cancer}) = P(\text{positive}, \text{cancer})/P(\text{cancer})$. In fact, this is exactly how such medical diagnostic tests are calibrated; you do a study on 8,520 women with breast cancer and see that there are 6,816 (or thereabouts) women with breast cancer and positive mammographies, then divide 6,816 by 8,520 to find that 80% of women with breast cancer had positive mammographies. (Incidentally, if you accidentally divide 8,520 by 6,816 instead of the other way around, your calcu-

lations will start doing strange things, such as insisting that 125% of women with breast cancer and positive mammographies have breast cancer. This is a common mistake in carrying out Bayesian arithmetic, in my experience.) And finally, if you know $P(\text{positive, cancer})$ and $P(\text{positive}|\text{cancer})$, you can deduce how many cancer patients there must have been originally. There are two degrees of freedom shared out among the three quantities; if we know any two, we can deduce the third.

How about $P(\text{positive})$, $P(\text{positive, cancer})$, and $P(\text{positive, } \neg\text{cancer})$? Again there are only two degrees of freedom among these three variables. The equation occupying the extra degree of freedom is

$$P(\text{positive}) = P(\text{positive, cancer}) + P(\text{positive, } \neg\text{cancer}).$$

This is how $P(\text{positive})$ is computed to begin with; we figure out the number of women with breast cancer who have positive mammographies, and the number of women without breast cancer who have positive mammographies, then add them together to get the total number of women with positive mammographies. It would be very strange to go out and conduct a study to determine the number of women with positive mammographies—just that one number and nothing else—but in theory you could do so. And if you then conducted another study and found the number of those women who had positive mammographies *and* breast cancer, you would also know the number of women with positive mammographies and *no* breast cancer—either a woman with a positive mammography has breast cancer or she doesn't. In general, $P(A, B) + P(A, \neg B) = P(A)$. Symmetrically, $P(A, B) + P(\neg A, B) = P(B)$.

What about $P(\text{positive, cancer})$, $P(\text{positive, } \neg\text{cancer})$, $P(\neg\text{positive, cancer})$, and $P(\neg\text{positive, } \neg\text{cancer})$? You might at first be tempted to think that there are only two degrees of freedom for these four quantities—that you can, for example, get $P(\text{positive, } \neg\text{cancer})$ by multiplying $P(\text{positive}) \times P(\neg\text{cancer})$, and thus that all four quantities can be found given only the two quantities $P(\text{positive})$ and $P(\text{cancer})$. This is not the

case! $P(\text{positive}, \neg\text{cancer}) = P(\text{positive}) \times P(\neg\text{cancer})$ only if the two probabilities are *statistically independent*—if the chance that a woman has breast cancer has no bearing on whether she has a positive mammography. This amounts to requiring that the two conditional probabilities be equal to each other—a requirement which would eliminate one degree of freedom. If you remember that these four quantities are the groups A , B , C , and D , you can look over those four groups and realize that, in theory, you can put any number of people into the four groups. If you start with a group of 80 women with breast cancer and positive mammographies, there's no reason why you can't add another group of 500 women with breast cancer and negative mammographies, followed by a group of 3 women without breast cancer and negative mammographies, and so on. So now it seems like the four quantities have four degrees of freedom. And they would, except that in expressing them as *probabilities*, we need to normalize them to *fractions* of the complete group, which adds the constraint that $P(\text{positive}, \text{cancer}) + P(\text{positive}, \neg\text{cancer}) + P(\neg\text{positive}, \text{cancer}) + P(\neg\text{positive}, \neg\text{cancer}) = 1$. This equation takes up one degree of freedom, leaving three degrees of freedom among the four quantities. If you specify the *fractions* of women in groups A , B , and D , you can deduce the fraction of women in group C .

Given the four groups A , B , C , and D , it is very straightforward to compute everything else:

$$P(\text{cancer}) = \frac{A + B}{A + B + C + D}$$

$$P(\neg\text{positive}|\text{cancer}) = \frac{B}{A + B},$$

and so on. Since $\{A, B, C, D\}$ contains three degrees of freedom, it follows that the entire set of probabilities relating cancer rates to test results contains only three degrees of freedom. Remember that in our problems we always needed *three* pieces of information—the prior probability and the two conditional probabilities—which, indeed, have three degrees of freedom among them. Actually, for Bayesian problems, *any* three quan-

tities with three degrees of freedom between them should logically specify the entire problem.

The probability that a test gives a true positive divided by *the probability that a test gives a false positive* is known as the *likelihood ratio* of that test. The likelihood ratio for a positive result summarizes how much a positive result will slide the prior probability. Does the likelihood ratio of a medical test then sum up everything there is to know about the usefulness of the test?

No, it does not! The likelihood ratio sums up everything there is to know about the *meaning* of a *positive* result on the medical test, but the meaning of a *negative* result on the test is not specified, nor is the frequency with which the test is useful. For example, a mammography with a hit rate of 80% for patients with breast cancer and a false positive rate of 9.6% for healthy patients has the same likelihood ratio as a test with an 8% hit rate and a false positive rate of 0.96%. Although these two tests have the same likelihood ratio, the first test is more useful in every way—it detects disease more often, and a negative result is stronger evidence of health.

Suppose that you apply *two* tests for breast cancer in succession—say, a standard mammography and also some other test which is *independent* of mammography. Since I don't know of any such test that is independent of mammography, I'll invent one for the purpose of this problem, and call it the Tams-Braylor Division Test, which checks to see if any cells are dividing more rapidly than other cells. We'll suppose that the Tams-Braylor gives a true positive for 90% of patients with breast cancer, and gives a false positive for 5% of patients without cancer. Let's say the prior prevalence of breast cancer is 1%. If a patient gets a positive result on her mammography *and* her Tams-Braylor, what is the revised probability she has breast cancer?

One way to solve this problem would be to take the revised probability for a positive mammography, which we already calculated as 7.8%, and plug that into the Tams-Braylор test as the new prior probability. If we do this, we find that the result comes out to 60%.

Suppose that the prior prevalence of breast cancer in a demographic is 1%. Suppose that we, as doctors, have a repertoire of three independent tests for breast cancer. Our first test, test A, a mammography, has a likelihood ratio of $80\%/9.6\% = 8.33$. The second test, test B, has a likelihood ratio of 18.0 (for example, from 90% versus 5%); and the third test, test C, has a likelihood ratio of 3.5 (which could be from 70% versus 20%, or from 35% versus 10%; it makes no difference). Suppose a patient gets a positive result on all three tests. What is the probability the patient has breast cancer?

Here's a fun trick for simplifying the bookkeeping. If the prior prevalence of breast cancer in a demographic is 1%, then 1 out of 100 women have breast cancer, and 99 out of 100 women do not have breast cancer. So if we rewrite the *probability* of 1% as an *odds ratio*, the odds are 1:99.

And the likelihood ratios of the three tests A, B, and C are:

$$8.33 : 1 = 25 : 3$$

$$18.0 : 1 = 18 : 1$$

$$3.5 : 1 = 7 : 2.$$

The *odds* for women with breast cancer who score positive on all three tests, versus women without breast cancer who score positive on all three tests, will equal:

$$1 \times 25 \times 18 \times 7 : 99 \times 3 \times 1 \times 2 = 3150 : 594.$$

To recover the probability from the odds, we just write:

$$3150/(3150 + 594) = 84\%.$$

This always works regardless of how the odds ratios are written; i.e., 8.33:1 is just the same as 25:3 or 75:9. It doesn't

matter in what order the tests are administered, or in what order the results are computed. The proof is left as an exercise for the reader.

E. T. Jaynes, in *Probability Theory With Applications in Science and Engineering*, suggests that credibility and evidence should be measured in decibels.^{6,7}

Decibels?

Decibels are used for measuring exponential differences of intensity. For example, if the sound from an automobile horn carries 10,000 times as much energy (per square meter per second) as the sound from an alarm clock, the automobile horn would be 40 decibels louder. The sound of a bird singing might carry 1,000 times less energy than an alarm clock, and hence would be 30 decibels softer. To get the number of decibels, you take the logarithm base 10 and multiply by 10:

$$\text{decibels} = 10 \log_{10}(\text{intensity})$$

or

$$\text{intensity} = 10^{\text{decibels}/10}.$$

Suppose we start with a prior probability of 1% that a woman has breast cancer, corresponding to an odds ratio of 1:99. And then we administer three tests of likelihood ratios 25:3, 18:1, and 7:2. You *could* multiply those numbers ...or you could just add their logarithms:

$$10 \log_{10}(1/99) \approx -20$$

$$10 \log_{10}(25/3) \approx 9$$

$$10 \log_{10}(18/1) \approx 13$$

$$10 \log_{10}(7/2) \approx 5.$$

6. Edwin T. Jaynes, "Probability Theory, with Applications in Science and Engineering," Unpublished manuscript (1974).

7. Edwin T. Jaynes, *Probability Theory, The Logic of Science*, pg 91, (Cambridge University Press, 2003)

It starts out as fairly unlikely that a woman has breast cancer—our credibility level is at -20 decibels. Then three test results come in, corresponding to 9, 13, and 5 decibels of evidence. This raises the credibility level by a total of 27 decibels, meaning that the prior credibility of -20 decibels goes to a posterior credibility of 7 decibels. So the odds go from 1:99 to 5:1, and the probability goes from 1% to around 83%.

You are a mechanic for gizmos. When a gizmo stops working, it is due to a blocked hose 30% of the time. If a gizmo's hose is blocked, there is a 45% probability that prodding the gizmo will produce sparks. If a gizmo's hose is unblocked, there is only a 5% chance that prodding the gizmo will produce sparks. A customer brings you a malfunctioning gizmo. You prod the gizmo and find that it produces sparks. What is the probability that a spark-producing gizmo has a blocked hose?

What is the sequence of arithmetical operations that you performed to solve this problem?

$$(45\% \times 30\%) / (45\% \times 30\% + 5\% \times 70\%)$$

Similarly, to find the chance that a woman with positive mammography has breast cancer, we computed:

$$\frac{P(\text{positive}|\text{cancer}) \times P(\text{cancer})}{\left(\begin{array}{l} P(\text{positive}|\text{cancer}) \times P(\text{cancer}) \\ + P(\text{positive}|\neg\text{cancer}) \times P(\neg\text{cancer}) \end{array} \right)}$$

which is

$$\frac{P(\text{positive, cancer})}{P(\text{positive, cancer}) + P(\text{positive, } \neg\text{cancer})}$$

which is

$$\frac{P(\text{positive, cancer})}{P(\text{positive})}$$

which is

$$P(\text{cancer}|\text{positive}).$$

The fully general form of this calculation is known as *Bayes's Theorem* or *Bayes's Rule*.

Bayes's Theorem:

$$P(A|X) = \frac{P(X|A) \times P(A)}{P(X|A) \times P(A) + P(X|\neg A) \times P(\neg A)}.$$

When there is some phenomenon A that we want to investigate, and an observation X that is evidence about A —for example, in the previous example, A is breast cancer and X is a positive mammography—Bayes's Theorem tells us how we should *update* our probability of A , given the *new evidence* X .

By this point, Bayes's Theorem may seem blatantly obvious or even tautological, rather than exciting and new. If so, this introduction has *entirely succeeded* in its purpose.

Bayes's Theorem describes what makes something “evidence” and how much evidence it is. Statistical models are judged by comparison to the *Bayesian method* because, in statistics, the Bayesian method is as good as it gets—the Bayesian method defines the maximum amount of mileage you can get out of a given piece of evidence, in the same way that thermodynamics defines the maximum amount of work you can get out of a temperature differential. This is why you hear cognitive scientists talking about *Bayesian reasoners*. In cognitive science, *Bayesian reasoner* is the technically precise code word that we use to mean *rational mind*.

There are also a number of general heuristics about human reasoning that you can learn from looking at Bayes's Theorem.

For example, in many discussions of Bayes's Theorem, you may hear cognitive psychologists saying that people *do not*

take prior frequencies sufficiently into account, meaning that when people approach a problem where there's some evidence X indicating that condition A might hold true, they tend to judge A 's likelihood solely by how well the evidence X seems to match A , without taking into account the prior frequency of A . If you think, for example, that under the mammography example, the woman's chance of having breast cancer is in the range of 70%–80%, then this kind of reasoning is insensitive to the prior frequency given in the problem; it doesn't notice whether 1% of women or 10% of women start out having breast cancer. "Pay more attention to the prior frequency!" is one of the many things that humans need to bear in mind to partially compensate for our built-in inadequacies.

A related error is to pay too much attention to $P(X|A)$ and not enough to $P(X|\neg A)$ when determining how much evidence X is for A . The degree to which a result X is *evidence for* A depends not only on the strength of the statement *we'd expect to see result X if A were true*, but also on the strength of the statement *we **wouldn't** expect to see result X if A weren't true*. For example, if it is raining, this very strongly implies the grass is wet— $P(\text{wetgrass}|\text{rain}) \approx 1$ —but seeing that the grass is wet doesn't necessarily mean that it has just rained; perhaps the sprinkler was turned on, or you're looking at the early morning dew. Since $P(\text{wetgrass}|\neg\text{rain})$ is substantially greater than zero, $P(\text{rain}|\text{wetgrass})$ is substantially less than one. On the other hand, if the grass was *never* wet when it wasn't raining, then knowing that the grass was wet would *always* show that it was raining, $P(\text{rain}|\text{wetgrass}) \approx 1$, even if $P(\text{wetgrass}|\text{rain}) = 50\%$; that is, even if the grass only got wet 50% of the times it rained. Evidence is always the result of the *differential* between the two conditional probabilities. *Strong* evidence is not the product of a very high probability that A leads to X , but the product of a very *low* probability that *not- A* could have led to X .

The *Bayesian revolution in the sciences* is fueled, not only by more and more cognitive scientists suddenly noticing that mental phenomena have Bayesian structure in them; not only by scientists in every field learning to judge their statistical methods by comparison with the Bayesian method; but also by

the idea that *science itself is a special case of Bayes's Theorem; experimental evidence is Bayesian evidence*. The Bayesian revolutionaries hold that when you perform an experiment and get evidence that “confirms” or “disconfirms” your theory, this confirmation and disconfirmation is governed by the Bayesian rules. For example, you have to take into account not only whether your theory predicts the phenomenon, but whether other possible explanations also predict the phenomenon.

Previously, the most popular philosophy of science was probably Karl Popper's *falsificationism*—this is the old philosophy that the Bayesian revolution is currently dethroning. Karl Popper's idea that theories can be definitely falsified, but never definitely confirmed, is yet another special case of the Bayesian rules; if $P(X|A) \approx 1$ —if the theory makes a definite prediction—then observing $\neg X$ very strongly falsifies A . On the other hand, if $P(X|A) \approx 1$, and we observe X , this doesn't definitely confirm the theory; there might be some other condition B such that $P(X|B) \approx 1$, in which case observing X doesn't favor A over B . For observing X to definitely confirm A , we would have to know, not that $P(X|A) \approx 1$, but that $P(X|\neg A) \approx 0$, which is something that we can't know because we can't range over all possible alternative explanations. For example, when Einstein's theory of General Relativity toppled Newton's incredibly well-confirmed theory of gravity, it turned out that all of Newton's predictions were just a special case of Einstein's predictions.

You can even formalize Popper's philosophy mathematically. The likelihood ratio for X , the quantity $P(X|A)/P(X|\neg A)$, determines how much observing X slides the probability for A ; the likelihood ratio is what says *how strong* X is as evidence. Well, in your theory A , you can predict X with probability 1, if you like; but you can't control the denominator of the likelihood ratio, $P(X|\neg A)$ —there will always be some alternative theories that also predict X , and while we go with the simplest theory that fits the current evidence, you may someday encounter some evidence that an alternative theory predicts but your theory does not. That's the hidden gotcha that toppled Newton's theory of gravity. So there's a limit on how much mileage you

can get from successful predictions; there's a limit on how high the likelihood ratio goes for *confirmatory* evidence.

On the other hand, if you encounter some piece of evidence Y that is definitely *not* predicted by your theory, this is *enormously* strong evidence against your theory. If $P(Y|A)$ is infinitesimal, then the likelihood ratio will also be infinitesimal. For example, if $P(Y|A)$ is 0.0001%, and $P(Y|\neg A)$ is 1%, then the likelihood ratio $P(Y|A)/P(Y|\neg A)$ will be 1:10,000. That's -40 decibels of evidence! Or, flipping the likelihood ratio, if $P(Y|A)$ is *very small*, then $P(Y|\neg A)/P(Y|A)$ will be very large, meaning that observing Y greatly favors $\neg A$ over A . Falsification is much stronger than confirmation. This is a consequence of the earlier point that *very strong* evidence is not the product of a very high probability that A leads to X , but the product of a very *low* probability that *not-A* could have led to X . This is the precise Bayesian rule that underlies the heuristic value of Popper's falsificationism.

Similarly, Popper's dictum that an idea must be falsifiable can be interpreted as a manifestation of the Bayesian conservation-of-probability rule; if a result X is positive evidence for the theory, then the result $\neg X$ would have disconfirmed the theory to some extent. If you try to interpret both X and $\neg X$ as "confirming" the theory, the Bayesian rules say this is impossible! To increase the probability of a theory you *must* expose it to tests that can potentially decrease its probability; this is not just a rule for detecting would-be cheaters in the social process of science, but a consequence of Bayesian probability theory. On the other hand, Popper's idea that there is *only* falsification and *no such thing* as confirmation turns out to be incorrect. Bayes's Theorem shows that falsification is *very strong* evidence compared to confirmation, but falsification is still probabilistic in nature; it is not governed by fundamentally different rules from confirmation, as Popper argued.

So we find that many phenomena in the cognitive sciences, plus the statistical methods used by scientists, plus the scientific method itself, are all turning out to be special cases of Bayes's Theorem. Hence the Bayesian revolution.

Having introduced Bayes's Theorem explicitly, we can explicitly discuss its components.

$$P(A|X) = \frac{P(X|A) \times P(A)}{P(X|A) \times P(A) + P(X|\neg A) \times P(\neg A)}$$

We'll start with $P(A|X)$. If you ever find yourself getting confused about what's A and what's X in Bayes's Theorem, start with $P(A|X)$ on the left side of the equation; that's the simplest part to interpret. In $P(A|X)$, A is the thing we want to know about. X is how we're observing it; X is the evidence we're using to make inferences about A . Remember that for every expression $P(Q|P)$, we want to know about the probability for Q given P , the degree to which P implies Q —a more sensible notation, which it is now too late to adopt, would be $P(Q \leftarrow P)$.

$P(Q|P)$ is closely related to $P(Q, P)$, but they are not identical. Expressed as a probability or a fraction, $P(Q, P)$ is the proportion of things that have property Q and property P among all things; e.g., the proportion of “women with breast cancer and a positive mammography” within the group of all women. If the total number of women is 10,000, and 80 women have breast cancer and a positive mammography, then $P(Q, P)$ is $80/10,000 = 0.8\%$. You might say that the absolute quantity, 80, is being normalized to a probability relative to the group of all women. Or to make it clearer, suppose that there's a group of 641 women with breast cancer and a positive mammography within a total sample group of 89,031 women. Six hundred and forty-one is the absolute quantity. If you pick out a random woman from the entire sample, then the probability you'll pick a woman with breast cancer and a positive mammography is $P(Q, P)$, or 0.72% (in this example).

On the other hand, $P(Q|P)$ is the proportion of things that have property Q and property P among *all things that have P* ; e.g., the proportion of women with breast cancer and a positive mammography within the group of *all women with positive mammographies*. If there are 641 women with breast cancer and positive mammographies, 7,915 women with positive mammographies, and 89,031 women, then $P(Q, P)$ is the probability

of getting one of those 641 women if you're picking at random from the entire group of 89,031, while $P(Q|P)$ is the probability of getting one of those 641 women if you're picking at random from the smaller group of 7,915.

In a sense, $P(Q|P)$ really means $P(Q, P|P)$, but specifying the extra P all the time would be redundant. You already *know* it has property P , so the property you're *investigating* is Q —even though you're looking at the size of group (Q, P) within group P , not the size of group Q within group P (which would be nonsense). This is what it means to take the property on the right-hand side as *given*; it means you know you're working only within the group of things that have property P . When you constrict your focus of attention to see only this smaller group, many other probabilities change. If you're taking P as *given*, then $P(Q, P)$ equals just $P(Q)$ —at least, *relative to the group P*. The *old* $P(Q)$, the frequency of “things that have property Q within the entire sample,” is revised to the new frequency of “things that have property Q within the subsample of things that have property P .” If P is *given*, if P is our entire world, then looking for (Q, P) is the same as looking for just Q .

If you constrict your focus of attention to only the population of eggs that are painted blue, then suddenly “the probability that an egg contains a pearl” becomes a different number; this proportion is different for the population of blue eggs than the population of all eggs. The *given*, the property that constricts our focus of attention, is always on the *right* side of $P(Q|P)$; the P becomes our world, the entire thing we see, and on the other side of the “given” P always has probability 1—that is what it means to take P as given. So $P(Q|P)$ means “If P has probability 1, what is the probability of Q ?” or “If we constrict our attention to only things or events where P is true, what is the probability of Q ?” The statement Q , on the other side of the given, is *not* certain—its probability may be 10% or 90% or any other number. So when you use Bayes's Theorem, and you write the part on the left side as $P(A|X)$ —how to *update* the probability of A after seeing X , the new probability of A *given* that we know X , the degree to which X *implies* A —you can tell that X is always the *observation* or the *evidence*, and A is

the property being investigated, the thing you want to know about.

The right side of Bayes's Theorem is derived from the left side through these steps:

$$\begin{aligned}
 P(A|X) &= P(A|X) \\
 P(A|X) &= \frac{P(X, A)}{P(X)} \\
 P(A|X) &= \frac{P(X, A)}{P(X, A) + P(X, \neg A)} \\
 P(A|X) &= \frac{P(X|A) \times P(A)}{P(X|A) \times P(A) + P(X|\neg A) \times P(\neg A)}.
 \end{aligned}$$

Once the derivation is finished, all the implications on the right side of the equation are of the form $P(X|A)$ or $P(X|\neg A)$, while the implication on the left side is $P(A|X)$. The symmetry arises because the elementary *causal relations* are generally implications from facts to observations, e.g., from breast cancer to positive mammography. The elementary *steps in reasoning* are generally implications from observations to facts, e.g., from a positive mammography to breast cancer. The left side of Bayes's Theorem is an elementary *inferential* step from the observation of positive mammography to the conclusion of an increased probability of breast cancer. Implication is written right-to-left, so we write $P(\text{cancer}|\text{positive})$ on the left side of the equation. The right side of Bayes's Theorem describes the elementary *causal* steps—for example, from breast cancer to a positive mammography—and so the implications on the right side of Bayes's Theorem take the form $P(\text{positive}|\text{cancer})$ or $P(\text{positive}|\neg\text{cancer})$.

And that's Bayes's Theorem. Rational inference on the left end, physical causality on the right end; an equation with mind on one side and reality on the other. Remember how the scientific method turned out to be a special case of Bayes's Theorem? If you wanted to put it poetically, you could say that Bayes's Theorem binds reasoning into the physical universe.

INTERLUDE: AN INTUITIVE EXPLANATION OF BAYES'S THEOREM

Okay, we're done.

Reverend Bayes says:



You are now an initiate of the Bayesian Conspiracy.

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